Active Directory and Policies

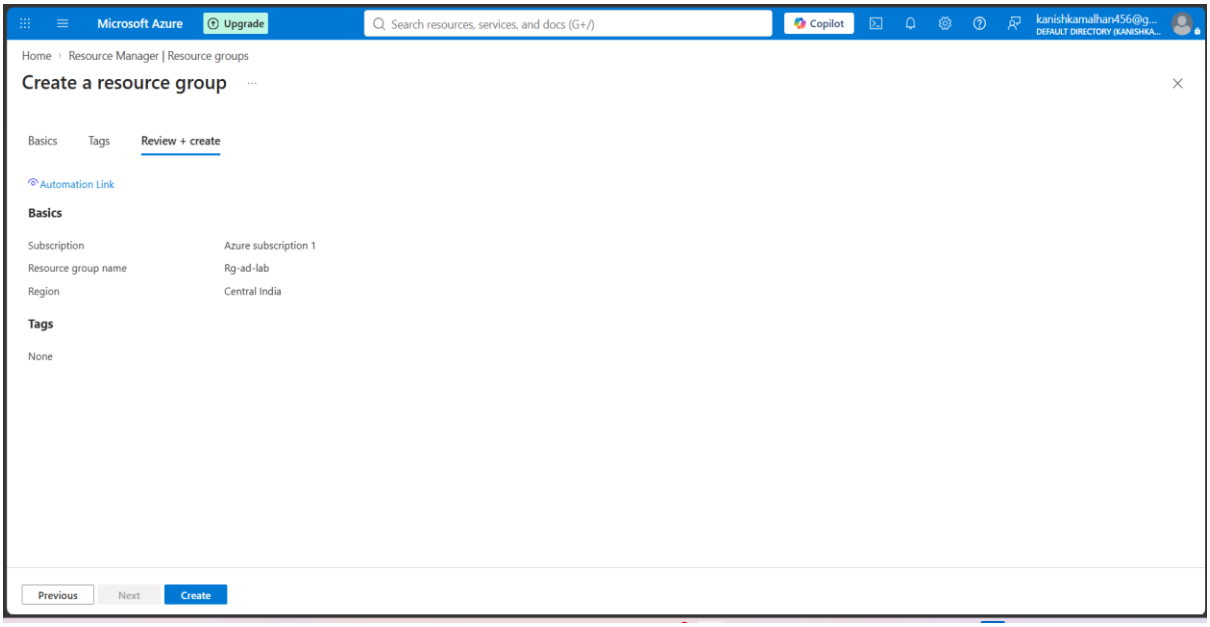
Part 1– Create the Azure Resources (Resource Group, Network, VMs)

Step 1: Create the resource group

In the Azure portal open Resource groups and click Create.

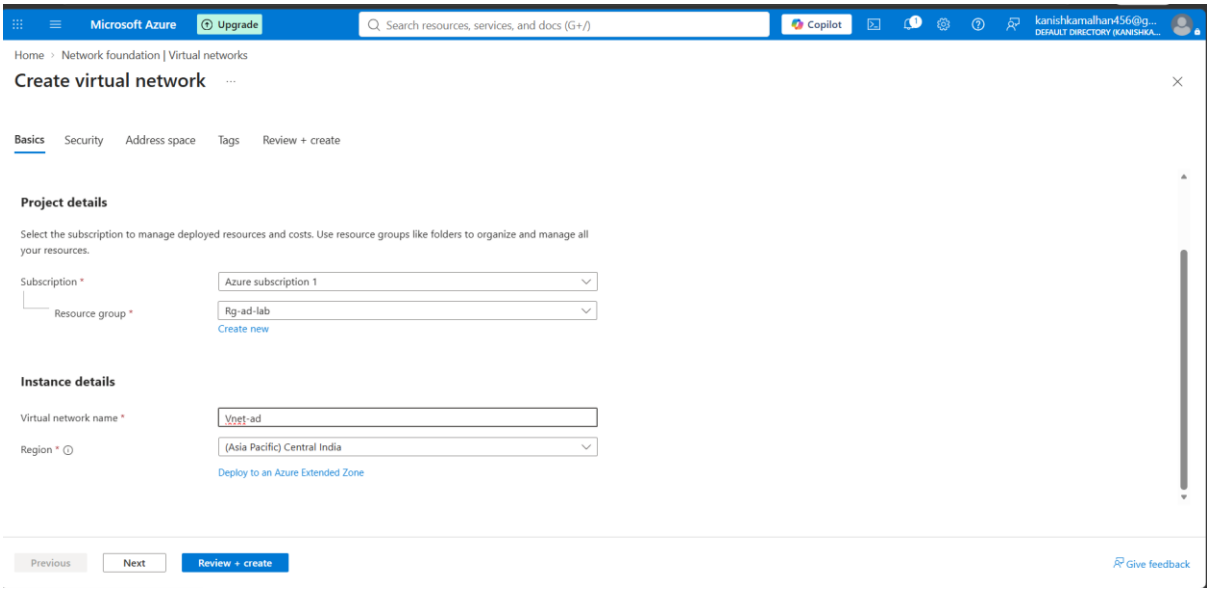
then go to Review + create and click Create.

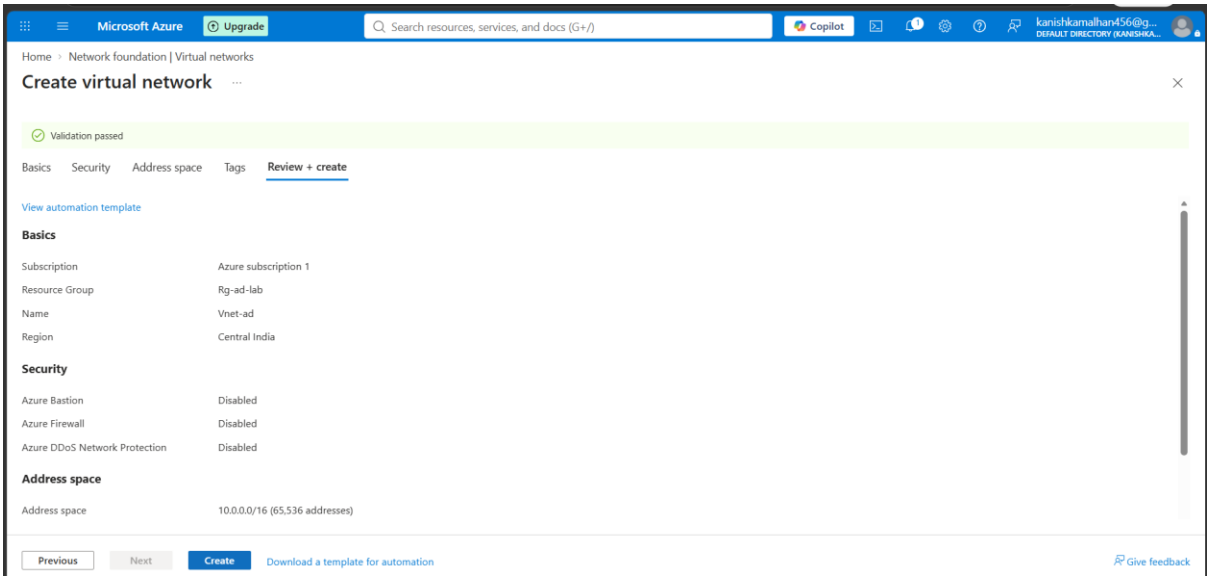
- Resource group name: Rg-ad-lab
- Region: Central India



Step 2: Start creating the virtual network

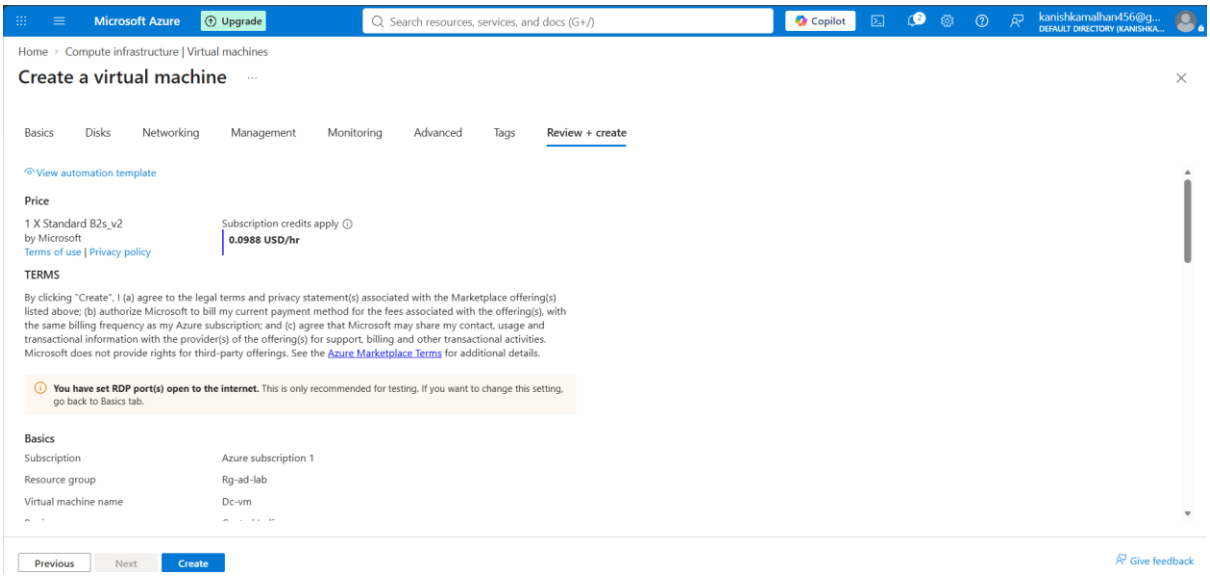
- Resource group: Rg-ad-lab
- Virtual network name: Vnet-ad
- Region: (Asia Pacific) Central India





Step 3: Create the domain controller VM (Dc-vm)

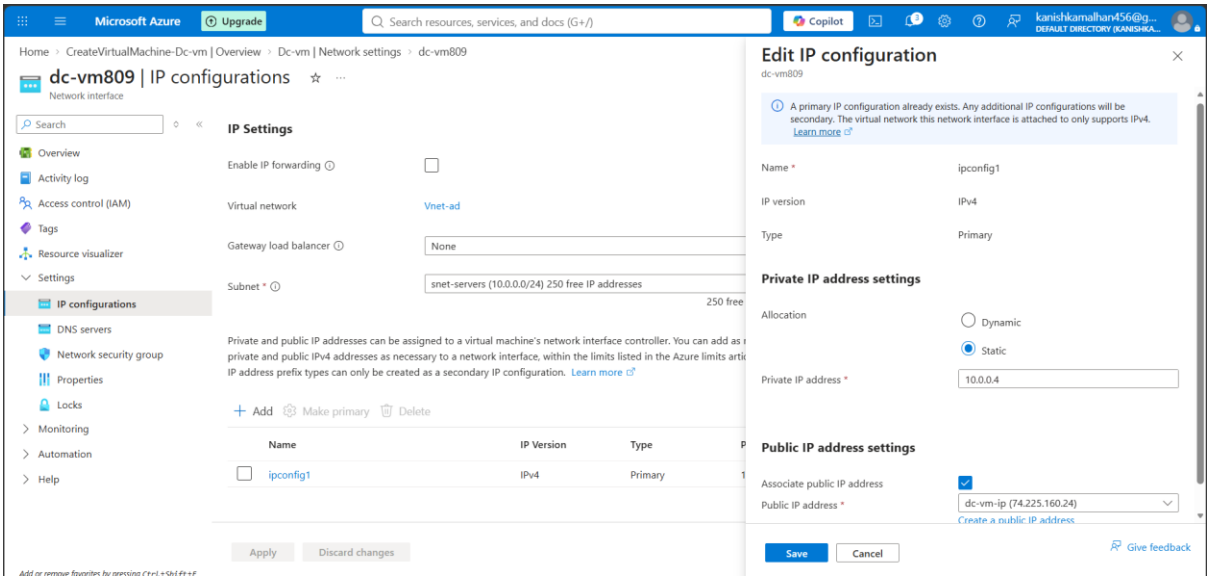
- Go to Virtual machines → Create → Azure virtual machine.
- Name the VM Dc-vm in
- resource group: Rg-ad-lab
- Windows Server 2022 Datacenter and size Standard B2s_v2.
- Review + create click Create.



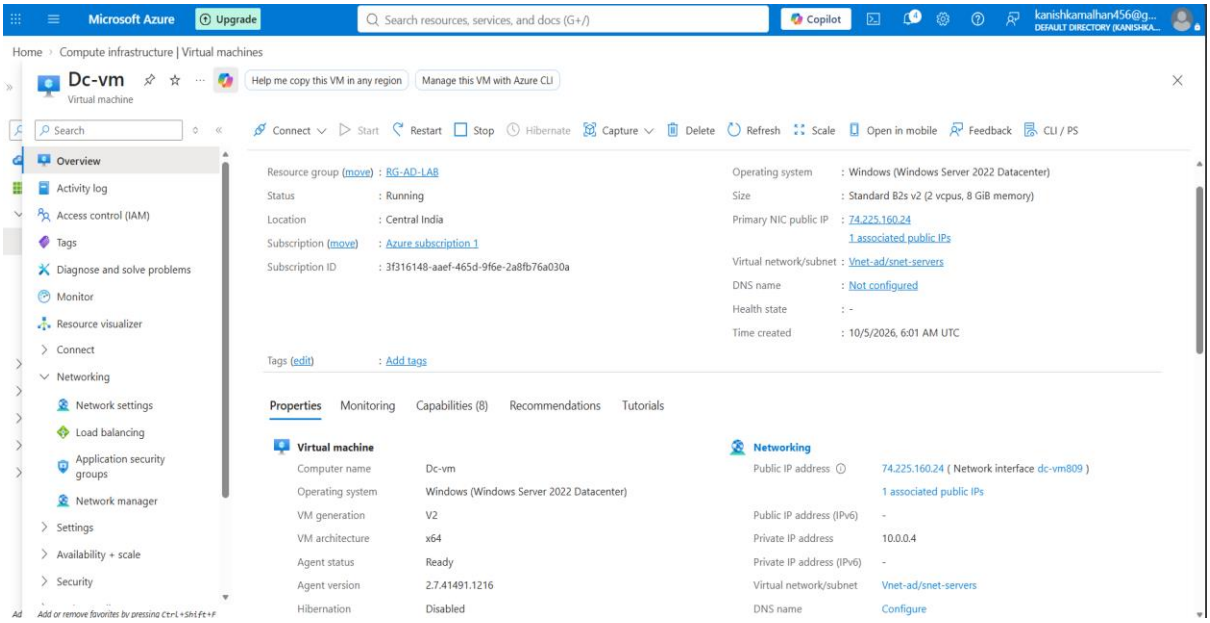
Step 4: Change the DC's private IP from Dynamic to Static

Open Dc-vm → Network settings → the network interface (dc-vm809) → IP configurations → ipconfig1. Under Private IP address settings set Allocation to Static, keep 10.0.0.4, then click Save.

- Allocation: Static
- Private IP address: 10.0.0.4
- Public IP: dc-vm-ip



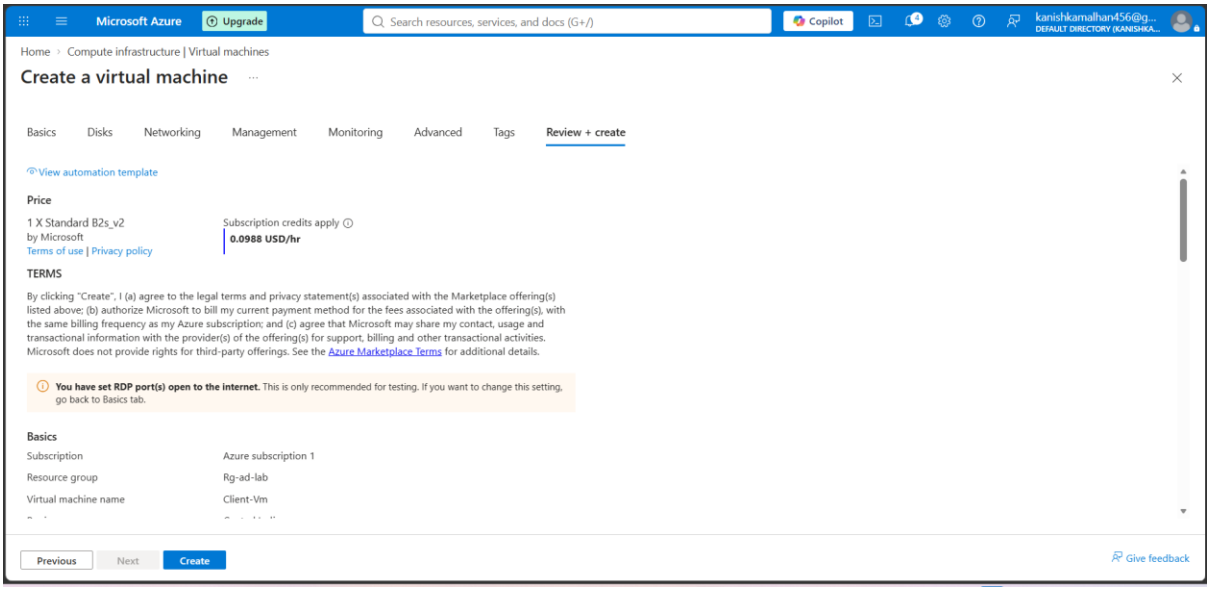
Step 5: Check that Dc-vm is running



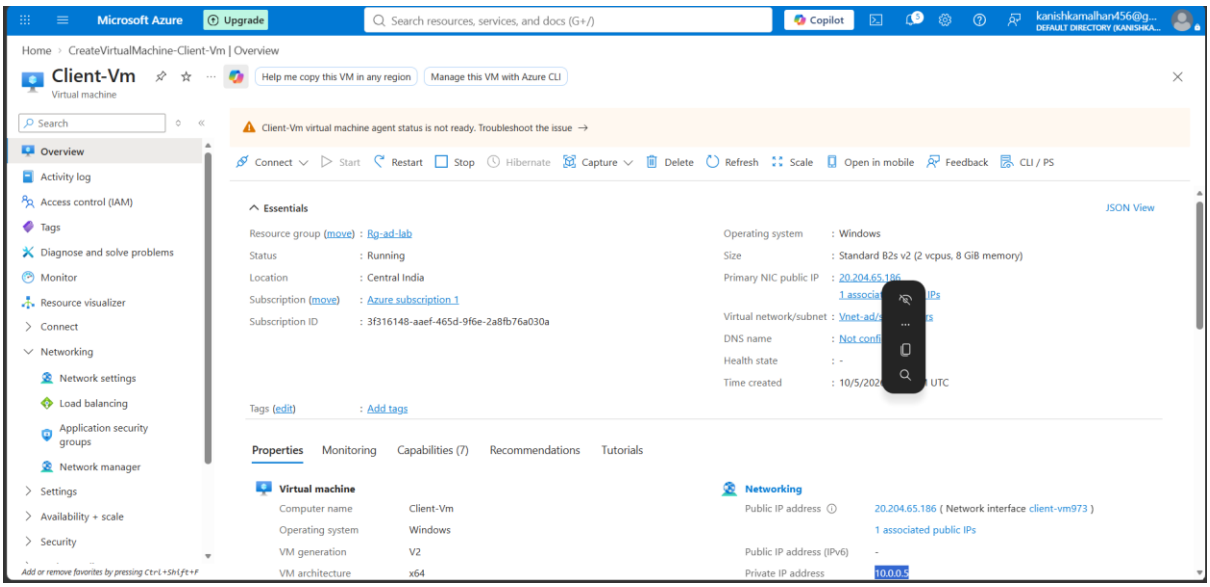
Step 6: Create the client VM (Client-Vm)

Repeat the VM creation for the client machine

Name it Client-Vm in the same resource group with the same size (Standard B2s_v2), then click Create on the Review + create tab.



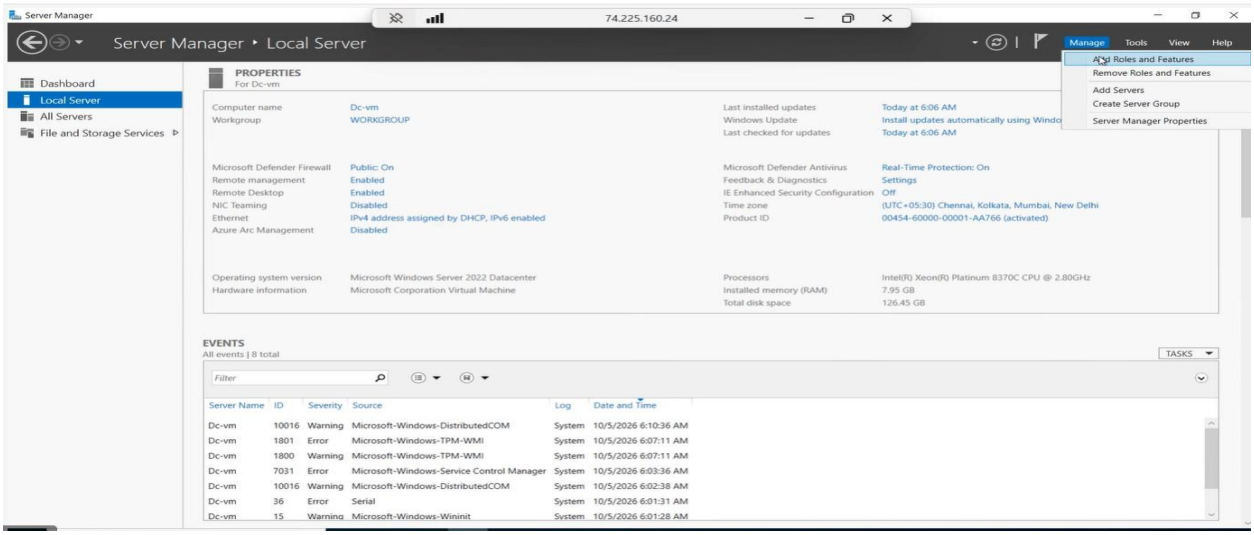
Step 7: Check that Client-Vm is running



Part 2 – Install the Active Directory Domain Services Role on Dc-vm

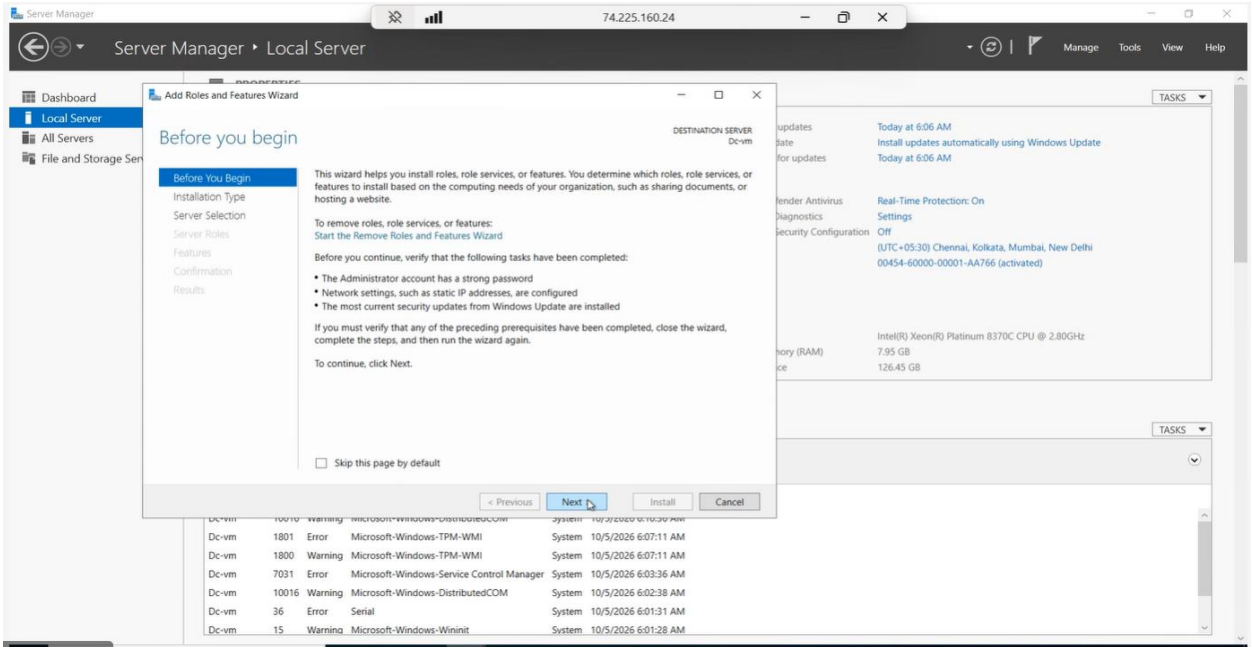
Step 8: Connect to Dc-vm and open Add Roles and Features

Connect to Dc-vm with Remote Desktop using its public IP (74.225.160.24) and the azureuser account. In Server Manager go to Manage → Add Roles and Features.



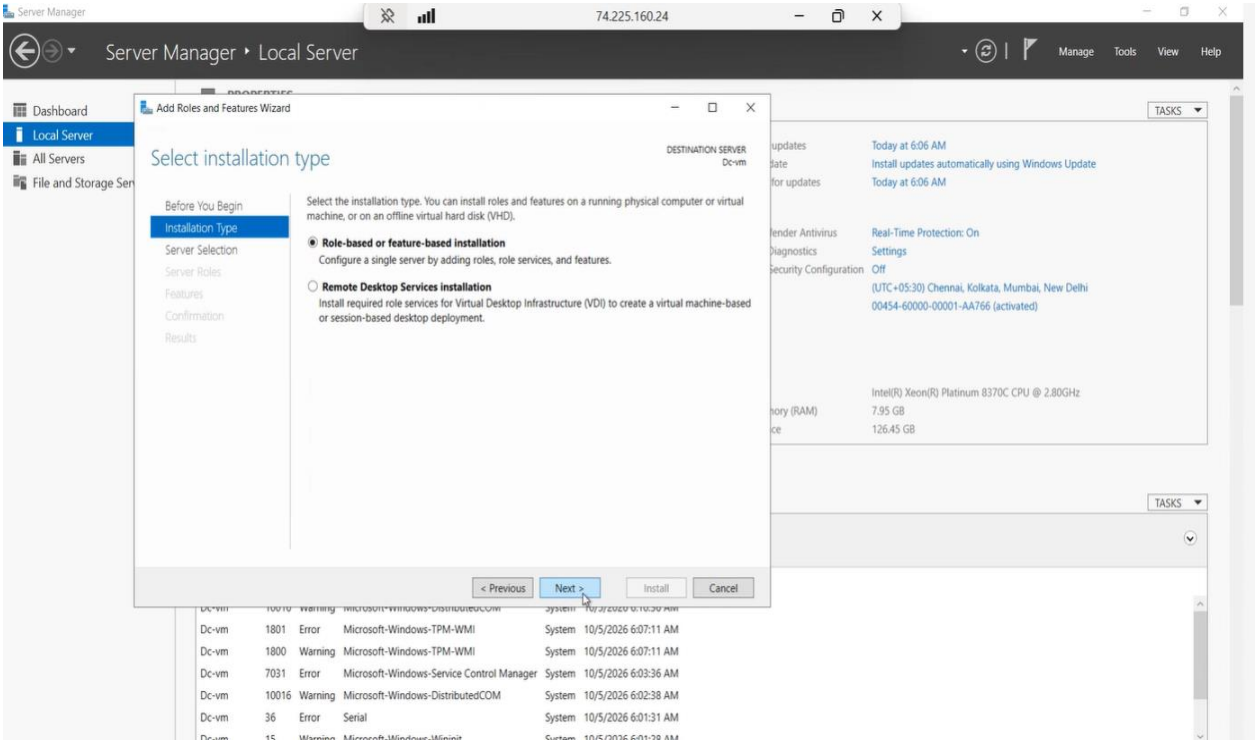
Step 9: Before you begin

Read the page and click Next.



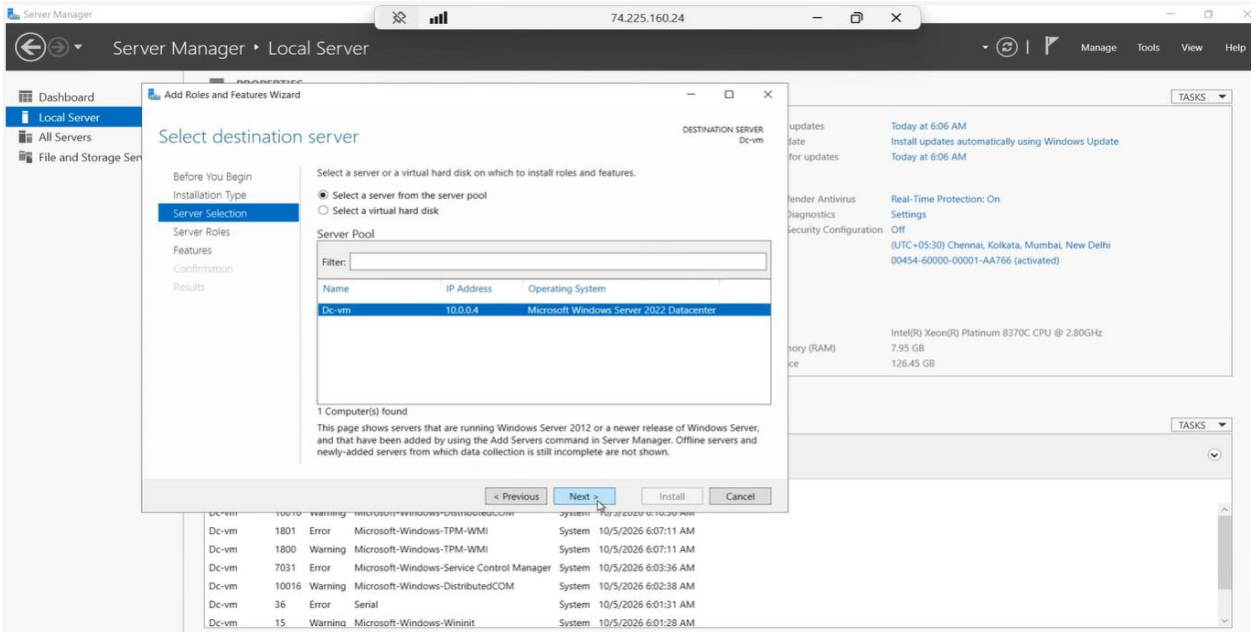
Step 10: Select installation type

Choose Role-based or feature-based installation and click Next.



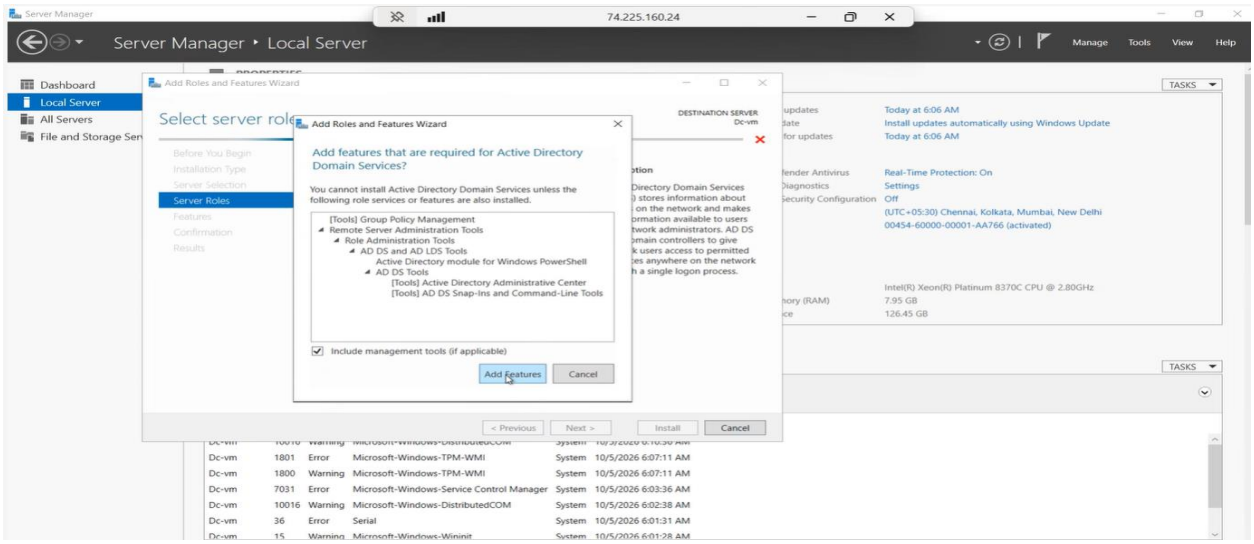
Step 11: Select the destination server

Choose Select a server from the server pool, pick Dc-vm (10.0.0.4) and click Next.

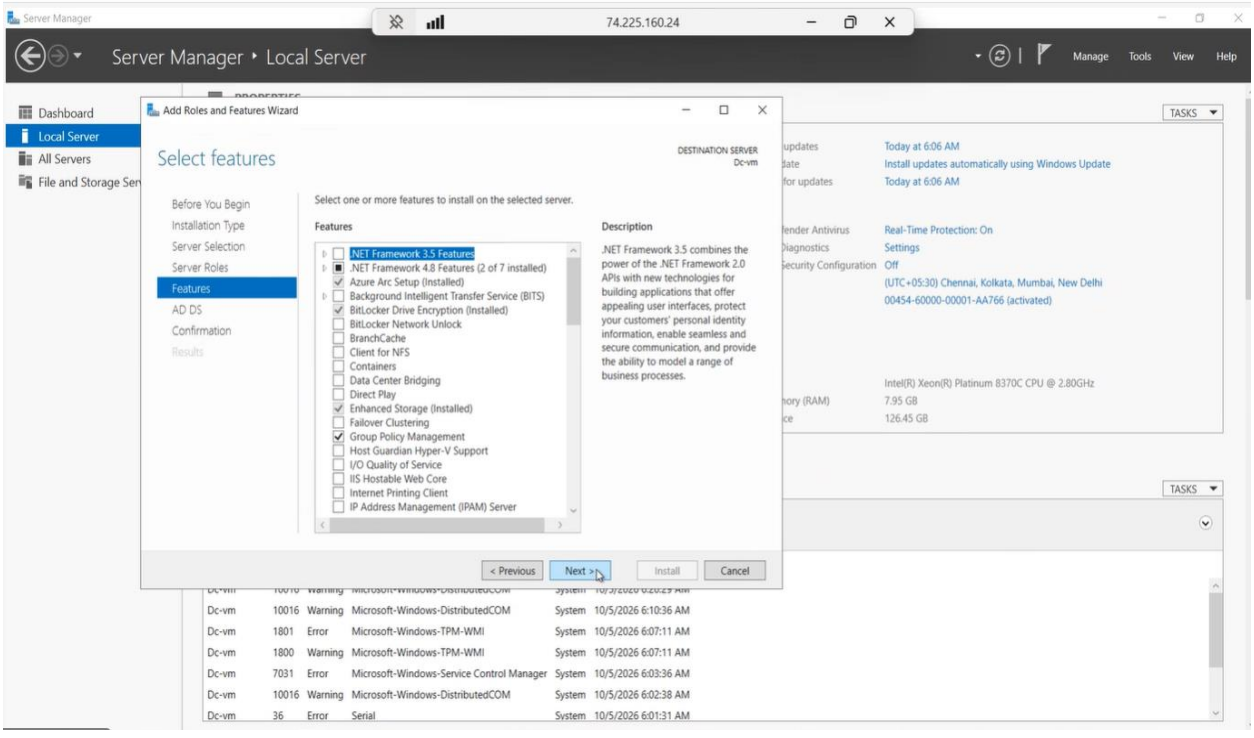


Step 12: Select Active Directory Domain Services

On the Server Roles page tick Active Directory Domain Services. In the pop-up keep Include management tools ticked and click Add Features, then click Next.

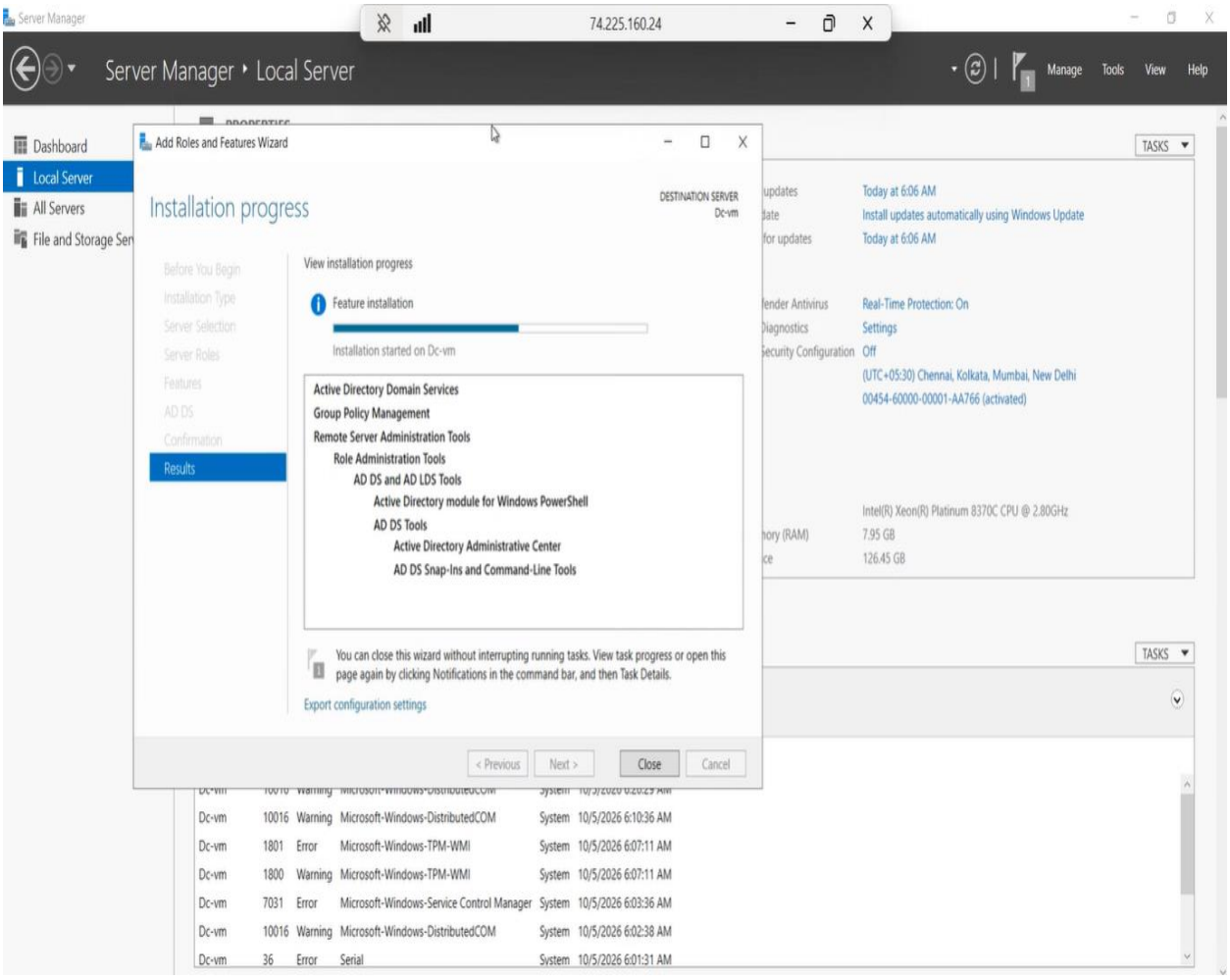


Step 13: Select features



Step 14: Install and wait for completion

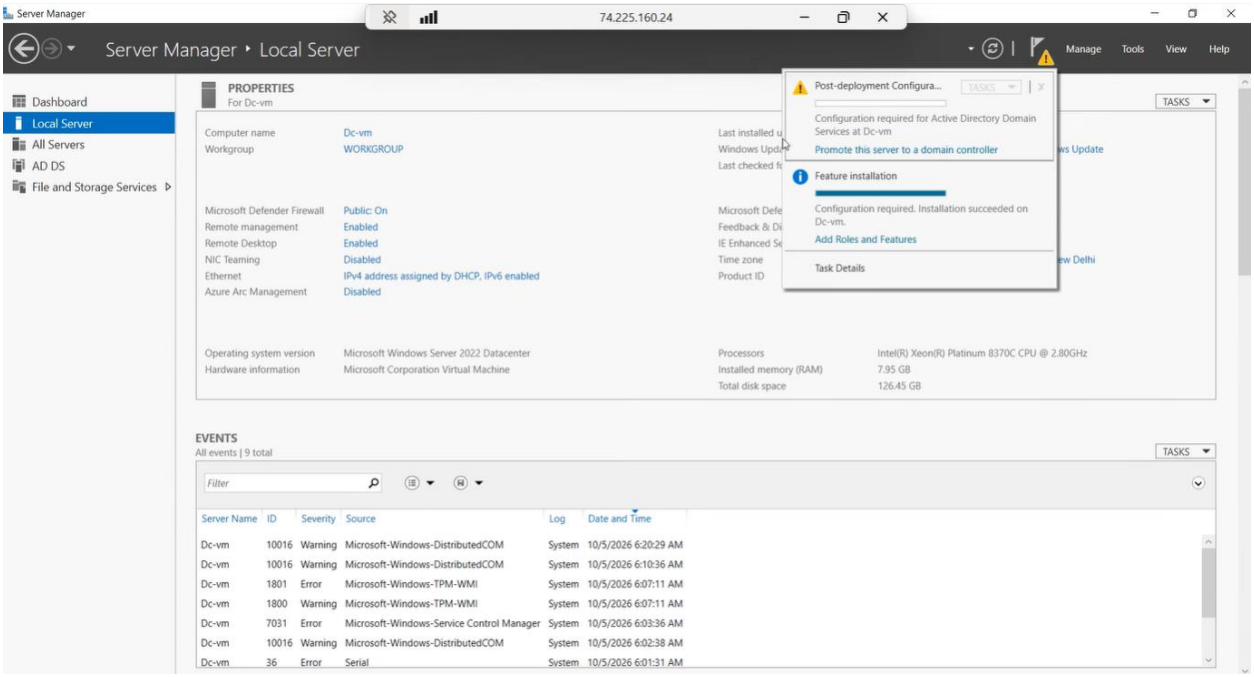
Click Install and wait on the Installation progress page until the role installation finishes. Then click Close



Part 3 – Promote Dc-vm to a Domain Controller

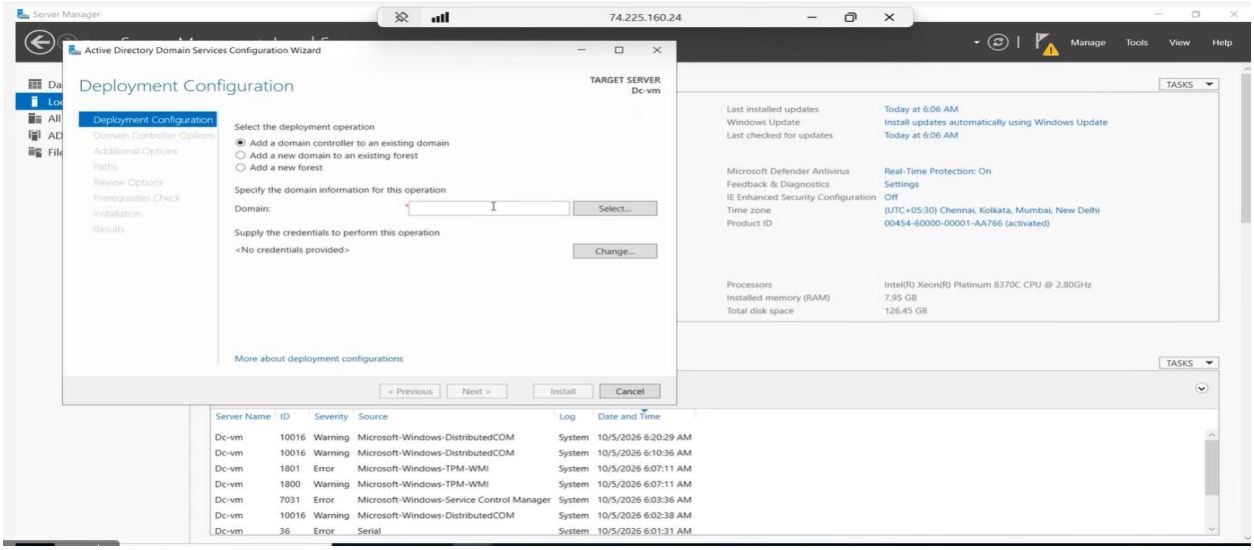
Step 15: Open the promotion wizard

In Server Manager click the yellow warning flag. Under Post-deployment Configuration click Promote this server to a domain controller.



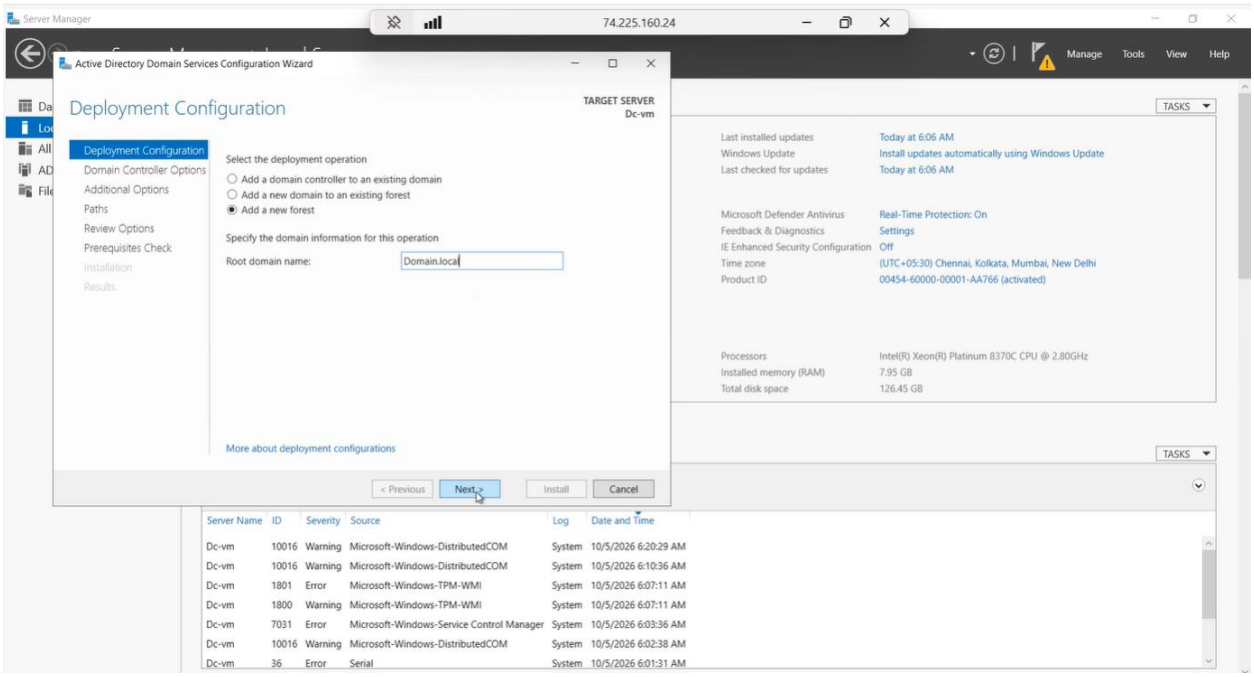
Step 16: Deployment Configuration

The wizard opens with Add a domain controller to an existing domain selected by default. Change it, because this is the first domain controller of a new network.



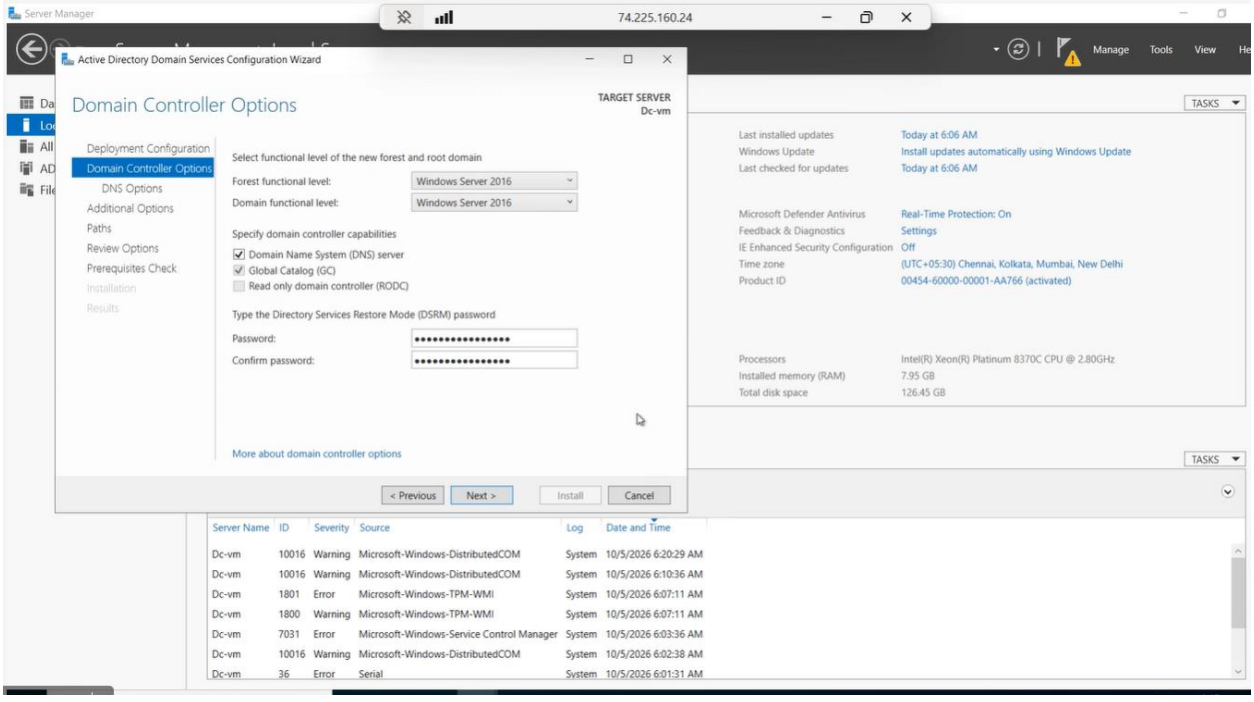
Step 17: Add a new forest

Select Add a new forest, type Domain.Local as the Root domain name and click Next.



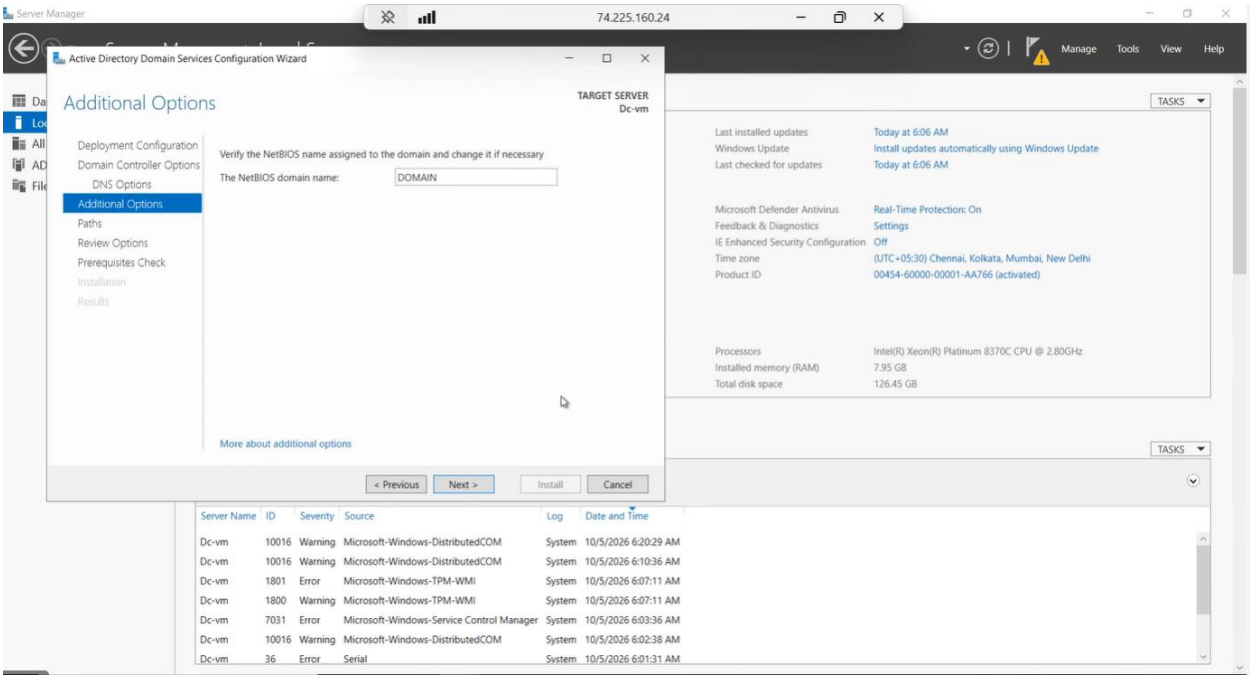
Step 18: Domain Controller Options

- Forest functional level: Windows Server 2016
- Domain functional level: Windows Server 2016



Step 19: Additional Options

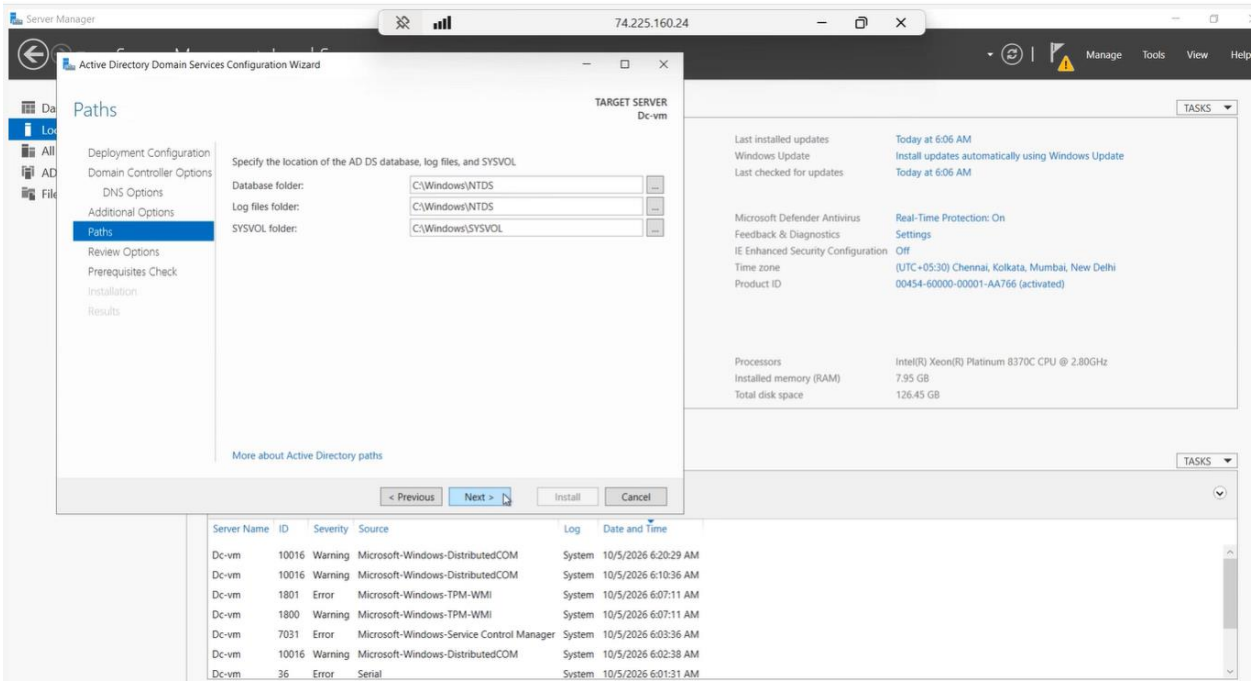
Check the NetBIOS domain name (DOMAIN) and click Next.



Step 20: Paths

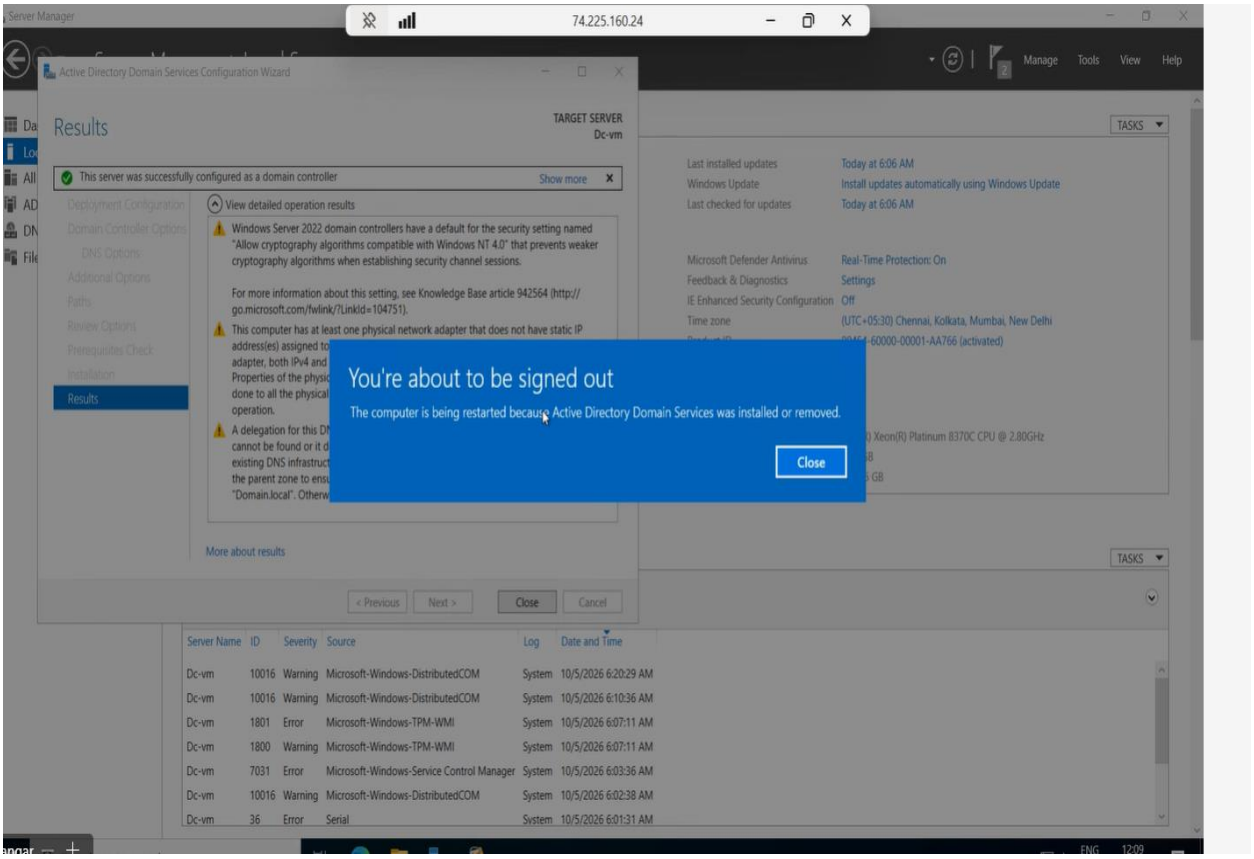
Keep the default locations and click Next. Then click Next on Review Options and Prerequisites Check, and click Install.

- Database and log folders: C:\Windows\NTDSSYSVOL folder: C:\Windows\SYSVOL



Step 21: Results – the server restarts

The wizard reports that the server was successfully configured as a domain controller. Windows then shows You're about to be signed out and restarts automatically. Click Close and wait for the restart



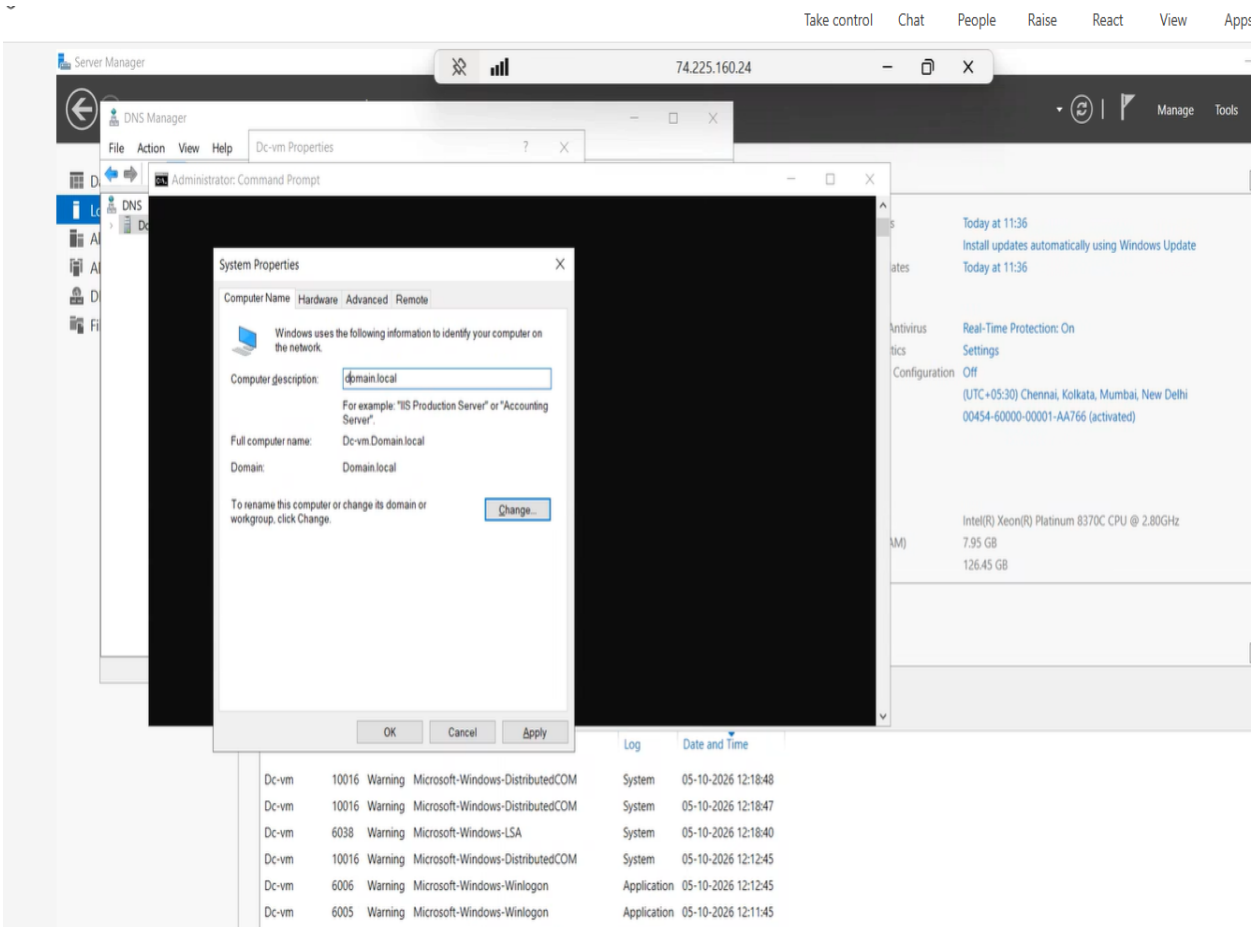
Step 22: Reconnect and verify the domain

After the restart connect to Dc-vm again, this time with the domain account DOMAIN\azureuser. Open System Properties and confirm the full computer name and domain.

- Full computer name: Dc-vm.Domain.local
- Domain: Domain.local

To connect the vm we have to use this

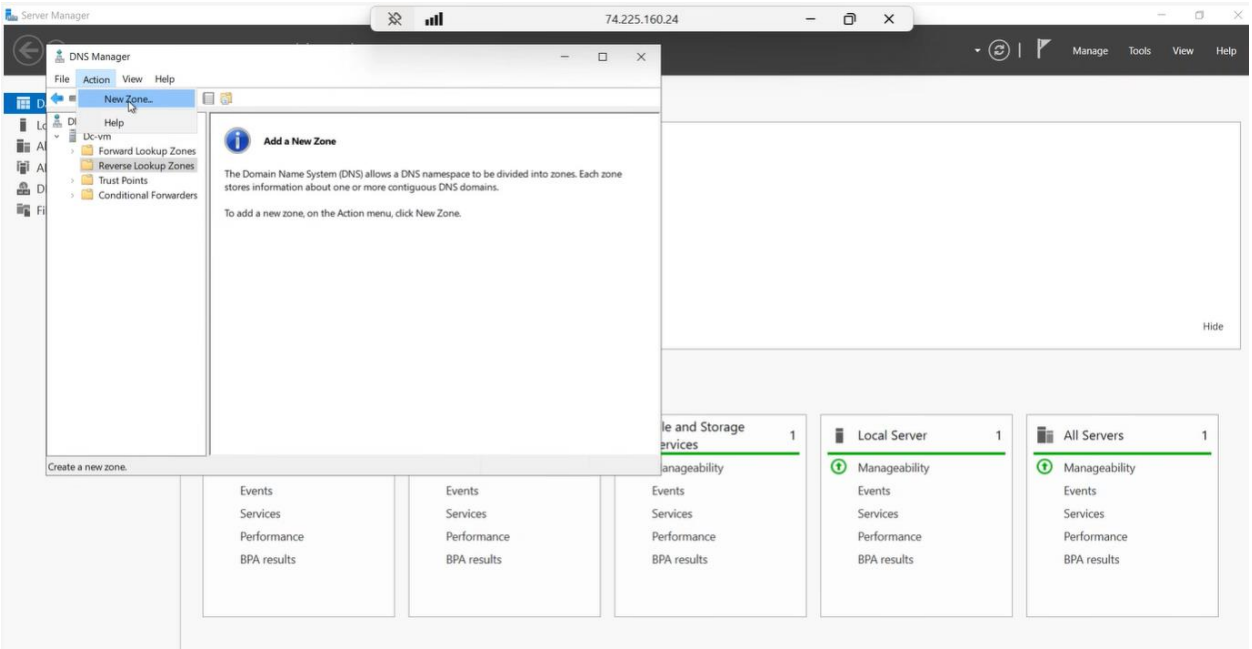
DOMAIN\azureuser


Take control
Chat
People
Raise
React
View
Apps
74.225.160.24
Server Manager
DNS Manager
File Action View Help
Dc-vm Properties
Administrator: Command Prompt
Manage
Tools
System Properties
Computer Name Hardware Advanced Remote
Windows uses the following information to identify your computer on the network.
Computer description:
domain.local
For example: "IIS Production Server" or "Accounting Server".
Full computer name:
Dc-vm.Domain.local
Domain:
Domain.local
To rename this computer or change its domain or workgroup, click Change.
Change...
OK
Cancel
Apply
Today at 11:36
Install updates automatically using Windows Update
Today at 11:36
Real-Time Protection: On
Settings
Off
(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
00454-60000-00001-AA766 (activated)
Intel(R) Xeon(R) Platinum 8370C CPU @ 2.80GHz
7.95 GB
126.45 GB
Log
Date and Time
Dc-vm 10016 Warning Microsoft-Windows-DistributedCOM System 05-10-2026 12:18:48
Dc-vm 10016 Warning Microsoft-Windows-DistributedCOM System 05-10-2026 12:18:47
Dc-vm 6038 Warning Microsoft-Windows-LSA System 05-10-2026 12:18:40
Dc-vm 10016 Warning Microsoft-Windows-DistributedCOM System 05-10-2026 12:12:45
Dc-vm 6006 Warning Microsoft-Windows-Winlogon Application 05-10-2026 12:12:45
Dc-vm 6005 Warning Microsoft-Windows-Winlogon Application 05-10-2026 12:11:45

Part 4 – Create the Reverse Lookup Zone in DNS

Step 23: Open DNS Manager and start a new zone

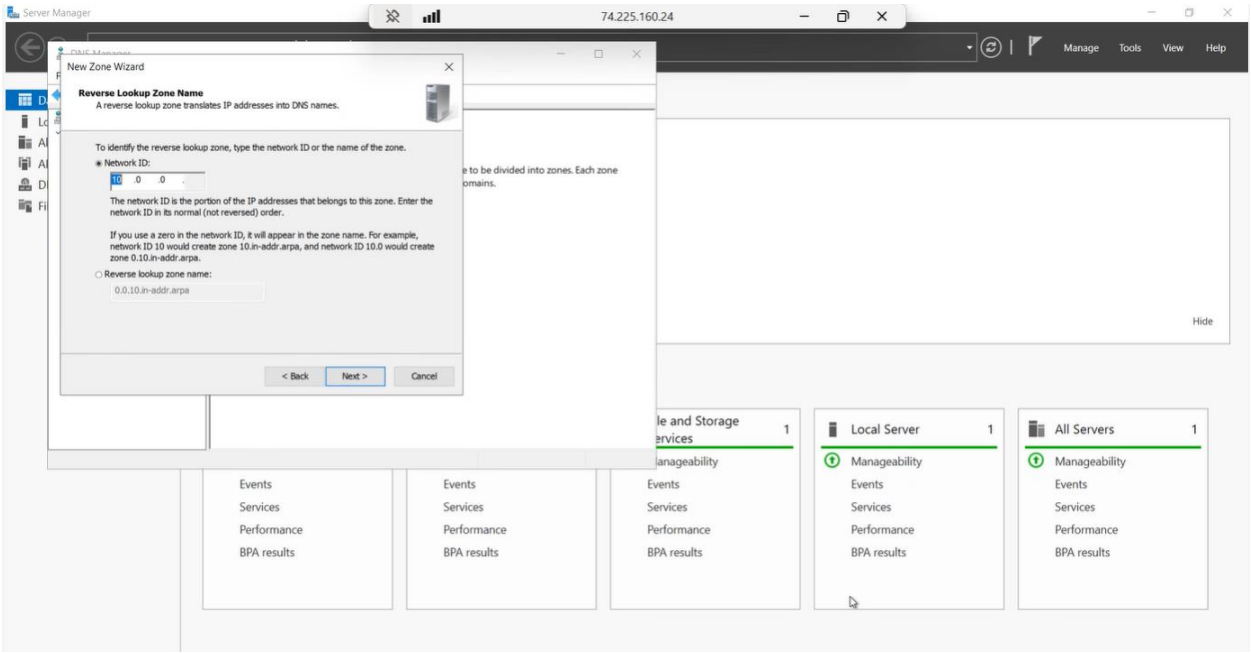
In Server Manager go to Tools → DNS. In DNS Manager select Reverse Lookup Zones (under Dc-vm), then click Action → New Zone.



Step 24: Enter the Network ID

In the New Zone Wizard choose Primary zone and the default replication/IPv4 options until you reach the Reverse Lookup Zone Name page. Under Network ID enter the first three parts of the DC's address, then click Next.

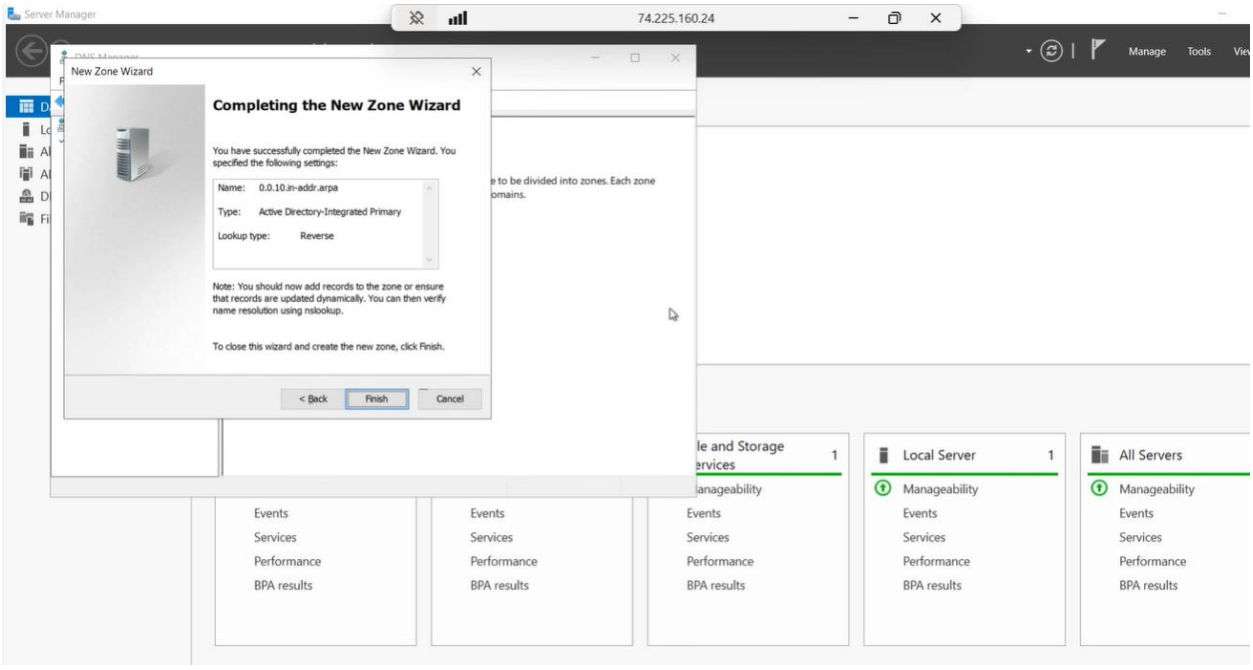
- Your DC IP is: 10.0.0.4
- Your subnet is: 10.0.0.0/24
- So enter the Network ID: 10.0.0



Step 25: Finish the New Zone Wizard

Check the summary and click Finish.

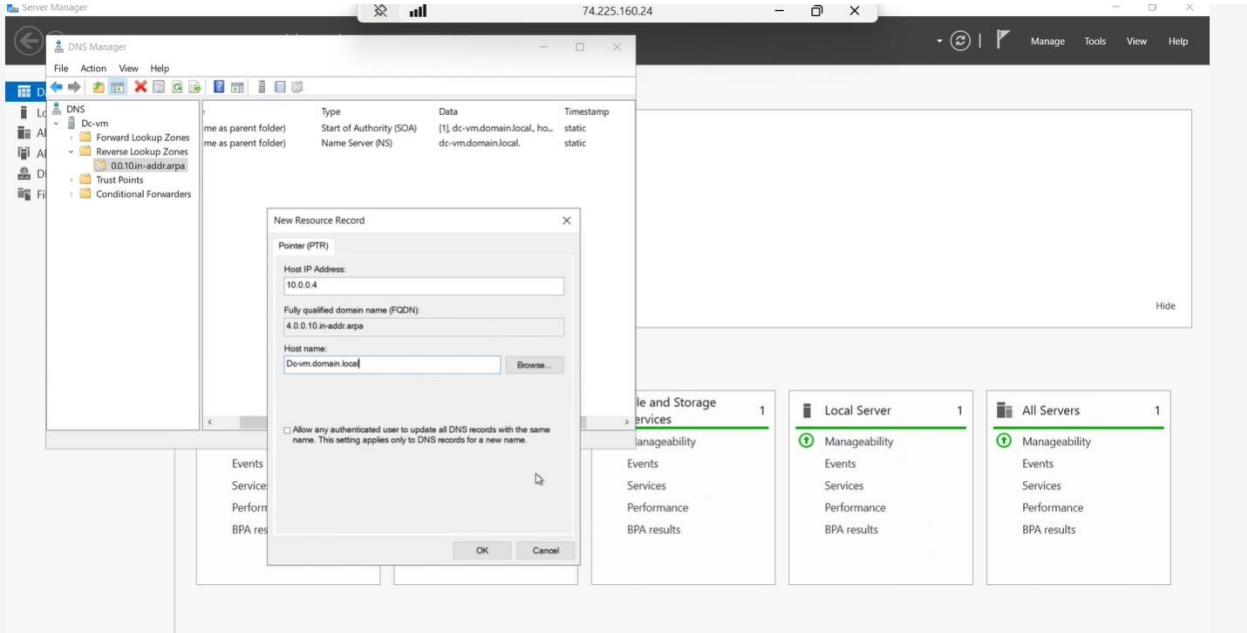
- Name: 0.0.10.in-addr.arpa
- Type: Active Directory-Integrated Primary
- Lookup type: Reverse



Step 26: Add the PTR record for the DC

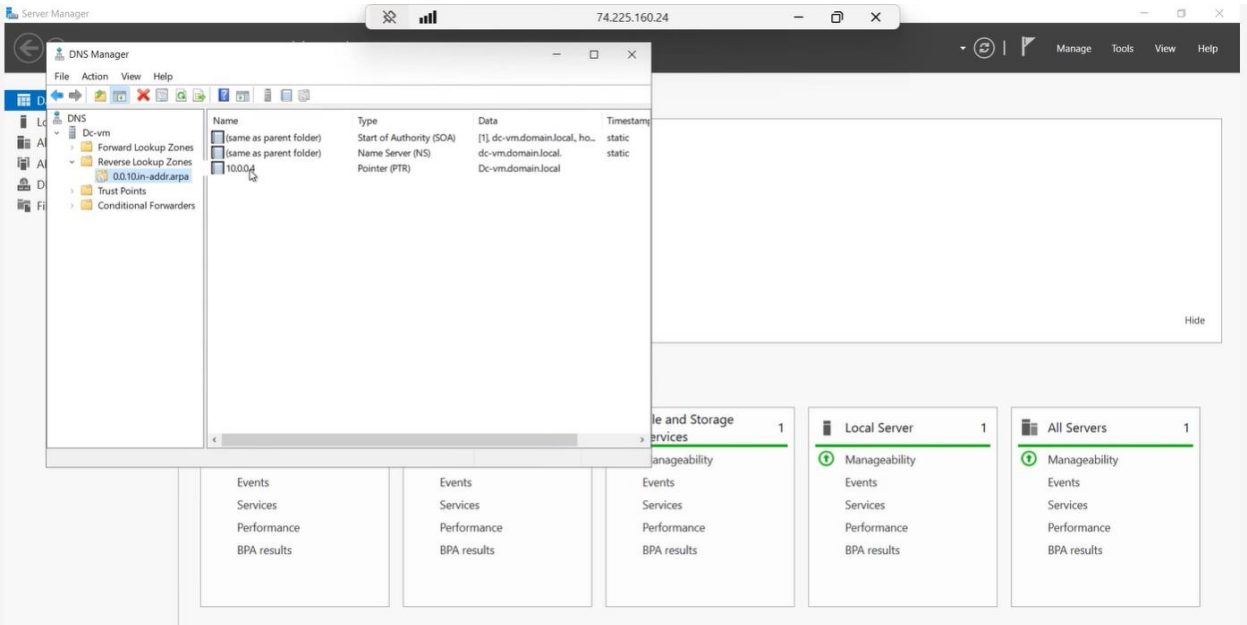
Select the 0.0.10.in-addr.arpa zone, right-click in the empty area and choose New Pointer (PTR). Fill in the record and click OK.

- Host IP Address: 10.0.0.4
- Host name: Dc-vm.domain.local



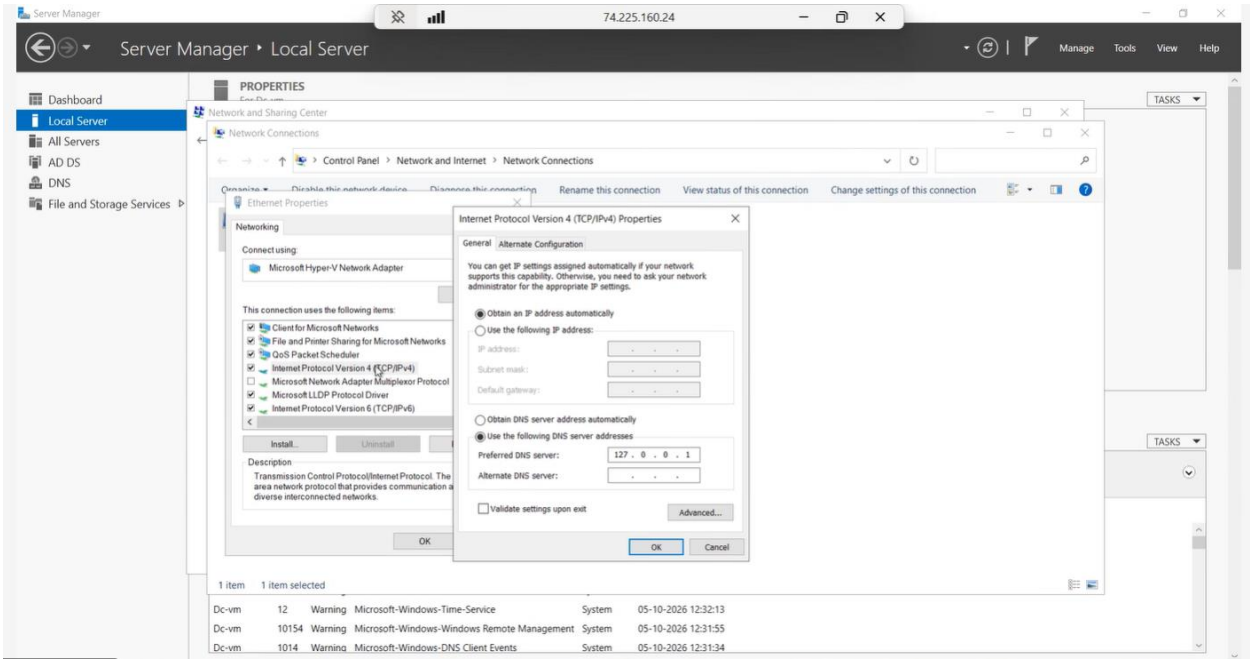
Step 27: Confirm the PTR record

The zone now lists the pointer record 10.0.0.4 → Dc-vm.domain.local, which confirms the reverse lookup zone works.



Step 28: Set the DC's own DNS server

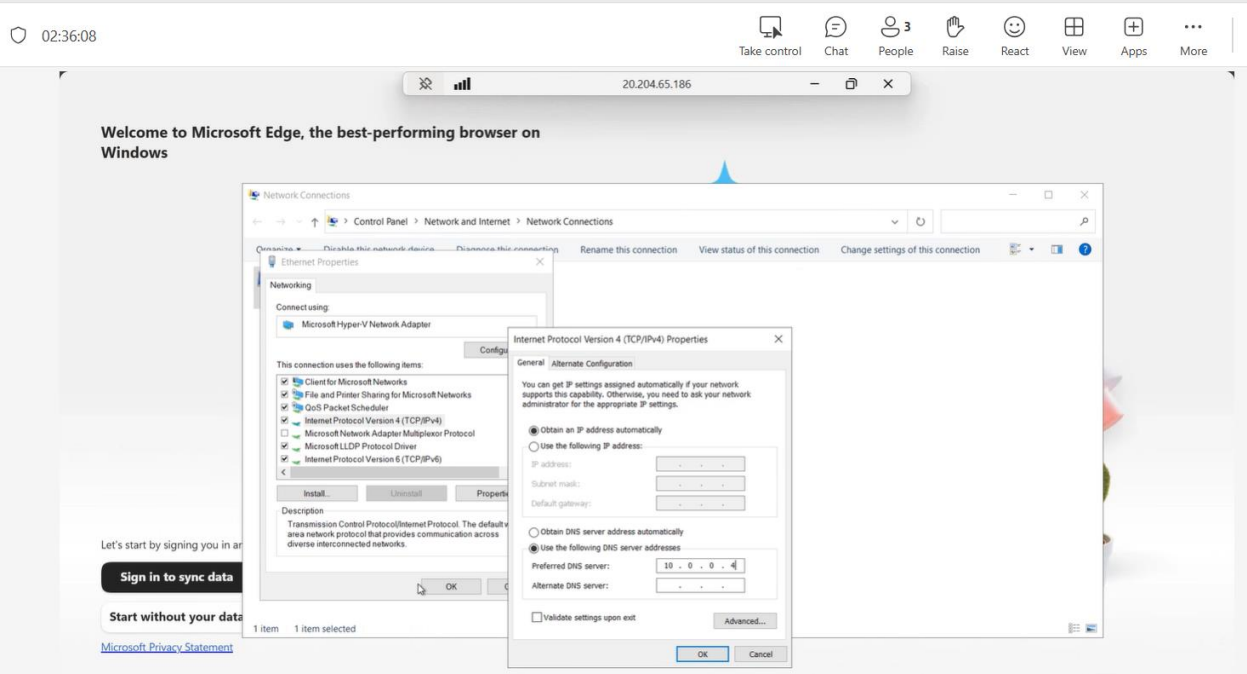
On Dc-vm open Network Connections → Ethernet → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties. Select Use the following DNS server addresses, enter 127.0.0.1 as the Preferred DNS server and click OK.



Part 5 – Join Client-Vm to the Domain

Step 29: Set the client's DNS server to the DC

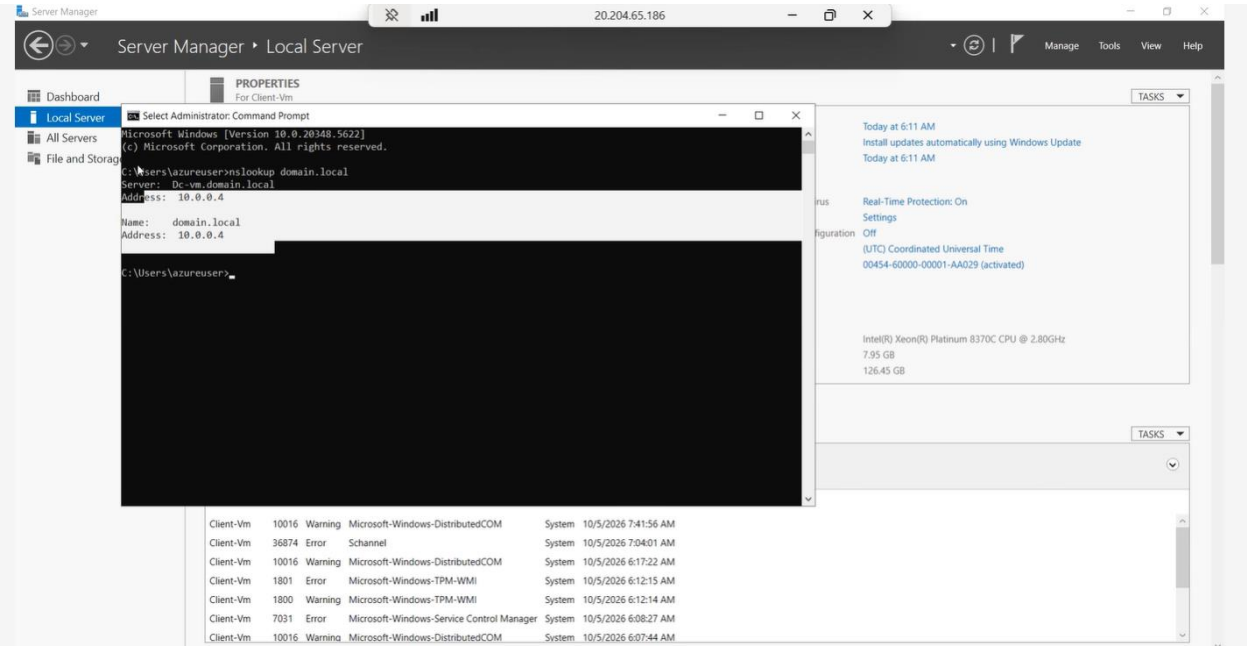
Connect to Client-Vm (public IP 20.204.65.186). Open Network Connections → Ethernet → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties. Set the Preferred DNS server to the DC's IP address 10.0.0.4 and click OK.



Step 30: Test DNS with nslookup

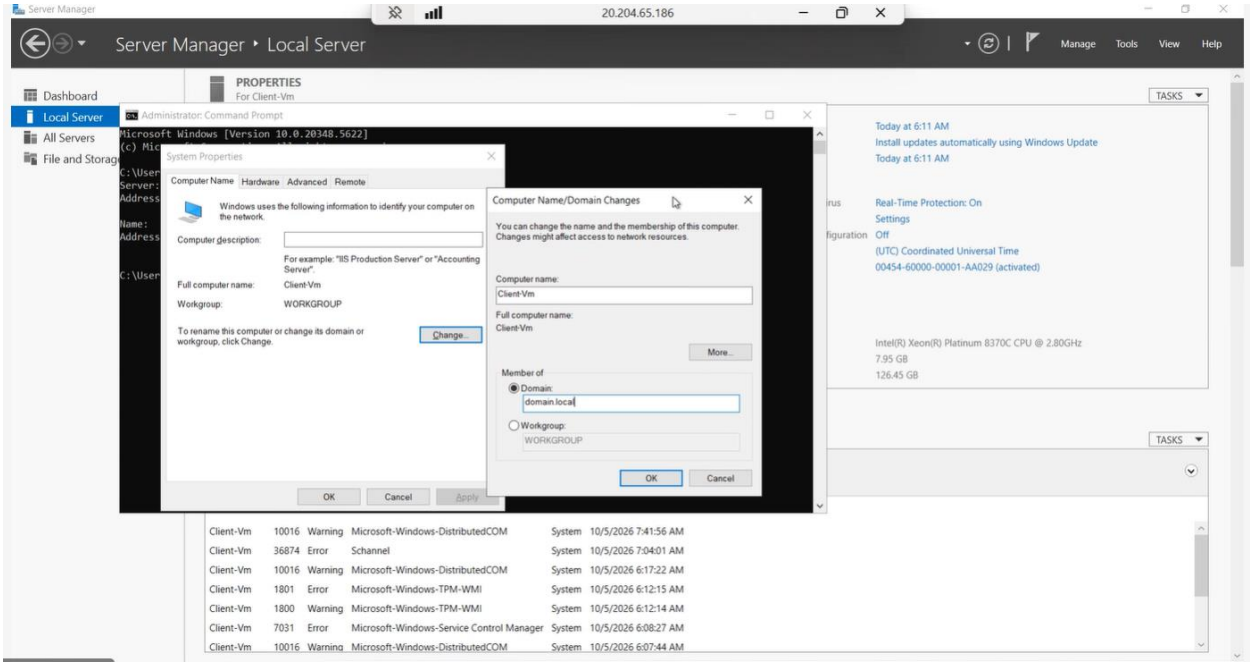
Open Command Prompt and run the command below. It should answer from Dc-vm.domain.local with the address 10.0.0.4.

- Command: nslookup domain.local



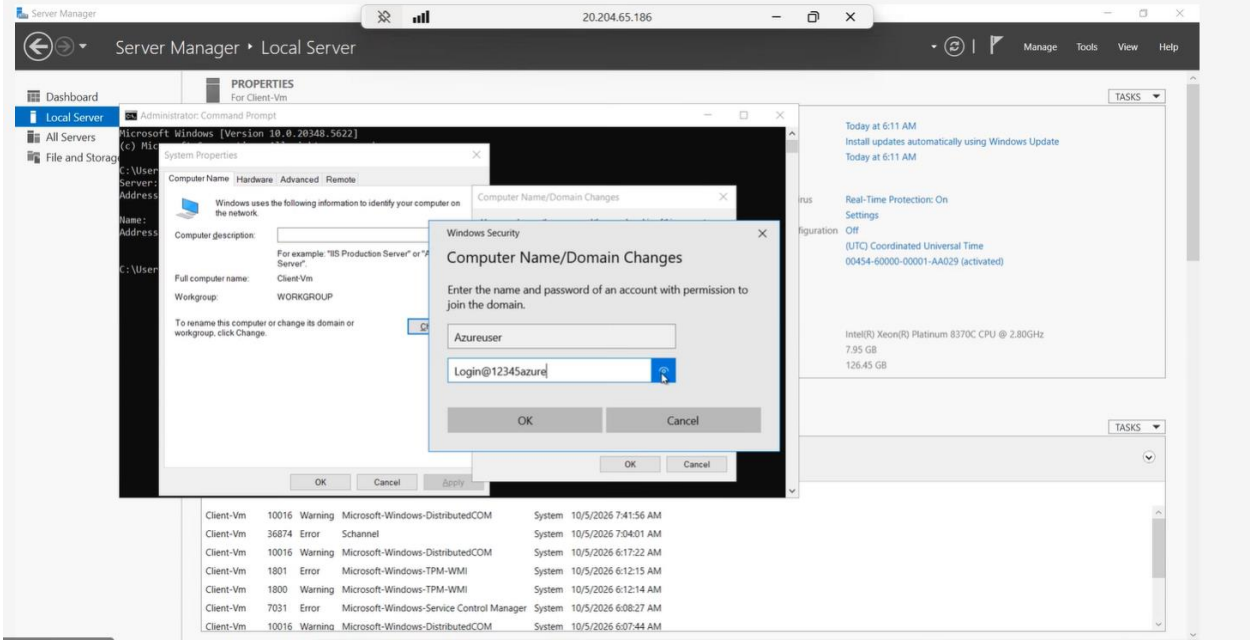
Step 31: Open the domain join dialog

Open System Properties → Computer Name → Change. Under Member of select Domain and type domain.local, then click OK.



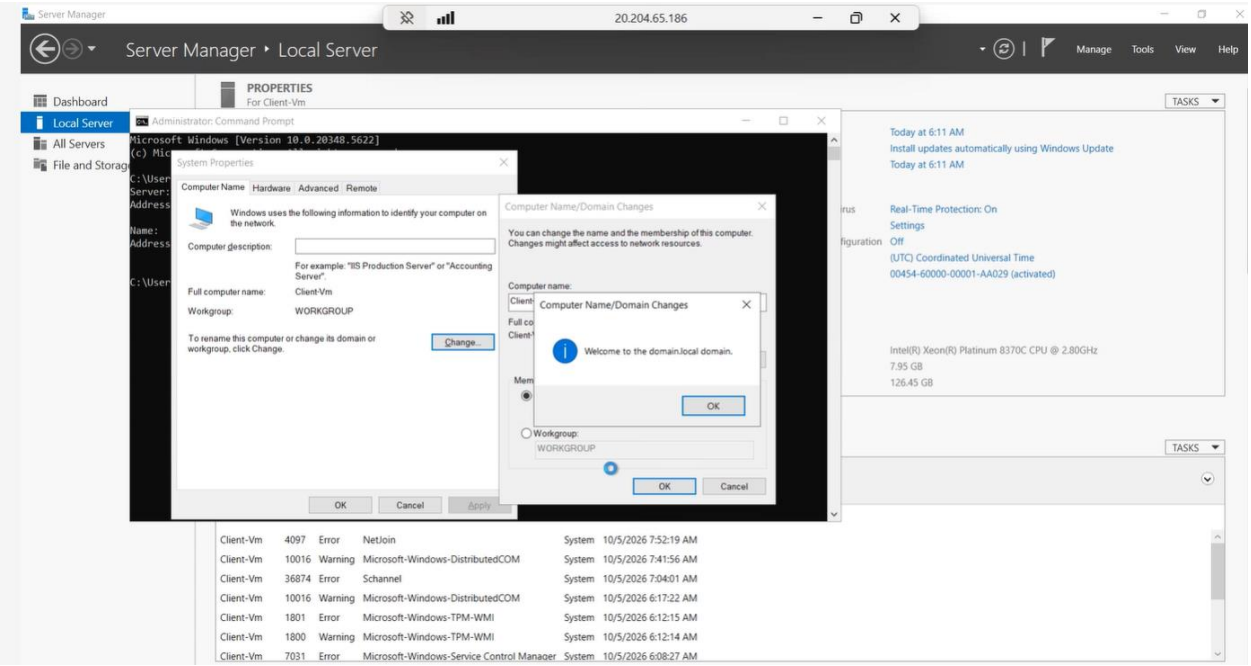
Step 32: Enter domain credentials

When Windows Security asks for the name and password of an account with permission to join the domain, enter the azureuser administrator credentials and click OK.



Step 33: Domain join succeeded

The message Welcome to the domain.local domain appears. Click OK and restart Client-Vm when asked.

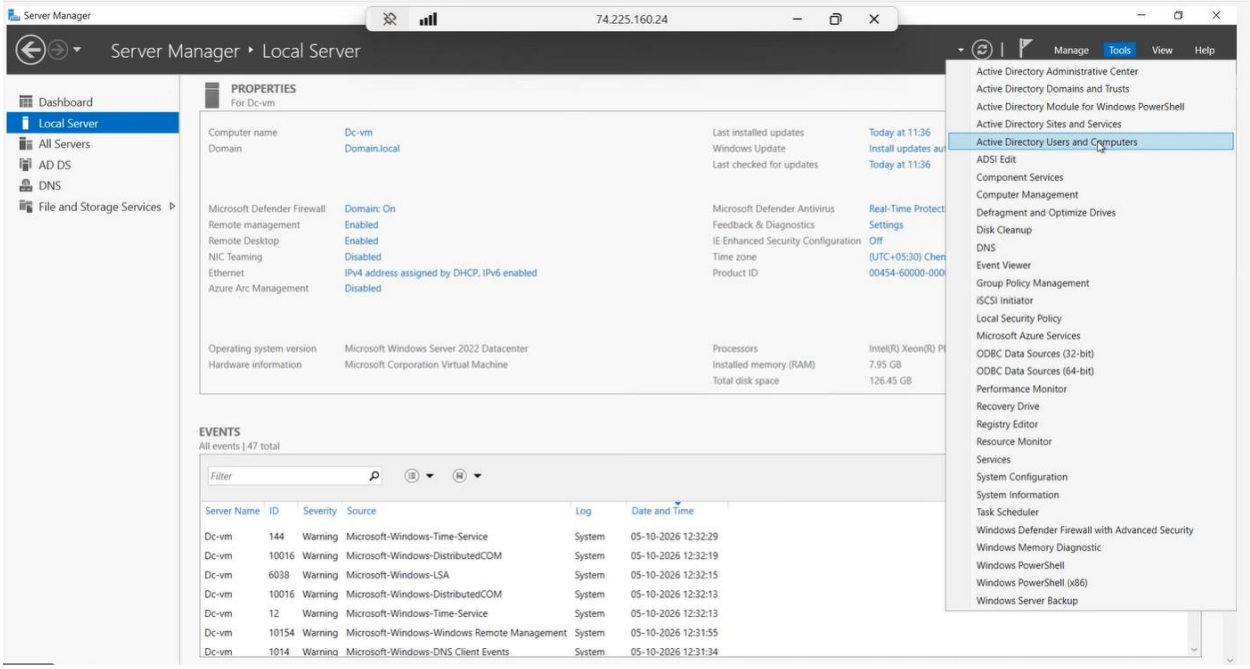


Part 6 – Create the OU and Users (on Dc-vm)

Now go to Dc-vm and create an Organizational Unit and the domain users.

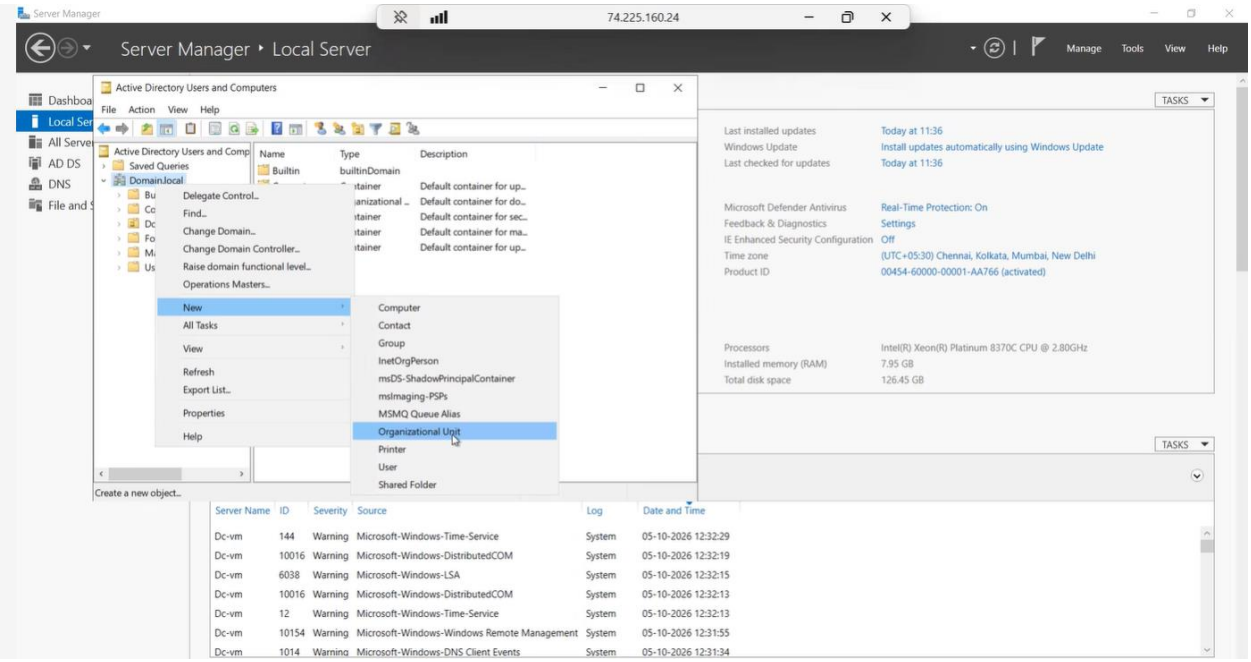
Step 34: Open Active Directory Users and Computers

In Server Manager click Tools → Active Directory Users and Computers.



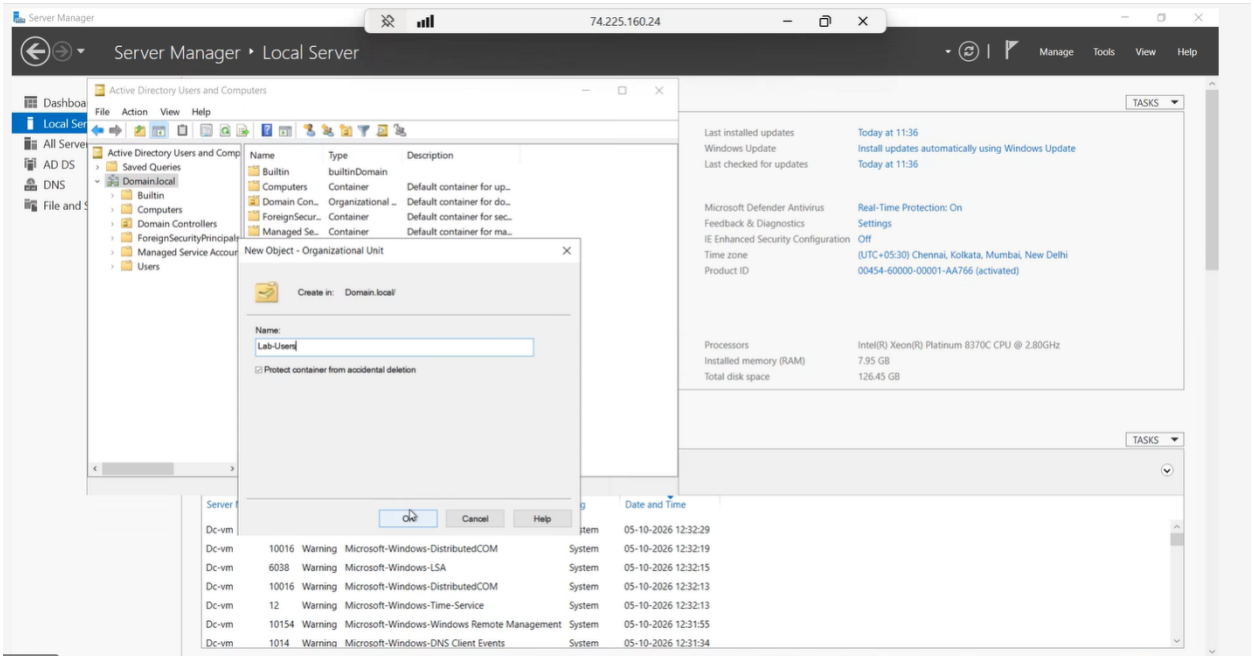
Step 35: Create an Organizational Unit

Right-click the domain Domain.local, then choose New → Organizational Unit.



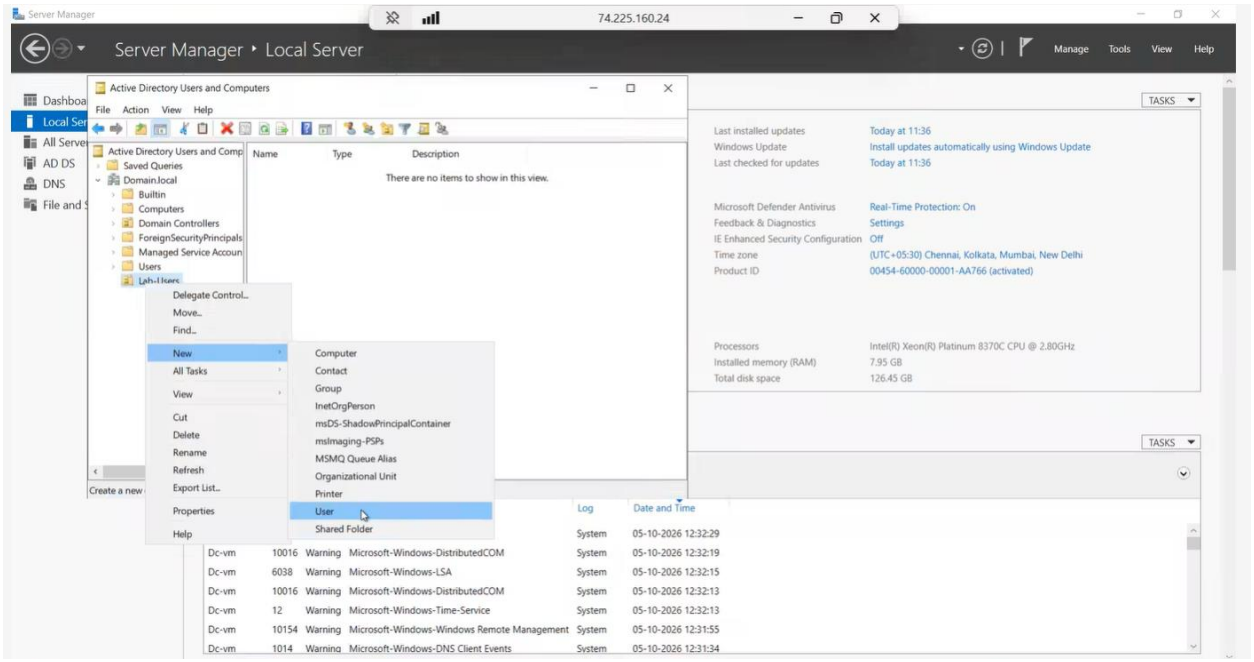
Step 36: Name the OU

Type Lab-Users as the name



Step 38: Start creating a user

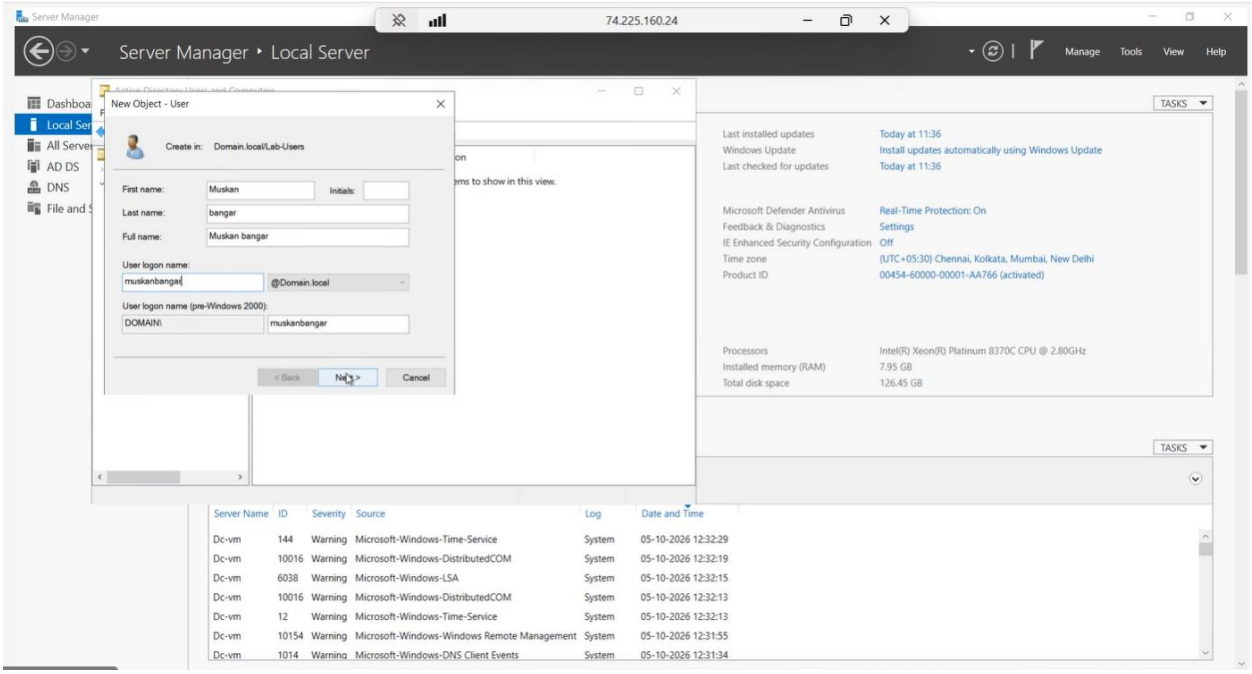
Right-click the Lab-Users OU and choose New → User.



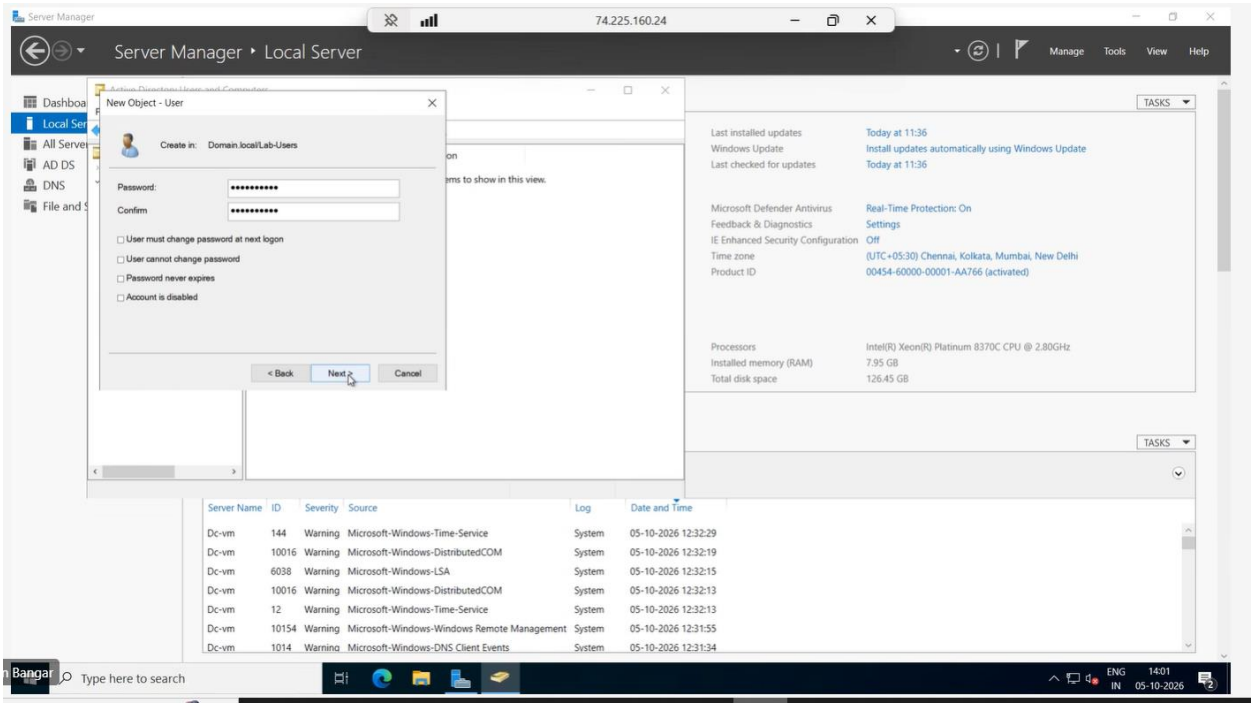
Step 39: Enter the user's details

Fill in the name fields and the logon name, then click Next.

- First name: Muskan, Last name: bangar
- User logon name: muskanbangar@Domain.local
- Pre-Windows 2000 name: DOMAIN\muskanbangar

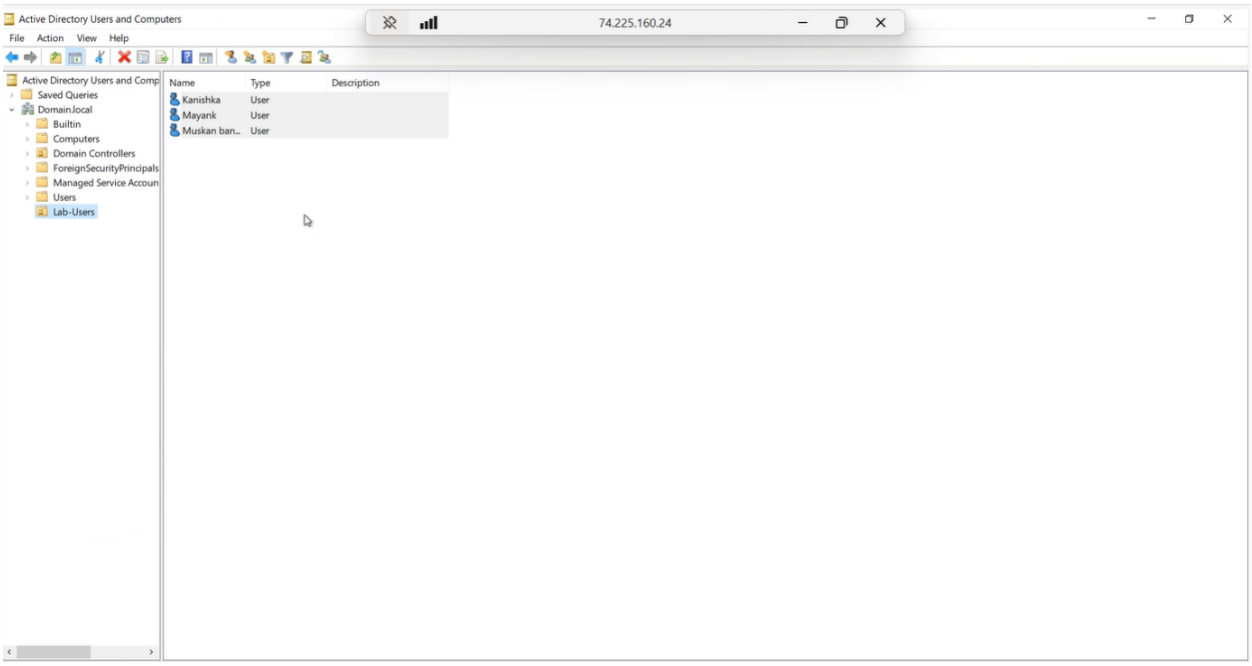


Step 40: Set the user's password



Step 40: Check the users

Select Lab-Users. The OU now lists the users Kanishka, Mayank and Muskan bangar. Repeat the same steps for each additional user.

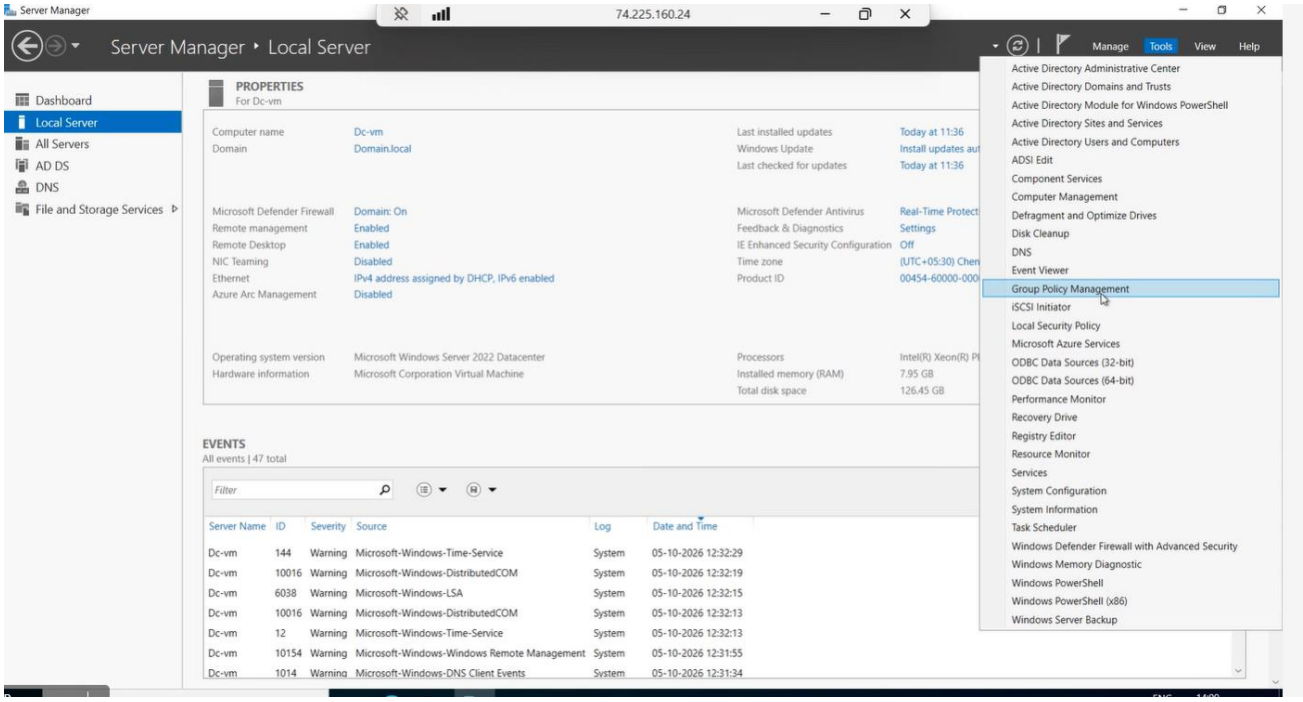


Part 7 – Password and Account Lockout Policy

Enable password expiry (password age) and account lockout policies in the Default Domain Policy. Do this after the users have been created.

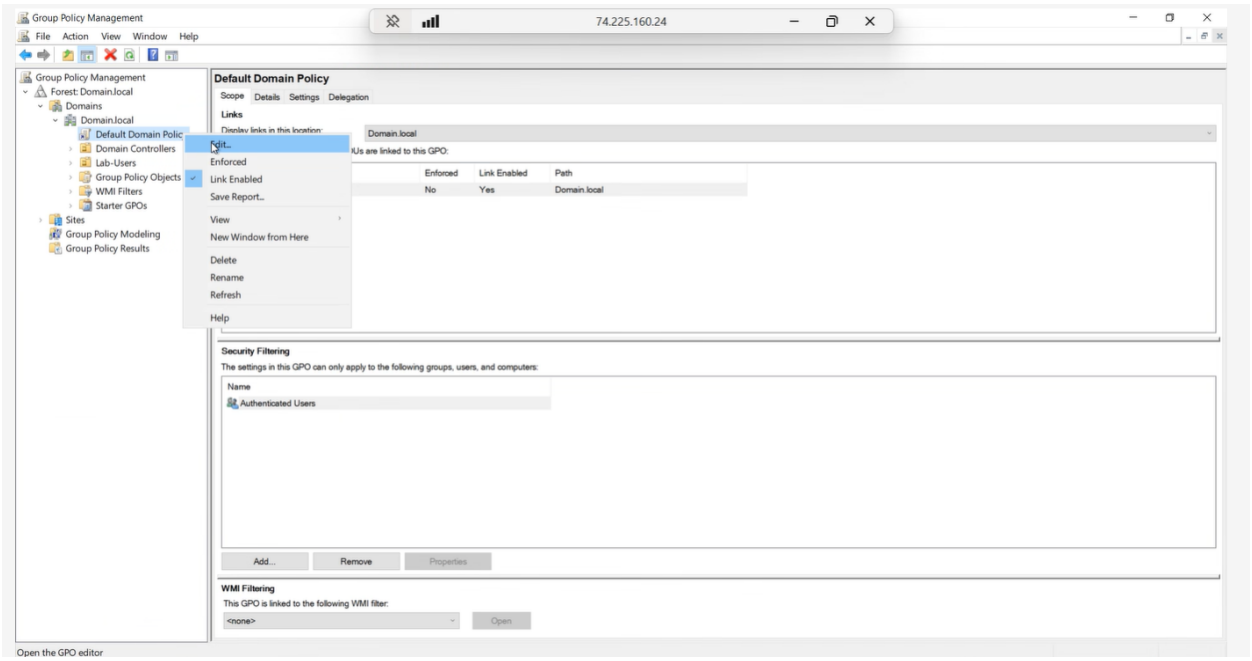
Step 41: Open Group Policy Management

In Server Manager click Tools → Group Policy Management.



Step 42: Edit the Default Domain Policy

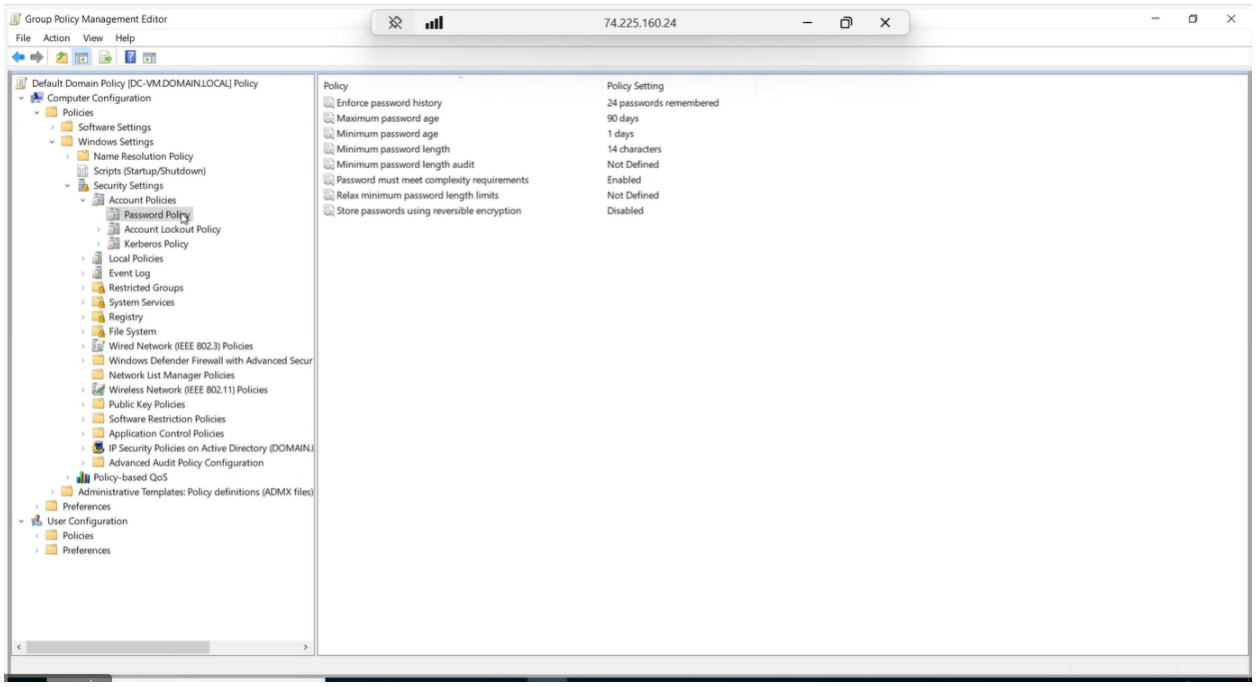
Expand Forest → Domains → Domain.local, right-click Default Domain Policy and choose Edit.


Group Policy Management
74.225.160.24
File Action View Window Help
Group Policy Management
Forest: Domain.local
Domains
Domain.local
Default Domain Polic
Domain Controllers
Lab-Users
Group Policy Objects
WMI Filters
Starter GPOs
Sites
Group Policy Modeling
Group Policy Results
Default Domain Policy
Scope Details Settings Delegation
Links
Domain.local
Edit...
Enforced
Link Enabled
Save Report...
View
New Window from Here
Delete
Rename
Refresh
Help
Enforced Link Enabled Path
No Yes Domain.local
Security Filtering
The settings in this GPO can only apply to the following groups, users, and computers:
Name
Authenticated Users
Add... Remove Properties
WMI Filtering
This GPO is linked to the following WMI filter:
<none>
Open
Open the GPO editor

Step 43: Configure the Password Policy

In the editor go to Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Password Policy and set the values below.

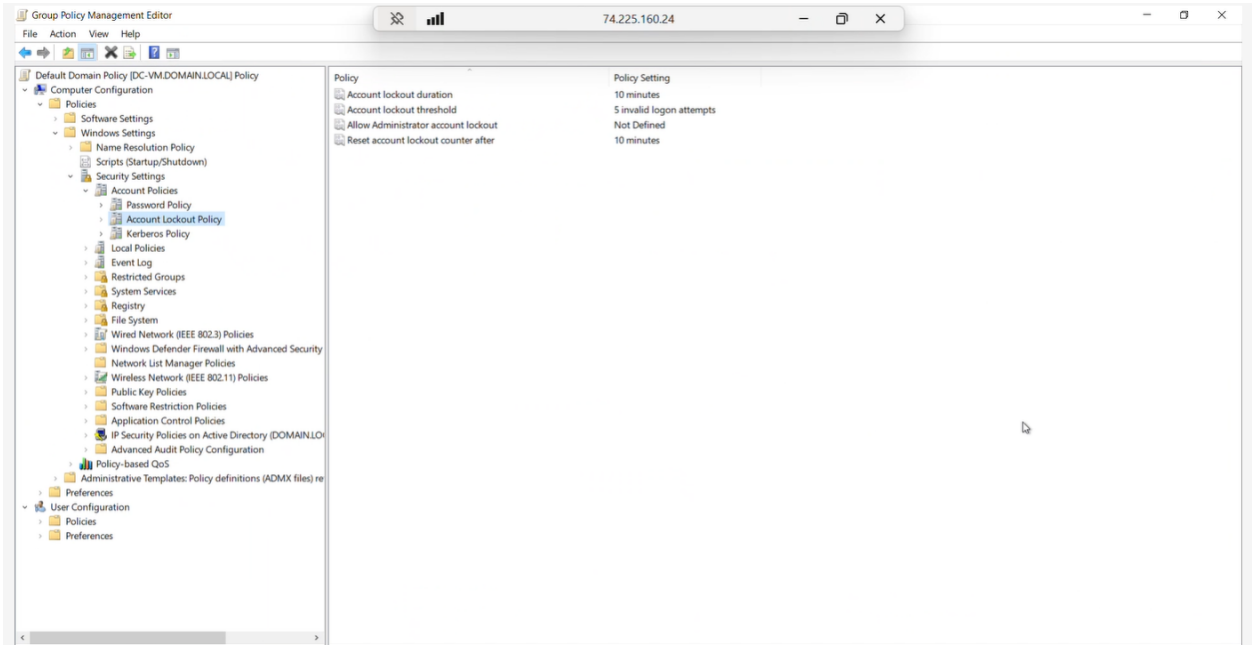
- Enforce password history: 24 passwords remembered
- Maximum password age: 90 days (password expiry)
- Minimum password age: 1 day
- Minimum password length: 14 characters
- Password must meet complexity requirements: Enabled
- Store passwords using reversible encryption: Disabled



Step 44: Configure the Account Lockout Policy

Select Account Lockout Policy (under Account Policies) and set the values below.

- Account lockout threshold: 5 invalid logon attempts
- Account lockout duration: 10 minutes
- Reset account lockout counter after: 10 minutes



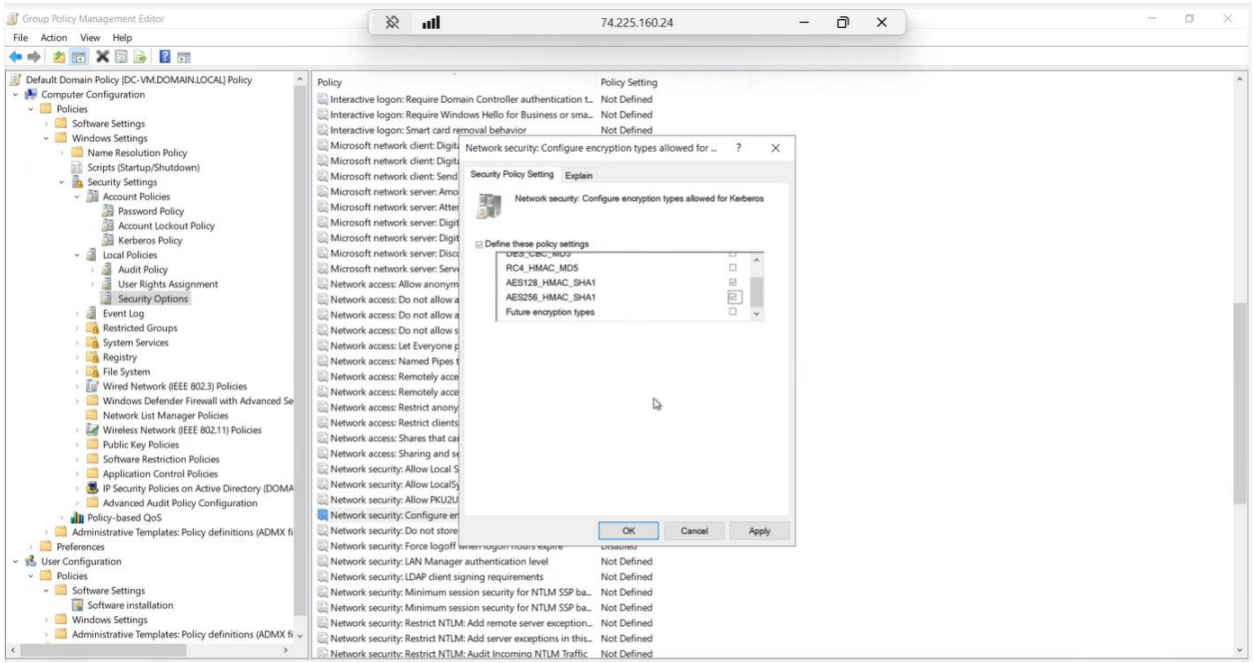
Part 8 – Enforce Kerberos and Harden NTLM

In the same Default Domain Policy editor go to Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → Security Options and configure the six settings below.

Setting	Value
Network security: Configure encryption types allowed for Kerberos	**Only AES128_HMAC_SHA1** aur **AES256_HMAC_SHA1**
Network security: LAN Manager authentication level	**Send NTLMv2 response only. Refuse LM & NTLM**
Network security: Restrict NTLM: Audit NTLM authentication in this domain	**Enable all**
Network security: Restrict NTLM: Outgoing NTLM traffic to remote servers	**Audit all**
Microsoft network server: Digitally sign communications (always)	**Enabled**
Microsoft network client: Digitally sign communications (always)	**Enabled**

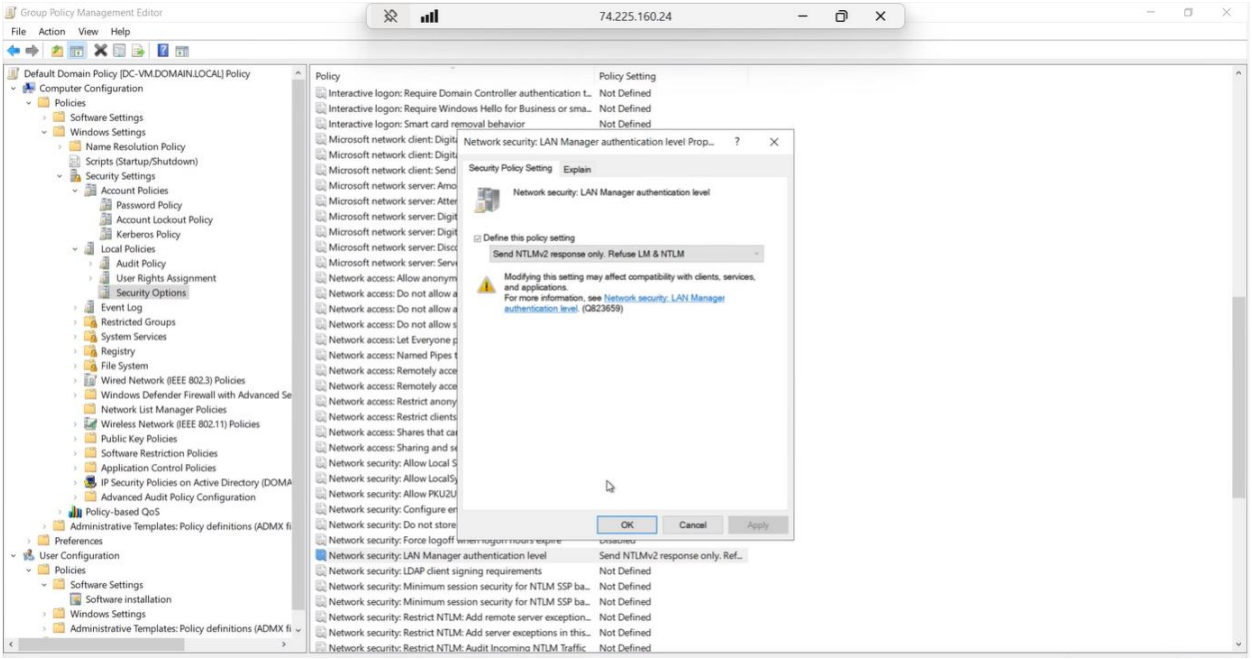
Step 45: Kerberos encryption types

Double-click Network security: Configure encryption types allowed for Kerberos. Tick Define these policy settings, tick only AES128_HMAC_SHA1 and AES256_HMAC_SHA1 (leave RC4 and DES unticked), then click OK



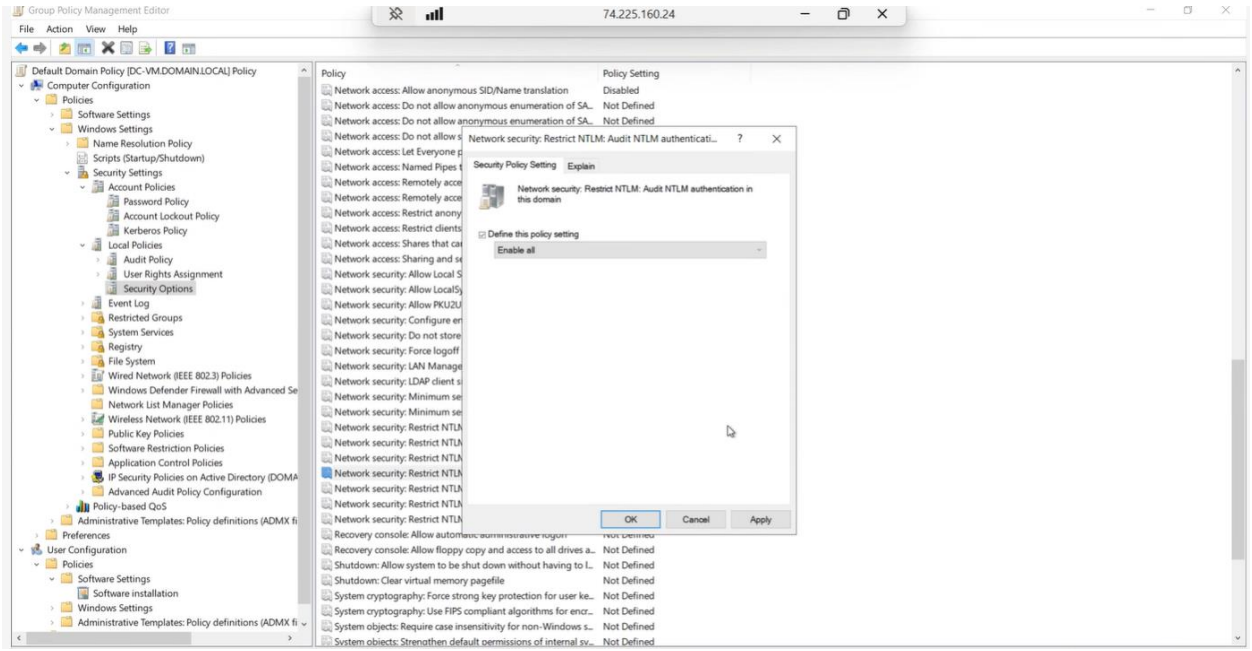
Step 46: LAN Manager authentication level

Double-click Network security: LAN Manager authentication level, tick Define this policy setting and choose Send NTLMv2 response only. Refuse LM & NTLM. Click OK.



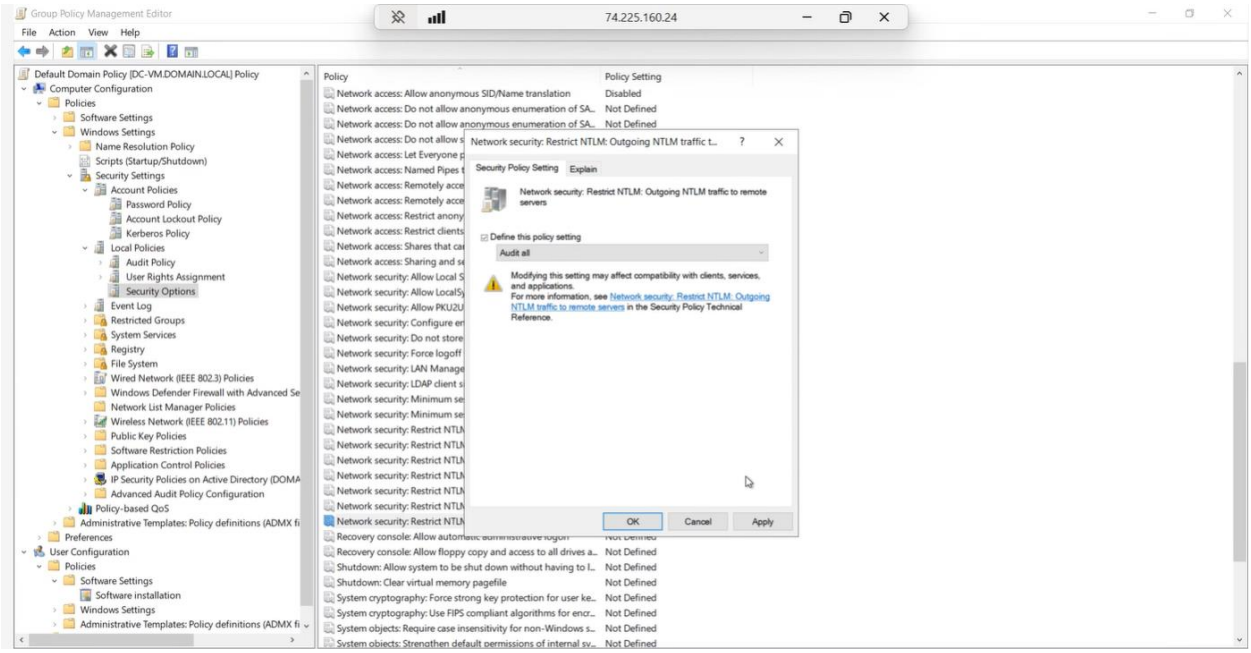
Step 47: Audit NTLM authentication in this domain

Double-click Network security: Restrict NTLM: Audit NTLM authentication in this domain, tick Define this policy setting and select Enable all. Click OK.



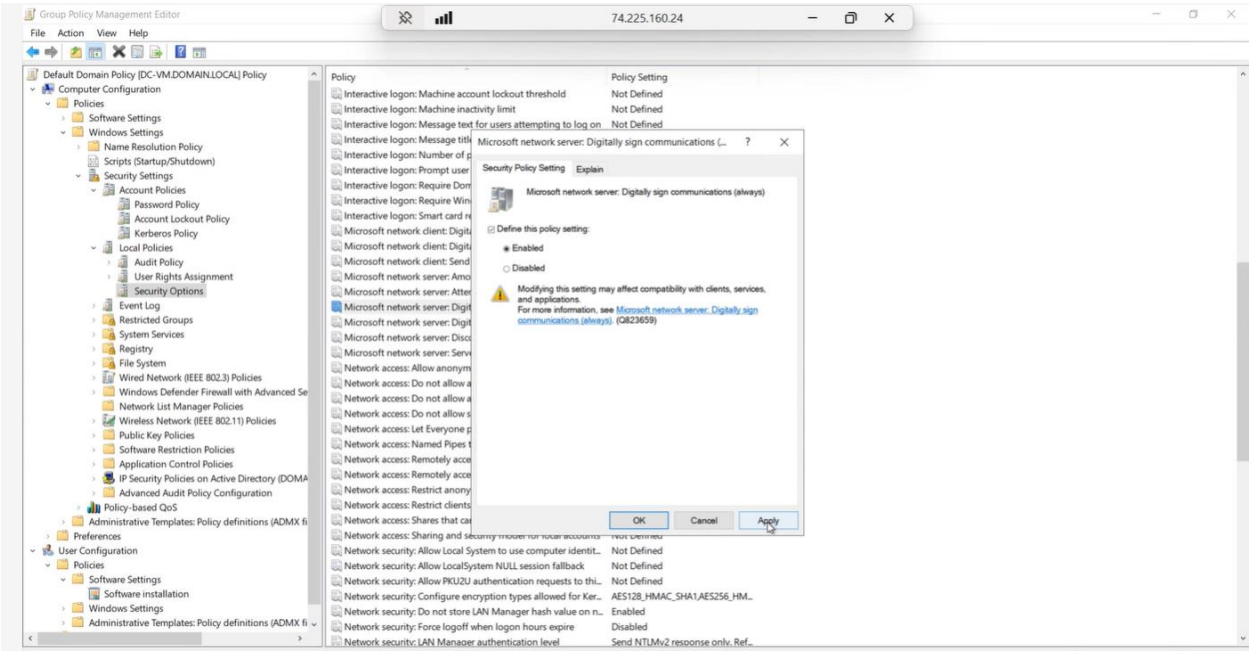
Step 48: Outgoing NTLM traffic to remote servers

Double-click Network security: Restrict NTLM: Outgoing NTLM traffic to remote servers, tick Define this policy setting and select Audit all. Click OK.



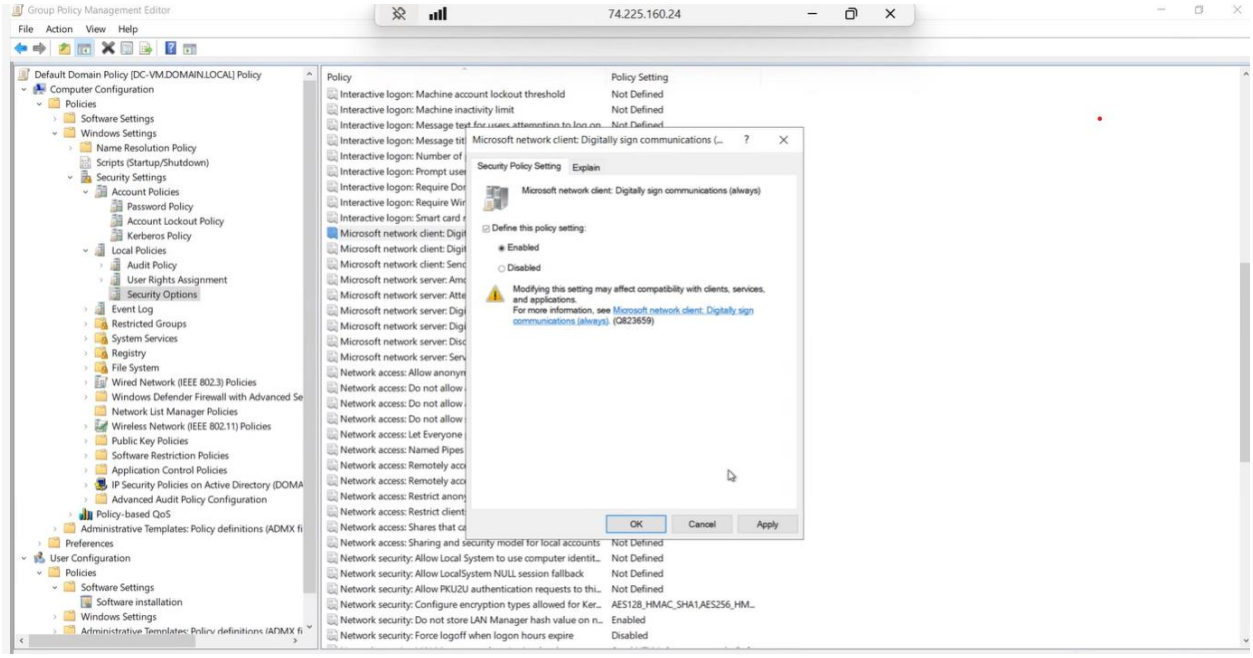
Step 49: Server – digitally sign communications (always)

Double-click Microsoft network server: Digitally sign communications (always), tick Define this policy setting, select Enabled and click Apply, then OK.



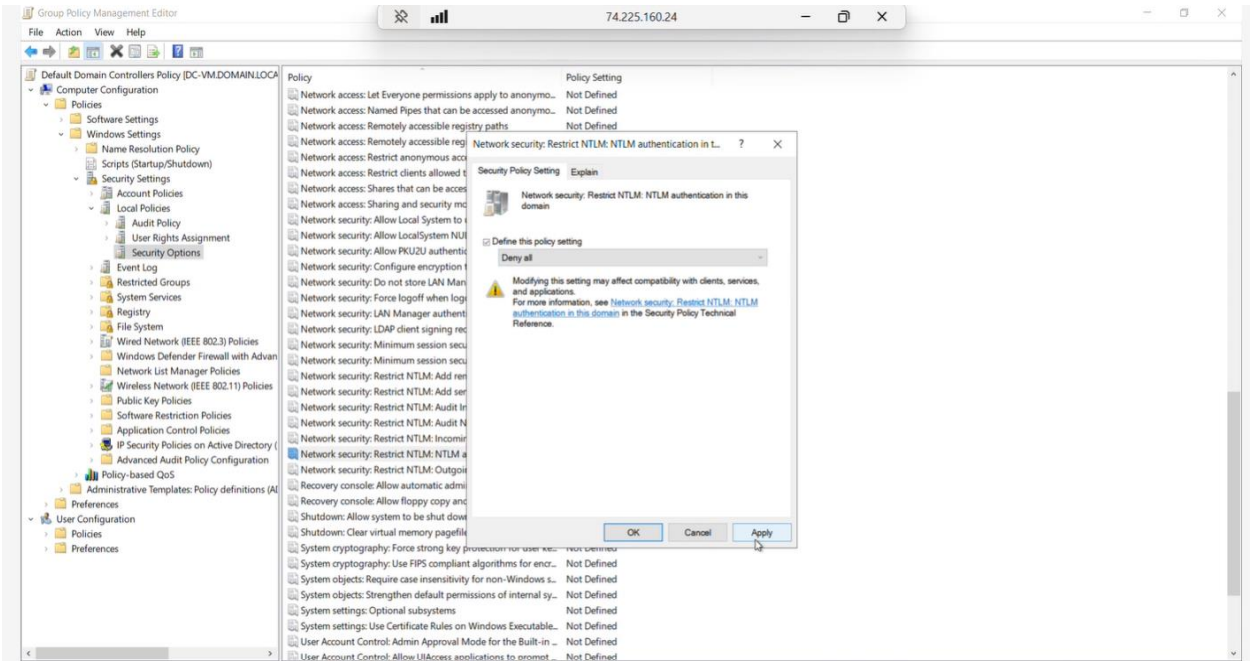
Step 50: Client – digitally sign communications (always)

Double-click Microsoft network client: Digitally sign communications (always), tick Define this policy setting, select Enabled and click Apply, then OK.



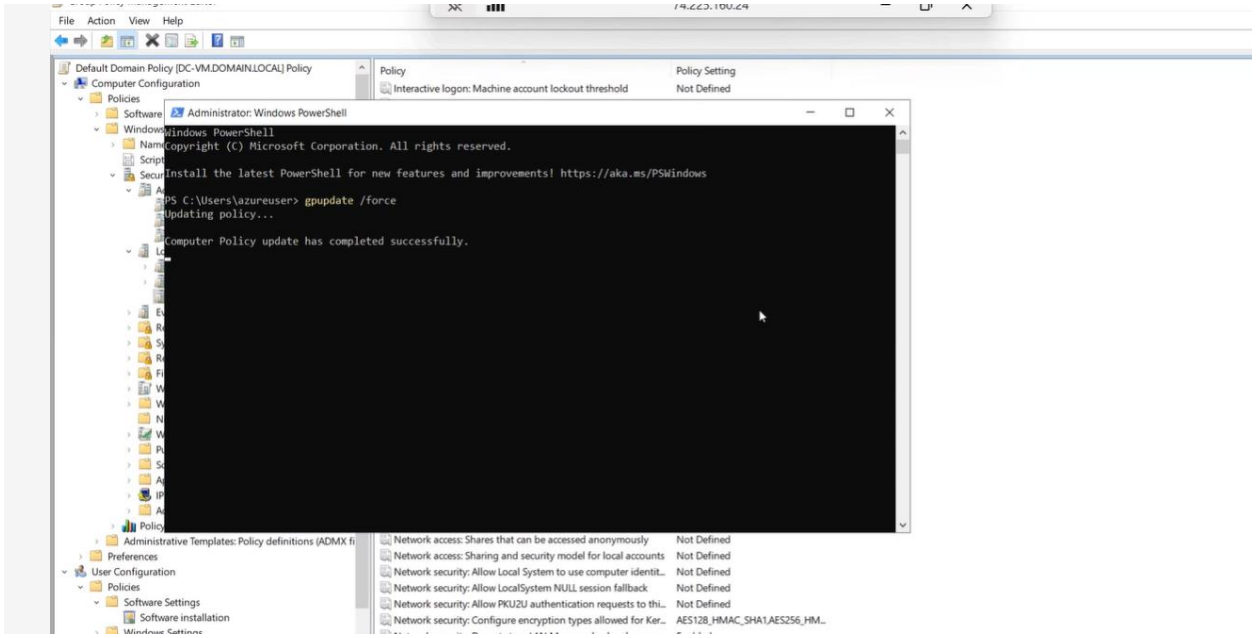
Step 51: Block NTLM authentication (Deny all)

1. On Dc-vm open Server Manager → Tools → **Group Policy Management**.
2. Expand Domain.local → Domain Controllers, right-click **Default Domain Controllers Policy** and choose **Edit**.
3. Go to Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → **Security Options**.
4. Double-click **Network security: Restrict NTLM: NTLM authentication in this domain**.
5. Tick **Define this policy setting** and select **Deny all**.
6. Click **Apply**, then **OK**.



Step 52: Apply the policy immediately

- Command: gpupdate /force



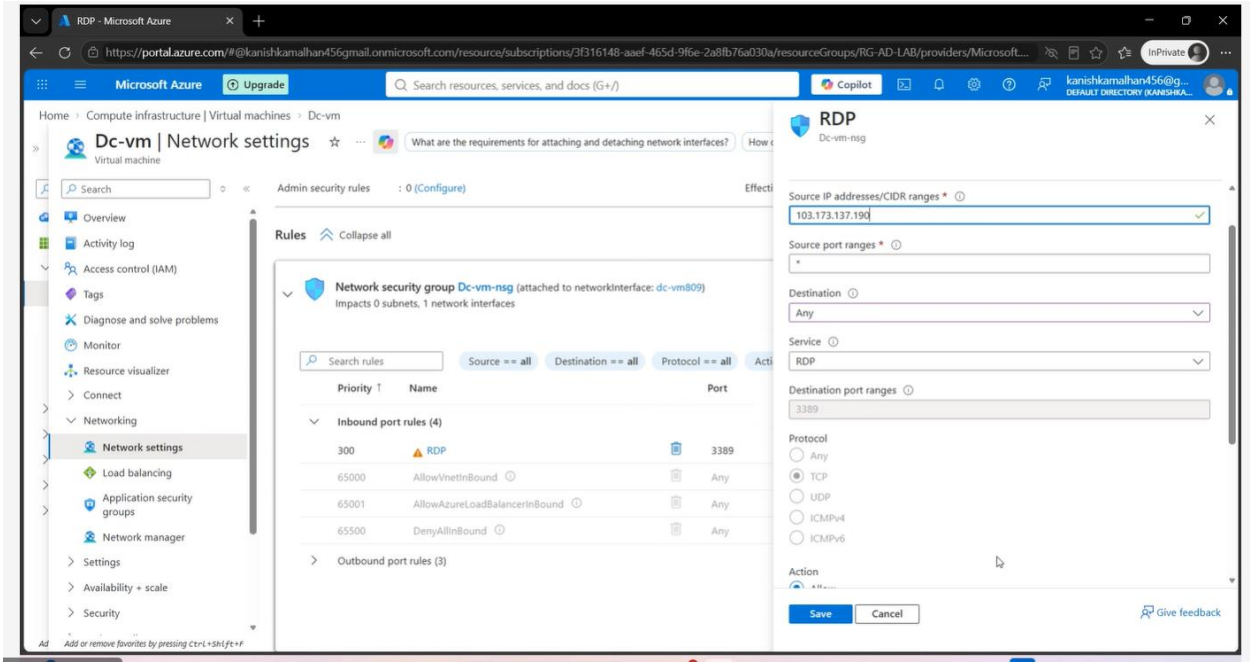
Part 9 – Brute-Force Protection (Restrict RDP in Azure)

Reduce brute-force attempts against Remote Desktop by allowing RDP only from your own public IP address.

Step 53: Restrict the RDP rule to your IP

In the Azure portal open Dc-vm → Network settings and click the inbound rule RDP (priority 300, port 3389). Change the Source IP addresses/CIDR ranges from Any to your public IP address, keep Service: RDP and Protocol: TCP, then click Save.

- Source IP addresses/CIDR ranges: 103.173.137.190 (your public IP)
- Service: RDP, Destination port: 3389, Protocol: TCP



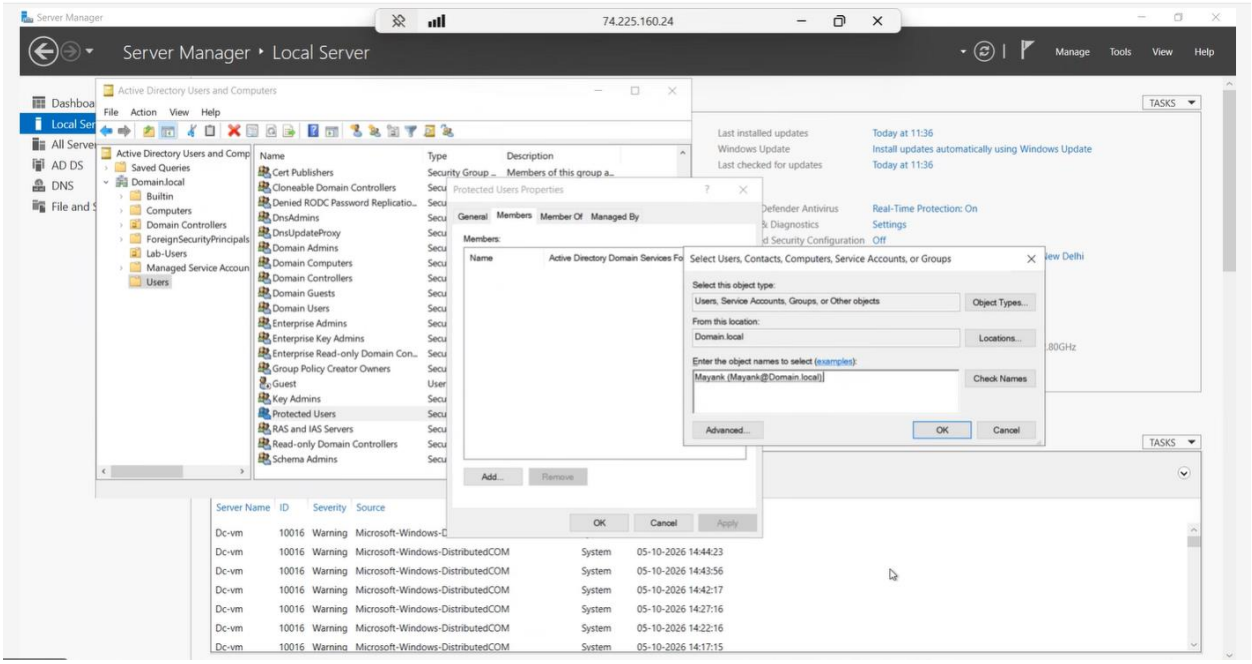
Part 10 – Pass-the-Hash Protection

Protect privileged accounts from pass-the-hash style attacks by adding them to the Protected Users group, then confirm that Kerberos with AES is being used.

Step 54: Add a user to the Protected Users group

In Active Directory Users and Computers open the Users container, double-click Protected Users and go to the Members tab. Click Add, type the user (Mayank@Domain.local), click Check Names and then OK.

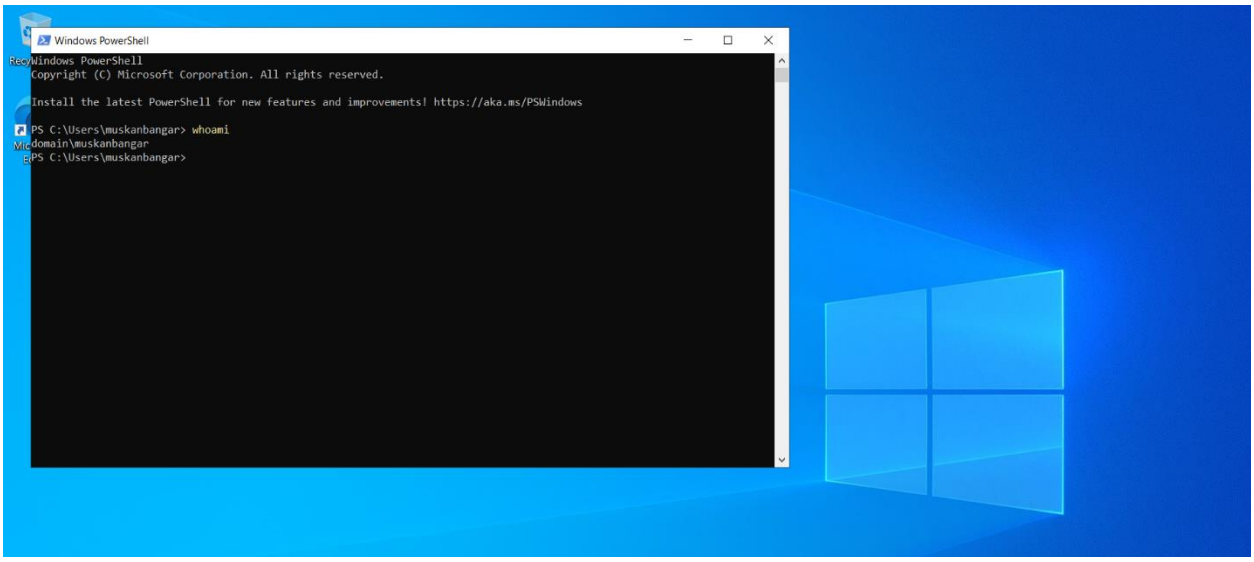
- Group: Protected Users
- Member added: Mayank (Mayank@Domain.local)



Step 55: Sign in as a domain user and check the identity (Client Vm)

Sign in as a domain user, open Windows PowerShell and run the command below. It returns domain\muskanbangar, which proves the session is a domain logon.

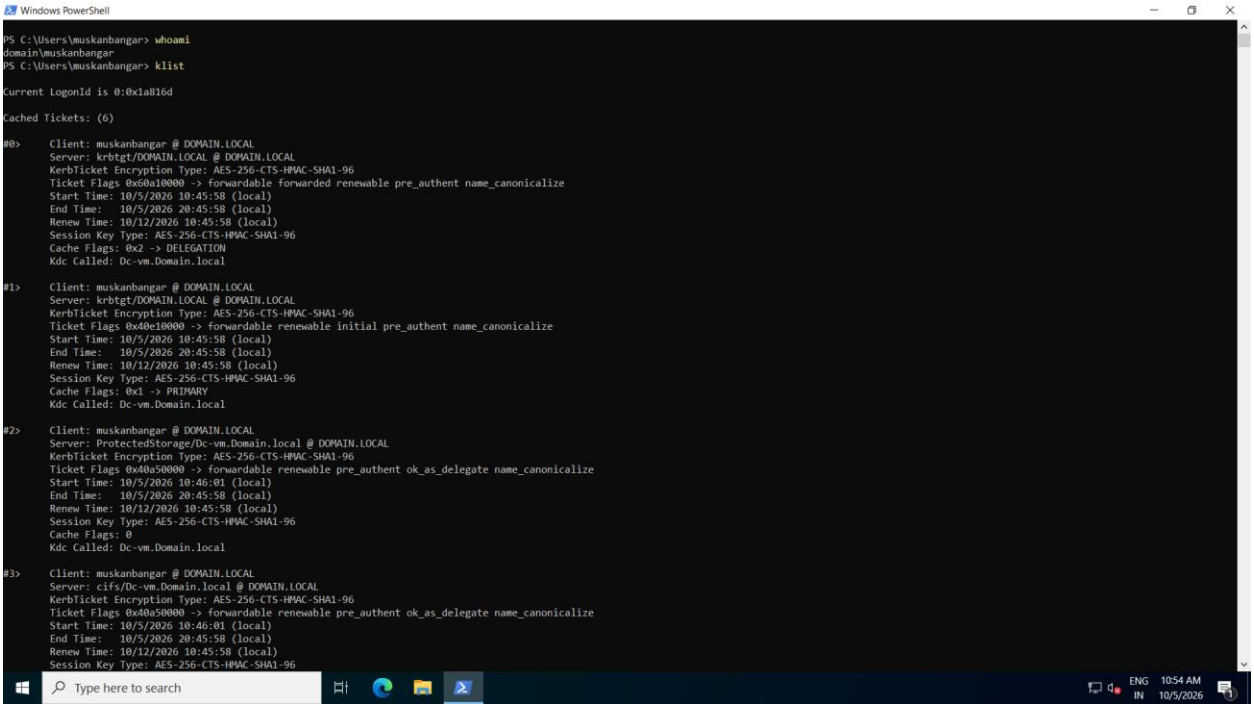
Command: whoami



Step 56: Check the Kerberos tickets (on client vm)

Run the command below. The cached tickets are issued by Dc-vm.Domain.local and use the AES-256 encryption type (AES-256-CTS-HMAC-SHA1-96), so Kerberos with AES is being used instead of NTLM.

- Command: klist



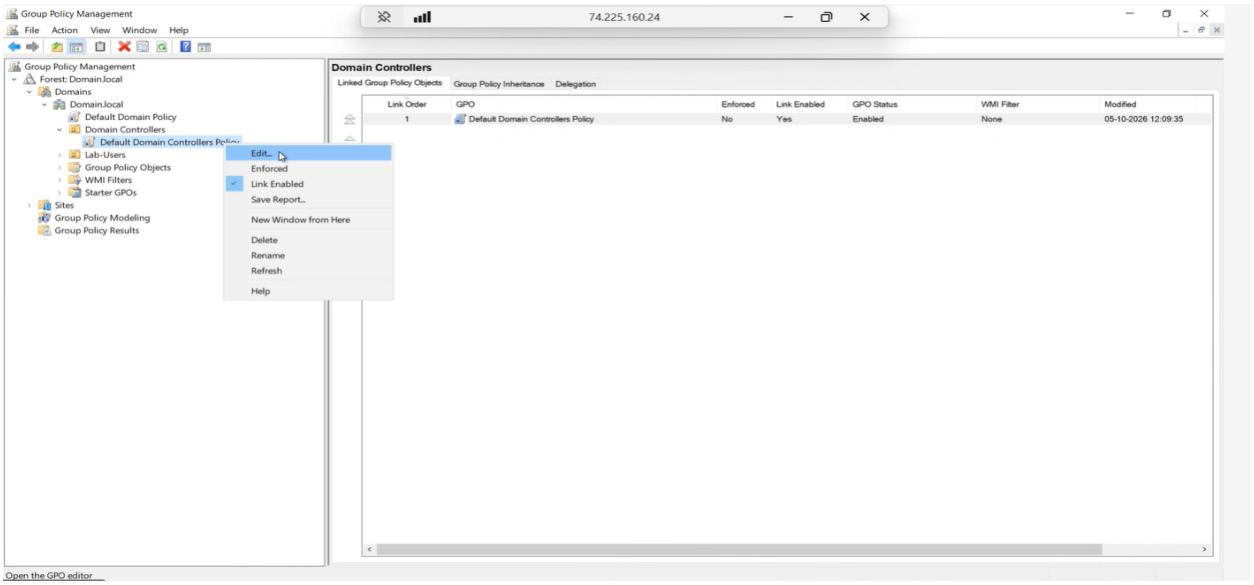
Part 11 – Audit Logging on the Domain Controller (Dc-vm)

These audit policies cover the logging requirements. Configure them in the Default Domain Controllers Policy. For each subcategory double-click it, tick Configure the following audit events, tick Success and Failure and click OK.

Requirement	Category	Subcategory
User logon/logoff	Logon/Logoff	Audit Logon, Audit Logoff, Audit Account Lockout
Failed authentication	Account Logon	Audit Credential Validation, Audit Kerberos Authentication Service, Audit Kerberos Service Ticket Operations
Privileged activity	Logon/Logoff	Audit Special Logon
Privileged activity	Privilege Use	Audit Sensitive Privilege Use
Account create/modify/delete	Account Management	Audit User Account Management, Audit Computer Account Management
Group membership changes	Account Management	Audit Security Group Management
AD object changes	DS Access	Audit Directory Service Changes
Audit policy tampering	Policy Change	Audit Audit Policy Change

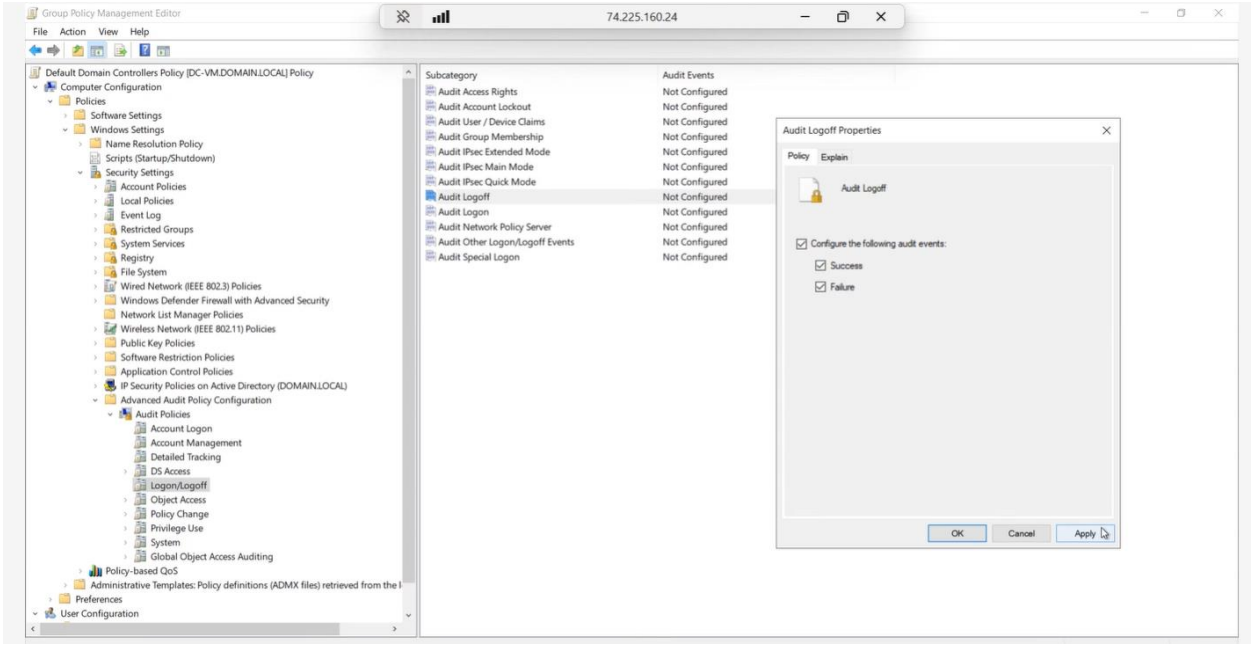
Step 57: Edit the Default Domain Controllers Policy

Open Tools → Group Policy Management → Domain.local → Domain Controllers. Right-click Default Domain Controllers Policy and choose Edit.



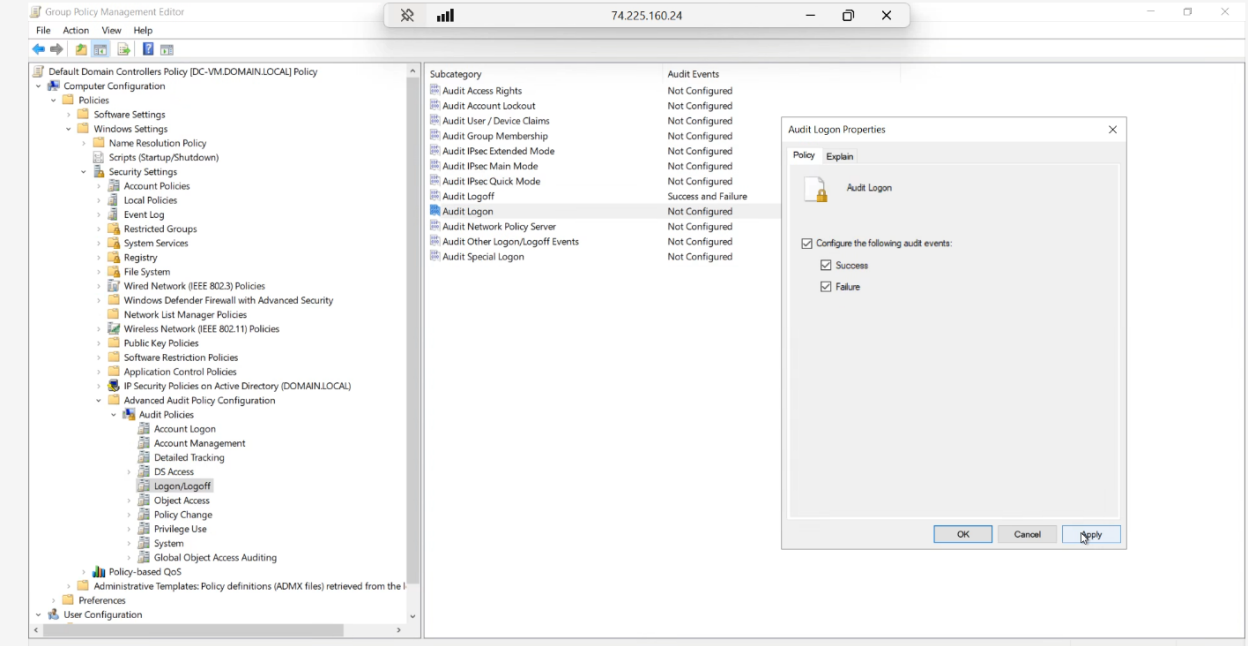
Step 58: Audit Logoff

Go to Computer Configuration → Policies → Windows Settings → Security Settings → Advanced Audit Policy Configuration → Audit Policies → Logon/Logoff. Double-click Audit Logoff, tick Configure the following audit events, tick Success and Failure and click Apply.



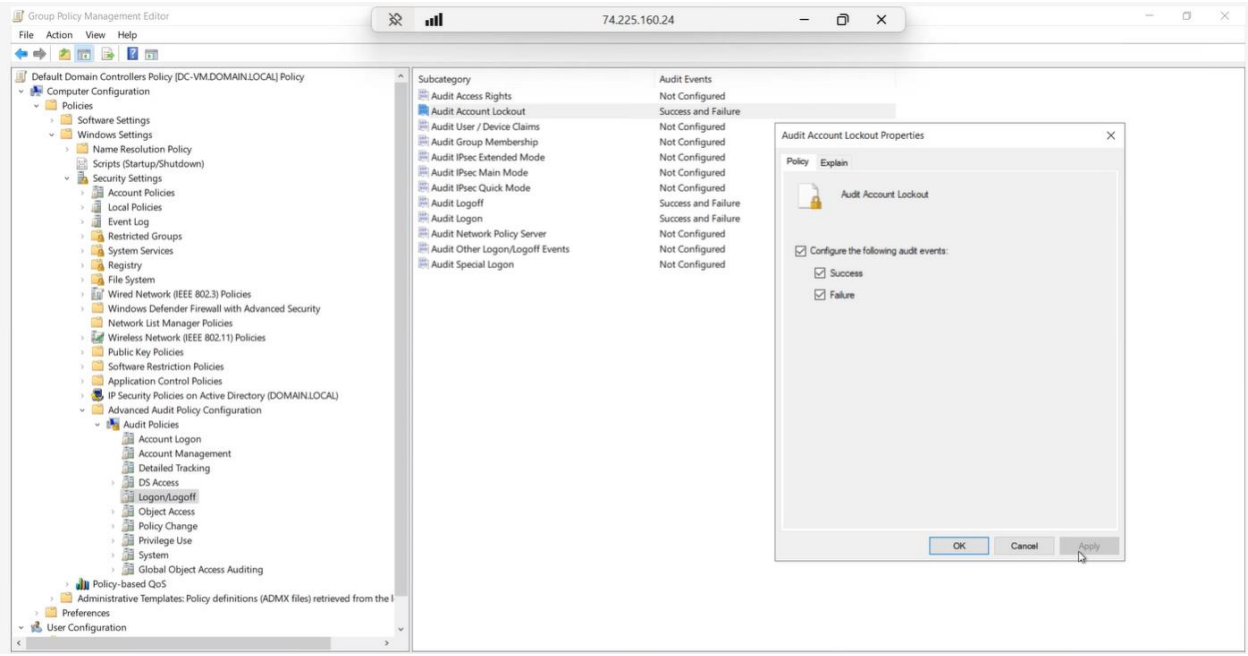
Step 59: Audit Logon

In the same Logon/Logoff category double-click Audit Logon, tick Configure the following audit events, tick Success and Failure and click Apply.



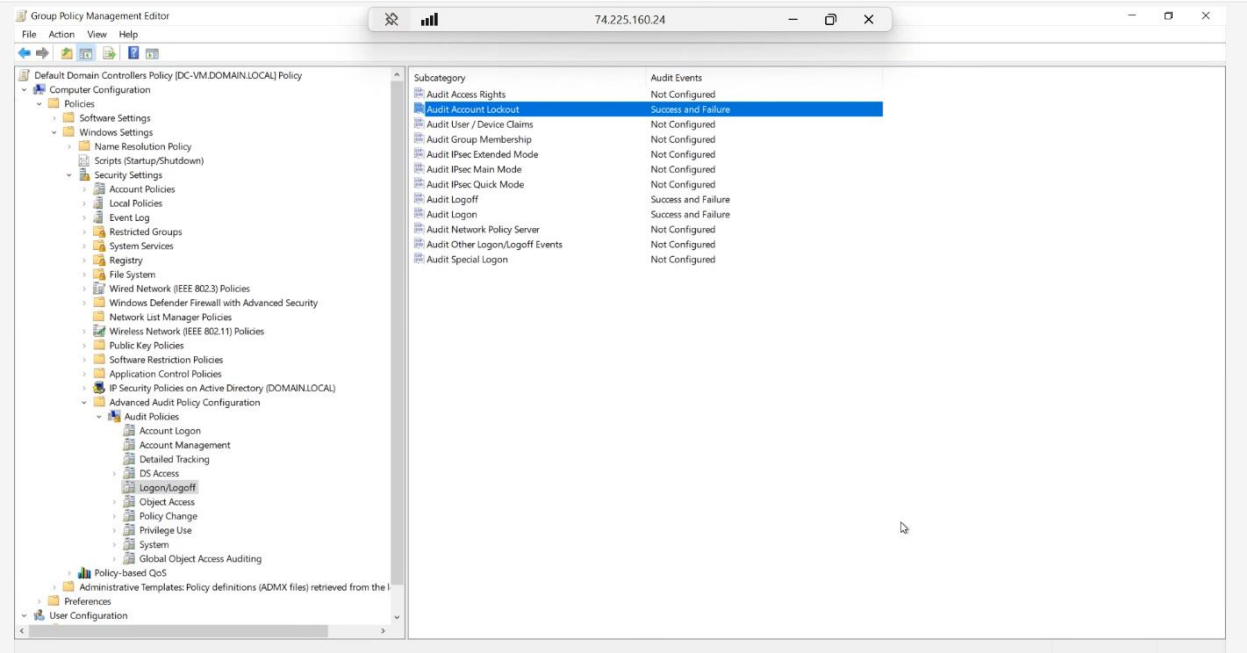
Step 60: Audit Account Lockout

Double-click Audit Account Lockout, tick Configure the following audit events, tick Success and Failure and click Apply.



Step 61: Check the Logon/Logoff results

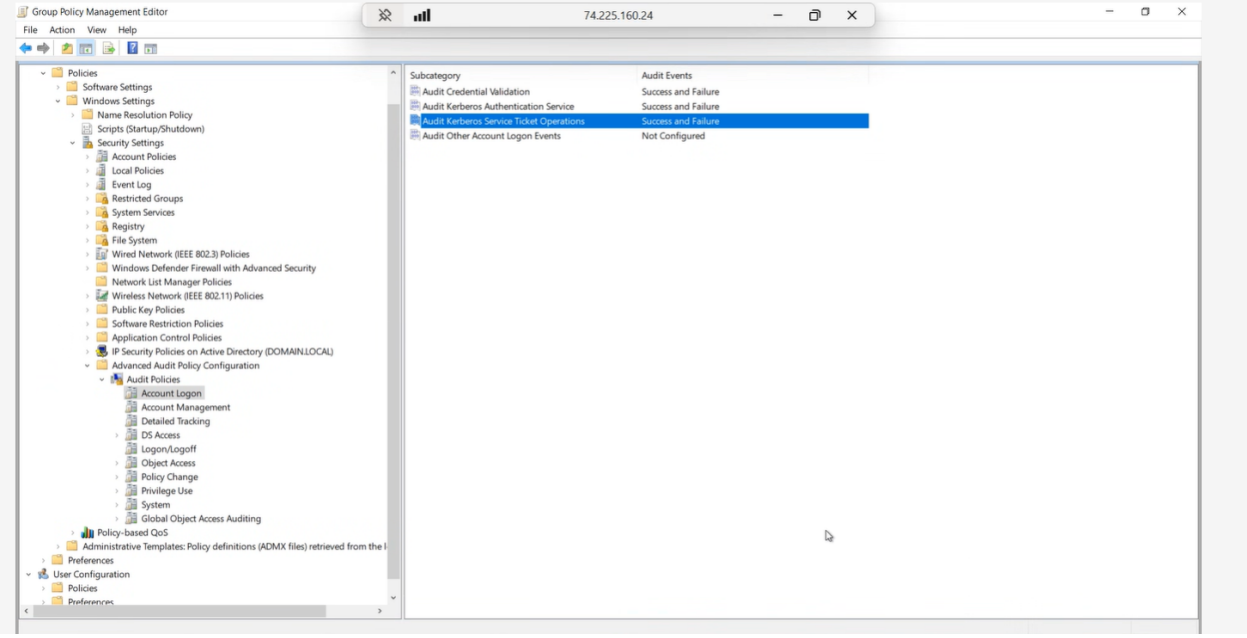
Audit Account Lockout, Audit Logoff and Audit Logon now show Success and Failure.



Step 62: Account Logon – failed authentication

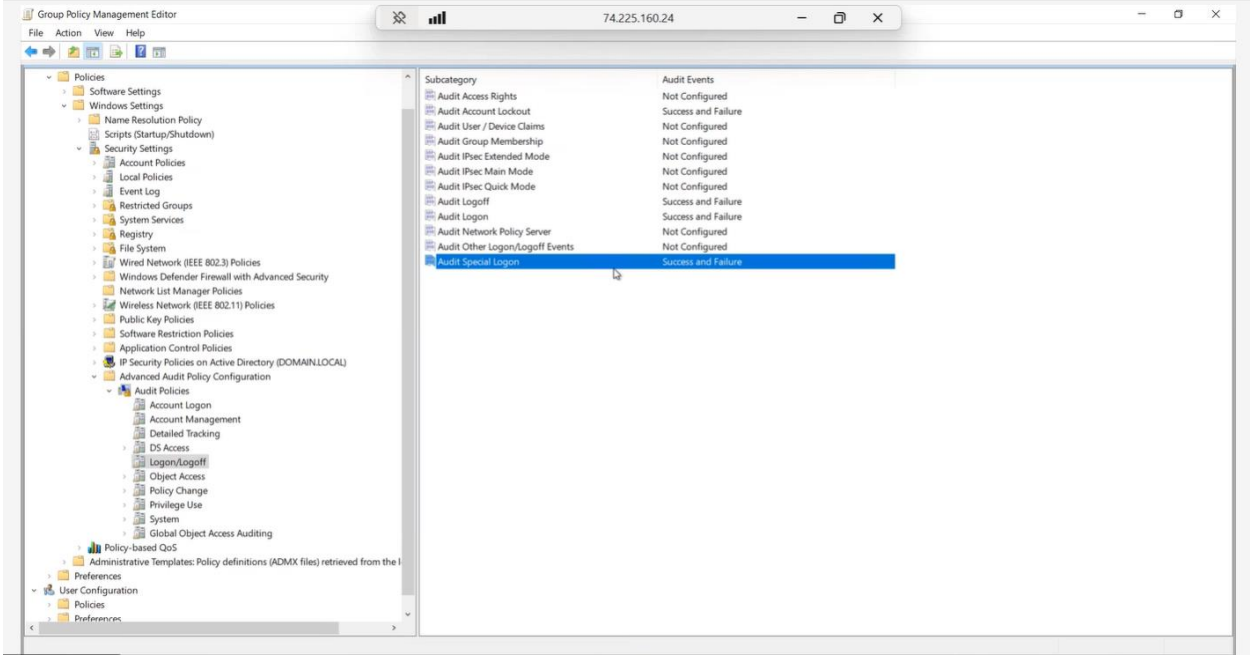
Select the Account Logon category and set Success and Failure for the three subcategories below.

- Audit Credential Validation
- Audit Kerberos Authentication Service
- Audit Kerberos Service Ticket Operations



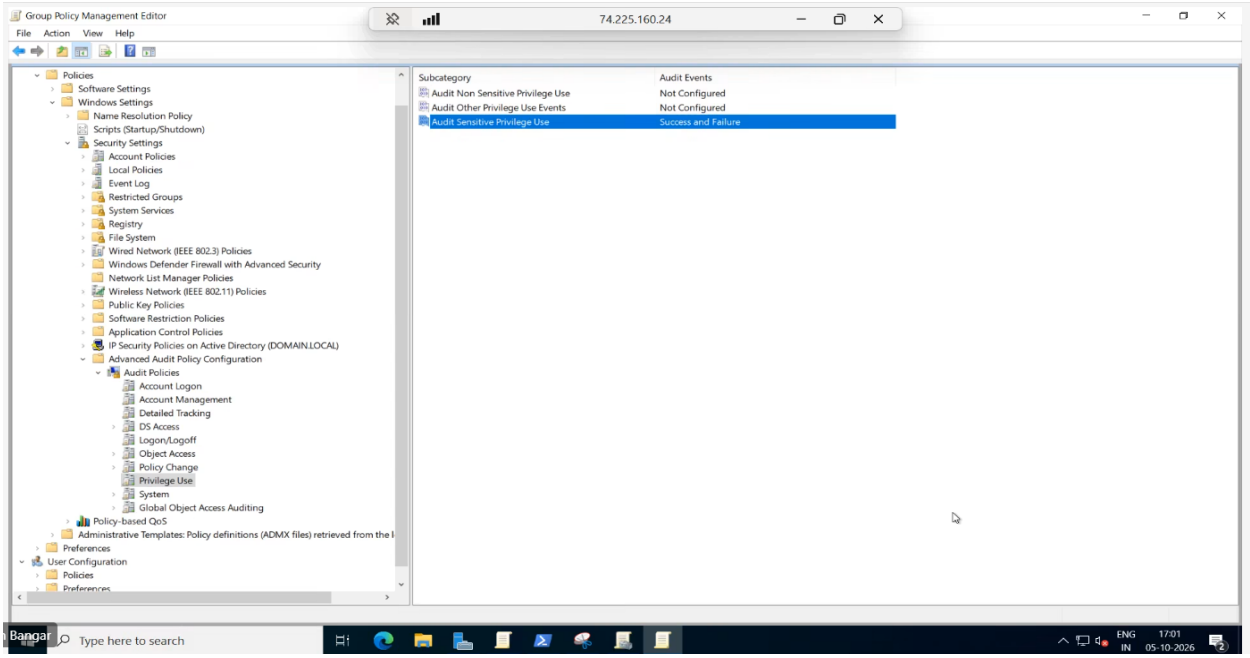
Step 63: Audit Special Logon (privileged activity)

Return to Logon/Logoff and set Audit Special Logon to Success and Failure.



Step 64: Audit Sensitive Privilege Use (privileged activity)

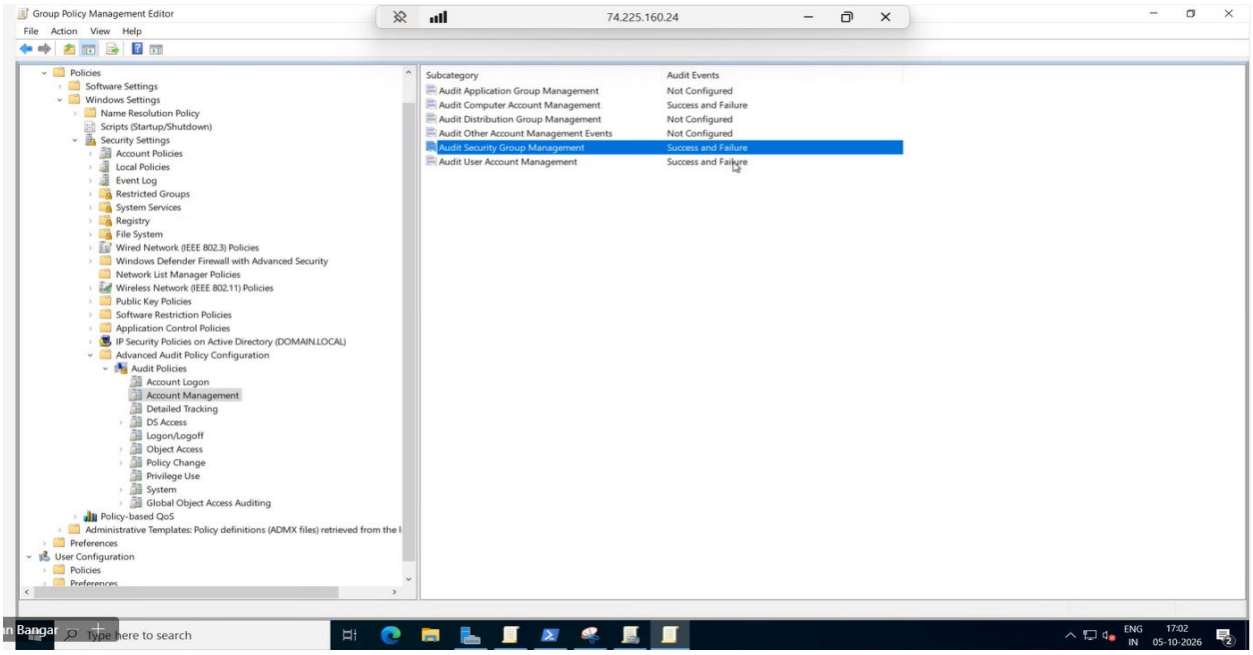
Select the Privilege Use category and set Audit Sensitive Privilege Use to Success and Failure.



Step 65: Account Management – accounts and groups

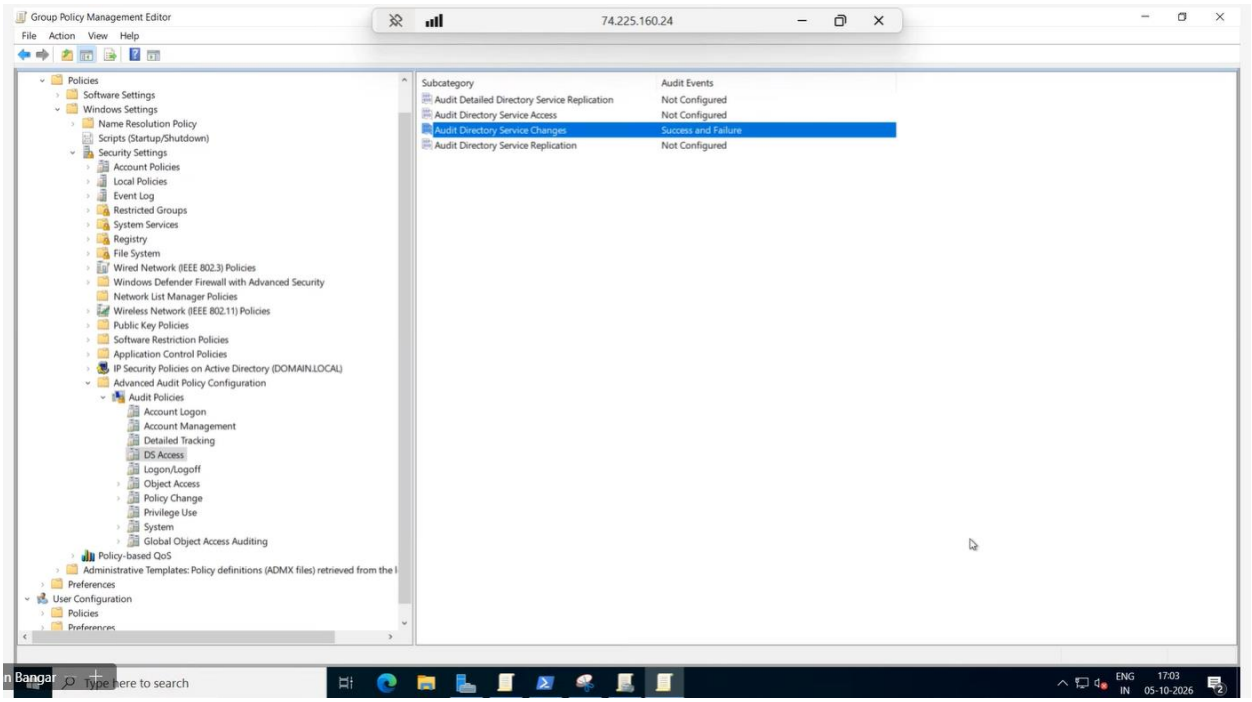
Select the Account Management category and set the subcategories below to Success and Failure.

- Audit User Account Management
- Audit Computer Account Management
- Audit Security Group Management



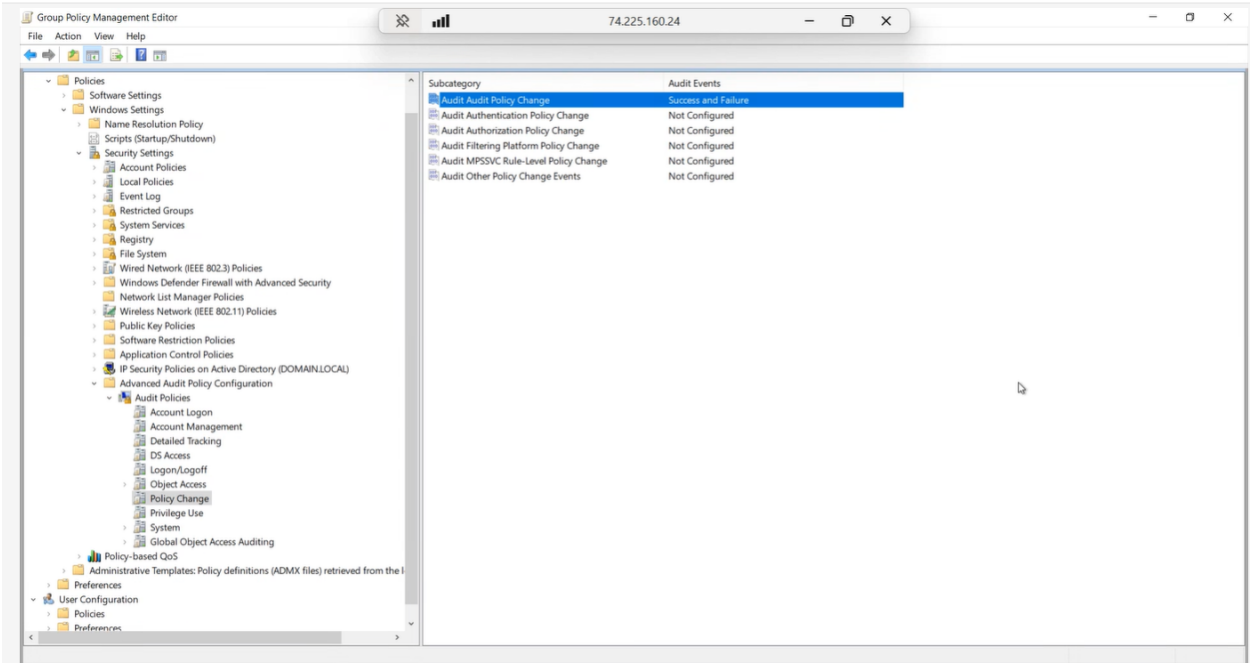
Step 66: DS Access – AD object changes

Select the DS Access category and set Audit Directory Service Changes to Success and Failure.



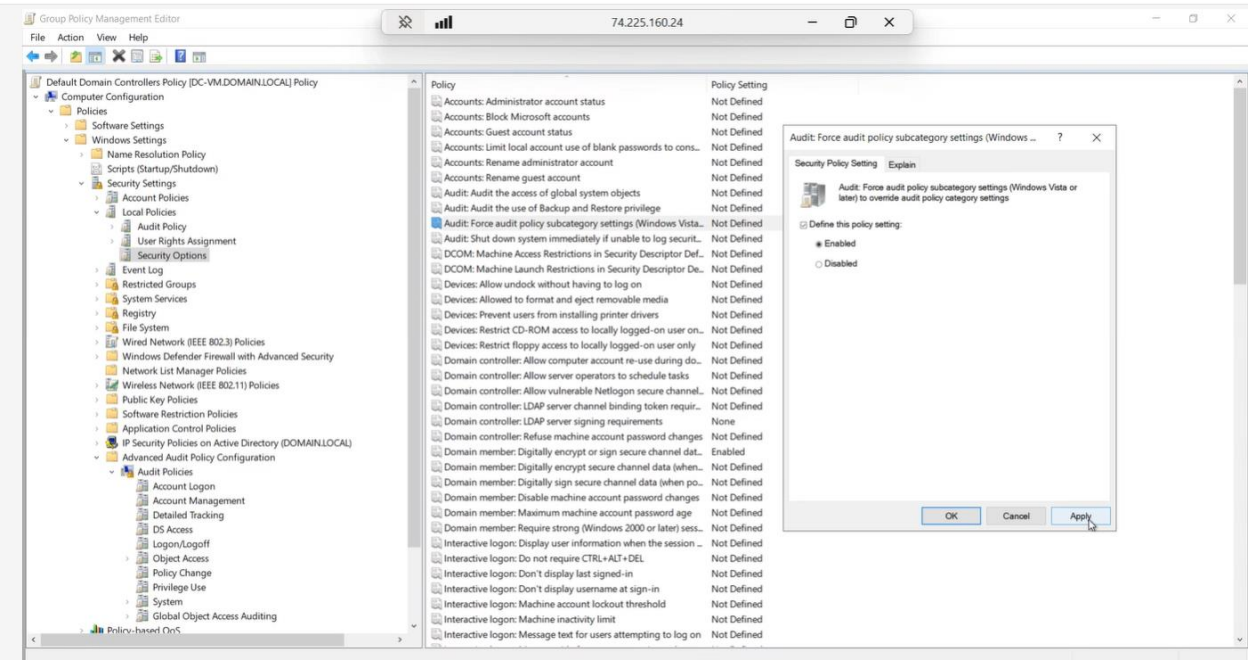
Step 67: Policy Change – audit policy tampering

Select the Policy Change category and set Audit Audit Policy Change to Success and Failure.



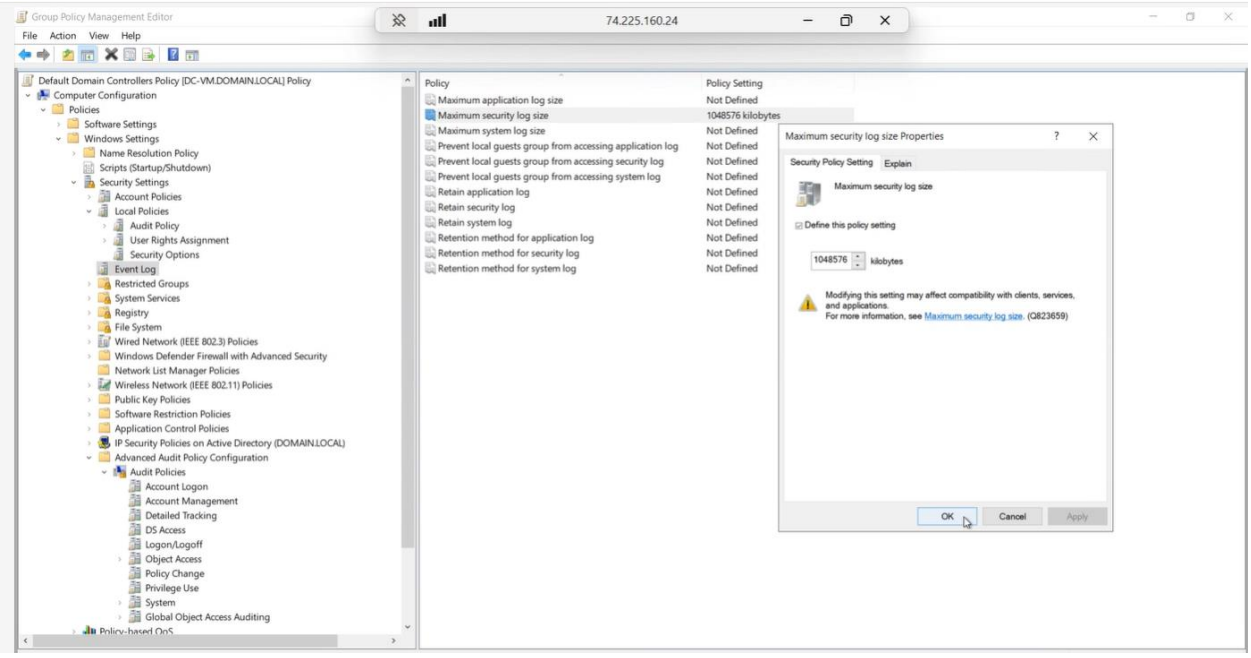
Step 68: Force audit subcategory settings

Go to Local Policies → Security Options. Double-click Audit: Force audit policy subcategory settings (Windows Vista or later) to override audit policy category settings, tick Define this policy setting, select Enabled and click Apply.



Step 69: Increase the security log size

Go to Event Log, double-click Maximum security log size, tick Define this policy setting, set it to 1048576 kilobytes and click OK.

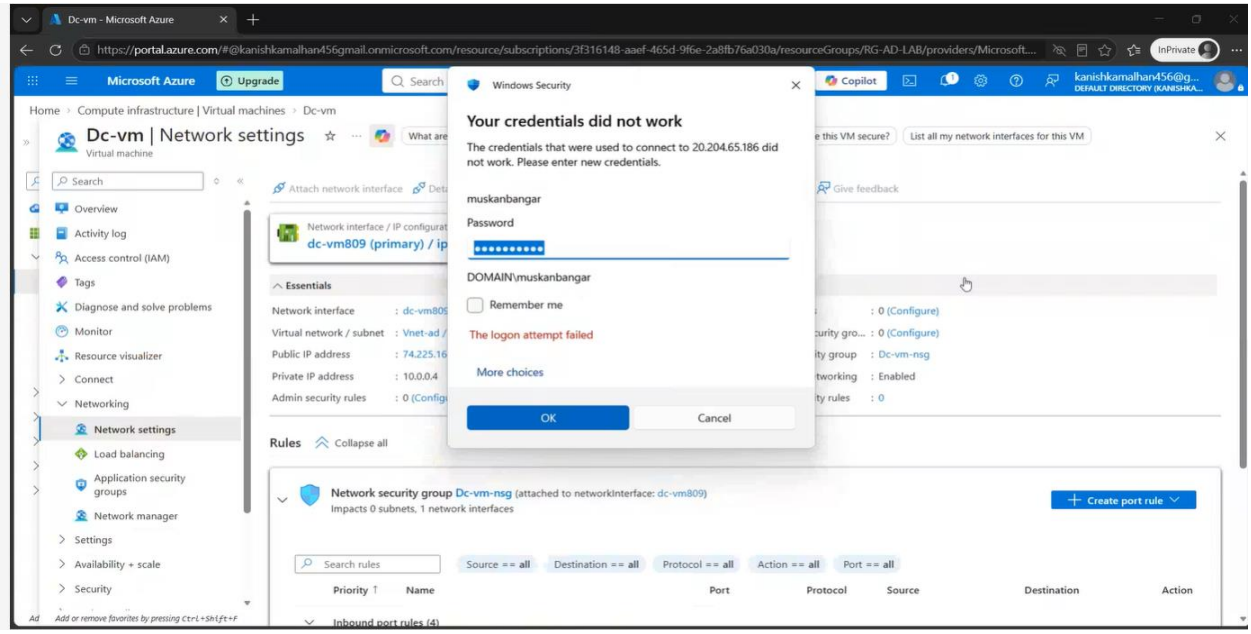


Part 12 – Verify the Policies and Audit Logs

Tested her all the lockout policy, password policy and audit logging is working fine

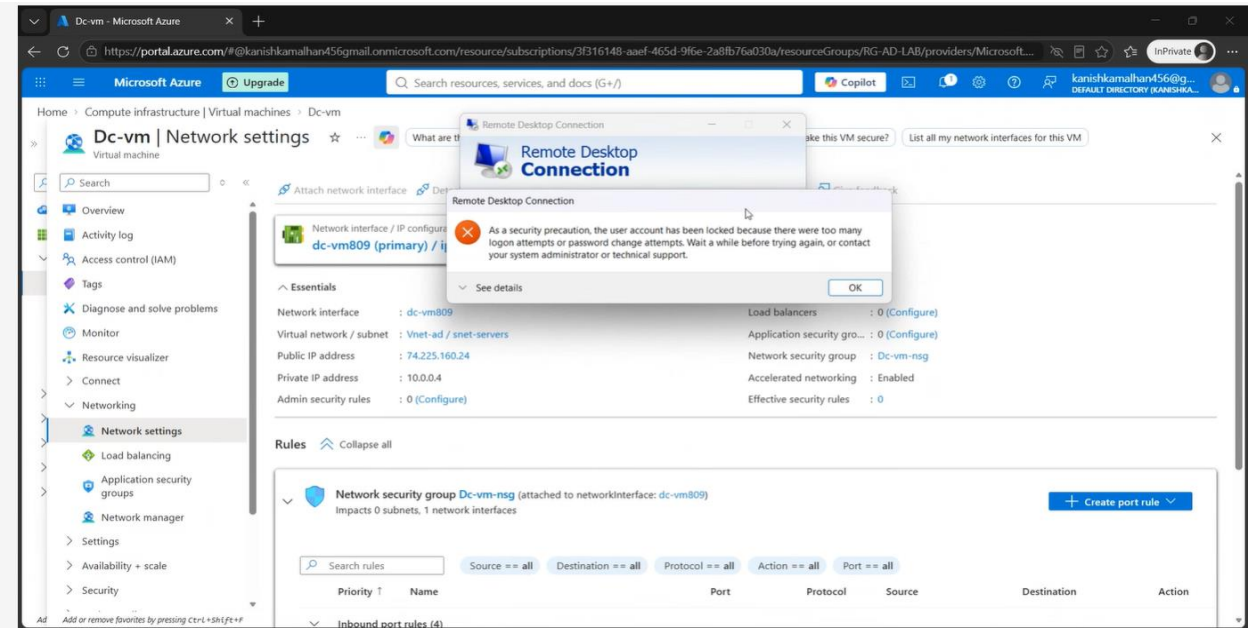
Step 70: Enter a wrong password

Tried to connect with Remote Desktop as DOMAIN\muskanbangar using a wrong password. And the Windows credentials did not work – The logon attempt failed. Repeat this several times.



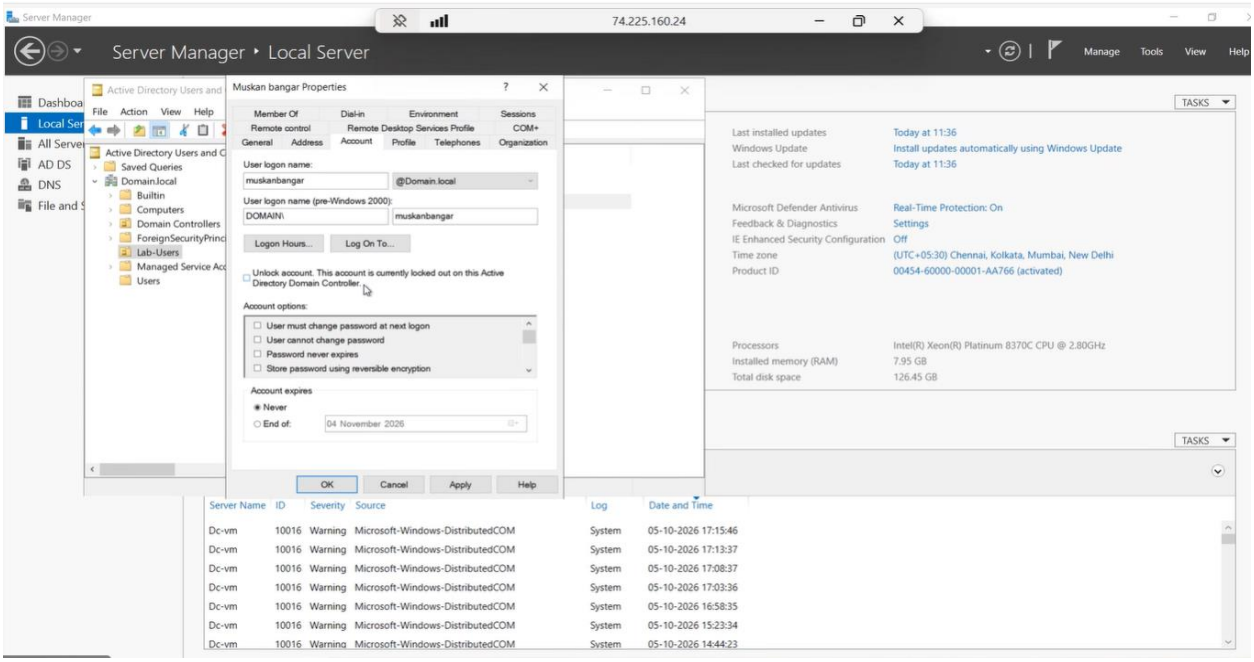
Step 71: The account gets locked

After the fifth wrong attempt Remote Desktop shows: As a security precaution, the user account has been locked because there were too many logon attempts or password change attempts. This proves the lockout threshold works.



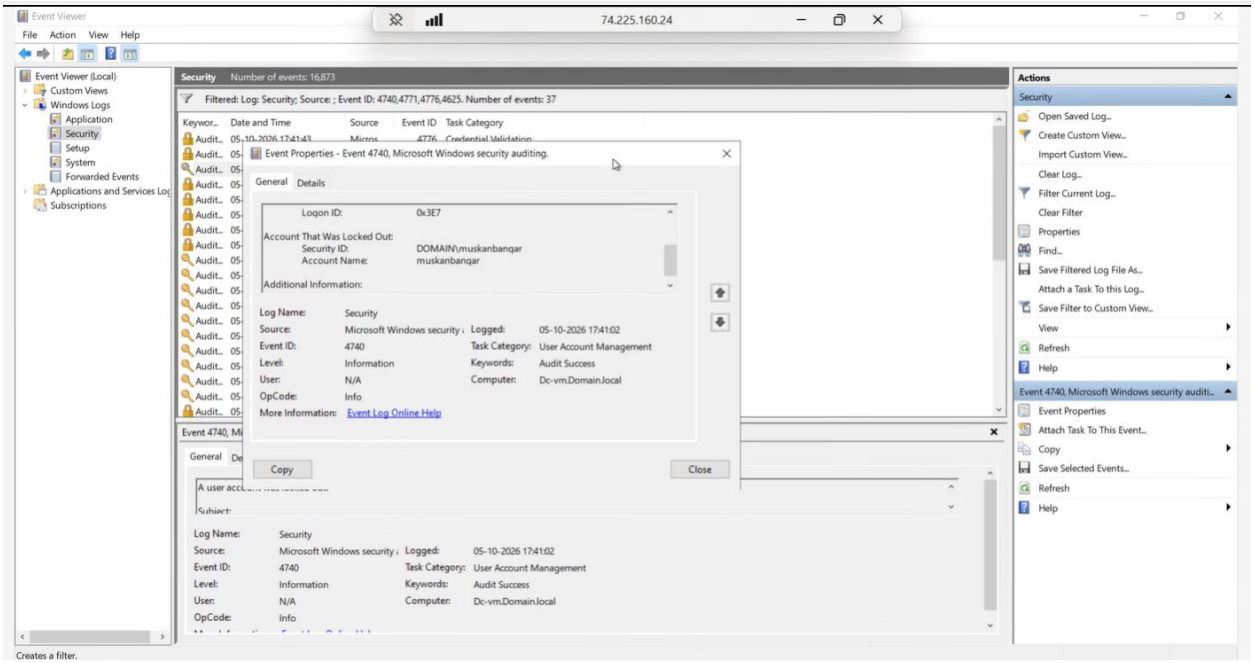
Step 72: Confirm the lockout in Active Directory

On Dc-vm open Active Directory Users and Computers → Lab-Users, double-click Muskan bangar and open the Account tab. The option Unlock account. This account is currently locked out on this Active Directory Domain Controller is shown. Tick it and click OK to unlock the user.



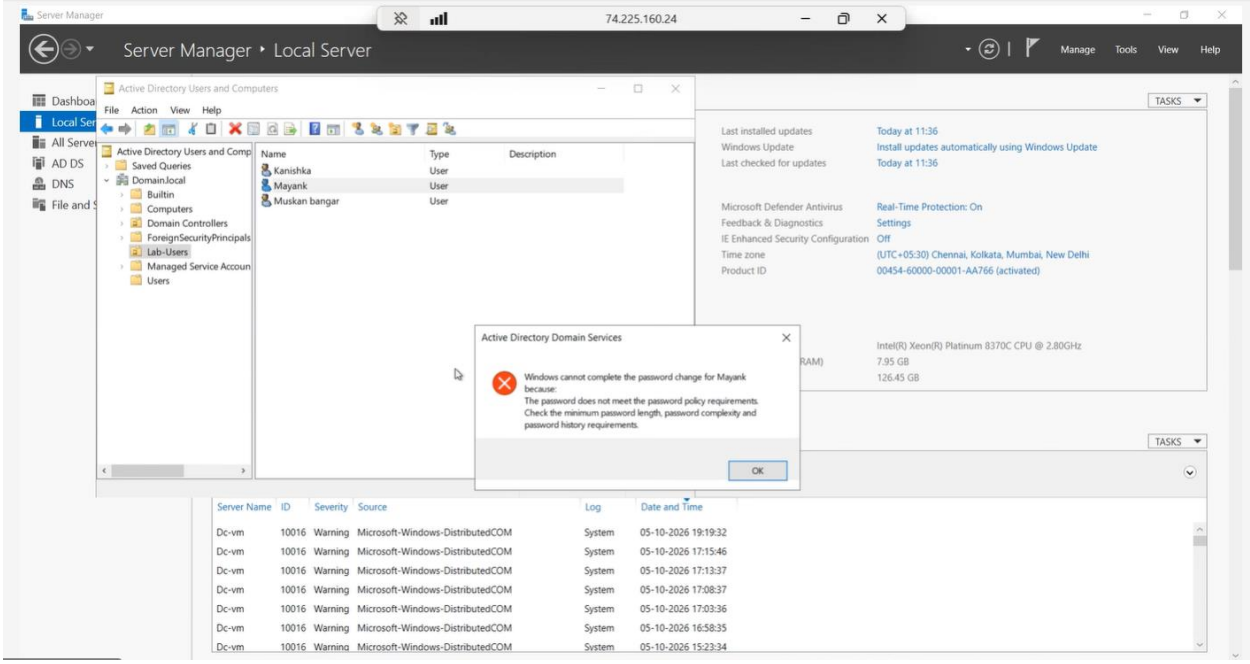
Step 73: Check the lockout event in Event Viewer

Open Event Viewer → Windows Logs → Security. Event ID 4740 (A user account was locked out) lists the locked account muskanbangar and the computer Dc-vm.Domain.local.



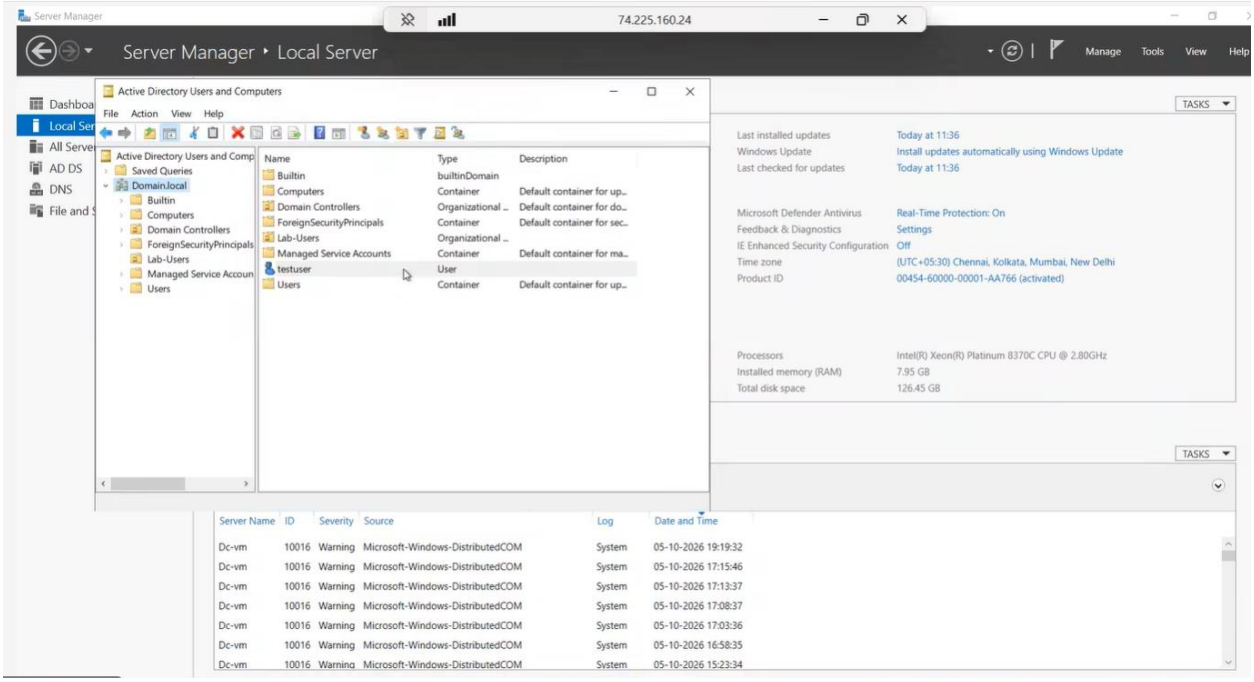
Step 74: Verify the password policy on reset

Right-click the user Mayank → Reset Password and enter a weak password. Active Directory refuses it: The password does not meet the password policy requirements. Check the minimum password length, password complexity and password history requirements.



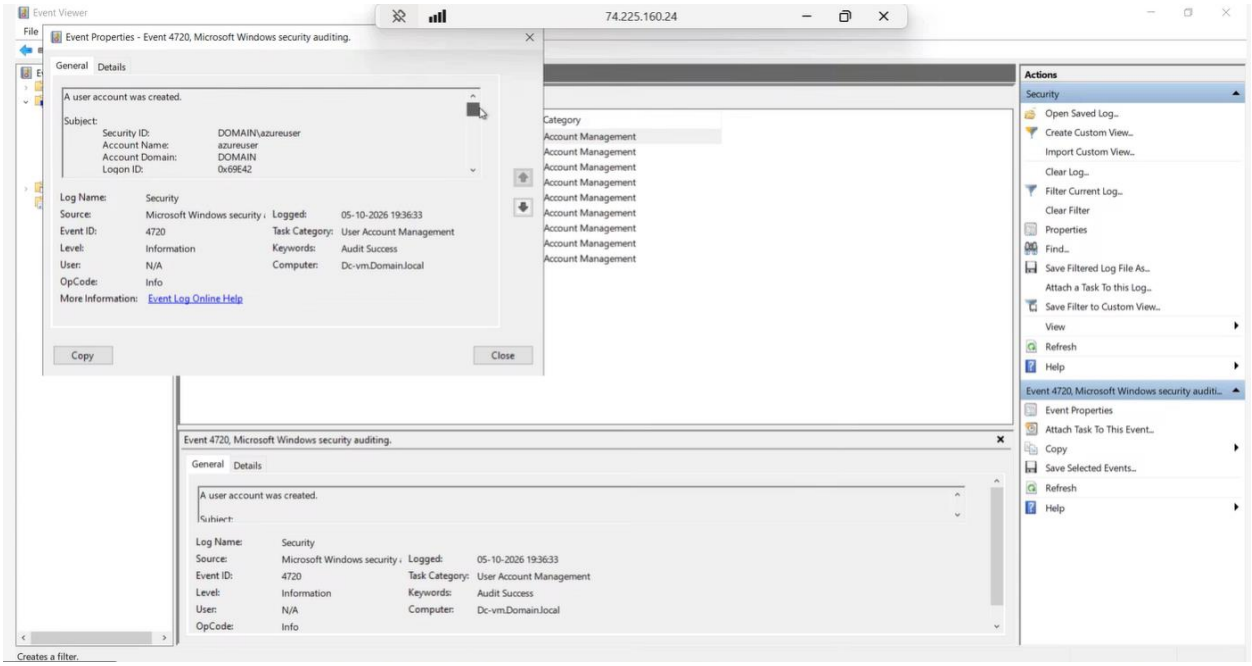
Step 75: Create a test user

To verify user-creation auditing, created a new user named testuser in Active Directory Users and Computers.



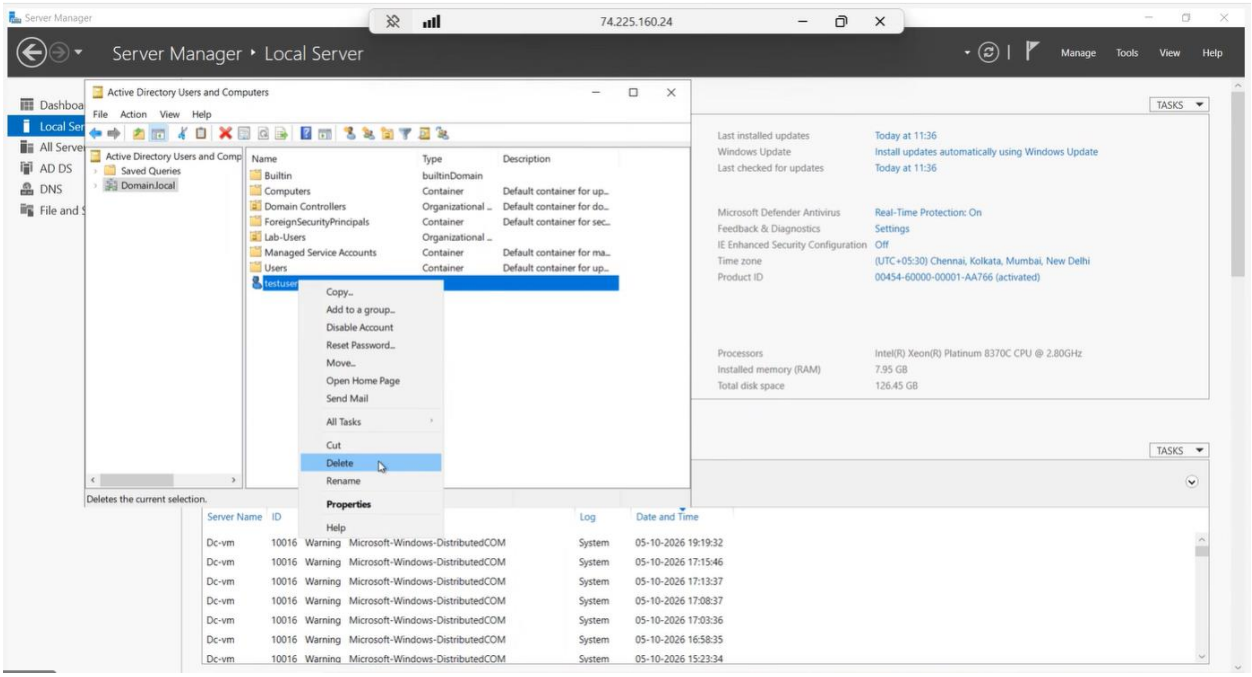
Step 76: Check the user-creation log (Event ID 4720)

In Event Viewer → Security open Event ID 4720. It reads A user account was created and shows the creator DOMAIN\azureuser.



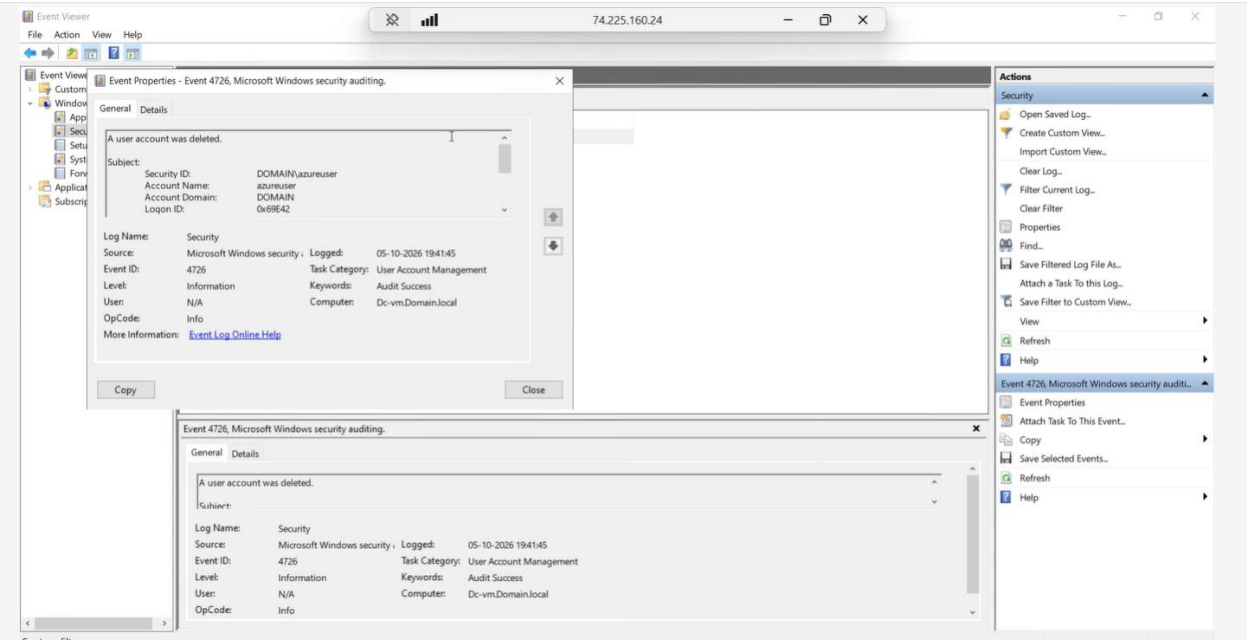
Step 77: Delete the test user

Right-click testuser and choose Delete, then confirm.



Step 78: Check the user-deletion log (Event ID 4726)

In Event Viewer → Security open Event ID 4726. It reads A user account was deleted and shows DOMAIN\azureuser as the actor.



Event IDs used for verification

Event ID	Meaning
4740	Account lockout auditing
4720	Account creation auditing
4726	Account deletion auditing